



IPNaCS & IPoPS

8th INTERNATIONAL CONFERENCE ON PHARMACEUTICAL,
NUTRACEUTICAL, AND COSMECEUTICAL SCIENCES

11th INTERNATIONAL POSTGRADUATE
CONFERENCE ON PHARMACEUTICAL SCIENCES

CONNECTED INNOVATION: DIGITAL TRANSFORMATION IN WELLNESS, LONGEVITY AND THERAPY

13-14 Aug 2026 | Mercure Kuala Lumpur Glenmarie, Shah Alam, Selangor, MALAYSIA

IPNaCS&IPoPS2026

Abstract Book

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**8th International Conference on Pharmaceutical, Nutraceutical, and
Cosmeceutical Sciences (IPNaCS) and the 11th International
Postgraduate Conference on Pharmaceutical Sciences (IPoPS) 2026**

**13-14 August 2026
Mercure Kuala Lumpur Glenmarie
Shah Alam, Malaysia**

Organised by: Faculty of Pharmacy, Universiti Teknologi MARA (UiTM)

Conference webpage: <https://ipnacs-ipops.com/>

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Welcome Message

Message from the Dean



**Professor Dato' Dr Abu
Bakar Abdul Majeed**

*Dean
Faculty of Pharmacy, UiTM*

Welcome to the 8th International Conference on Pharmaceutical, Nutraceutical, and Cosmeceutical Sciences (IPNaCS) and the 11th International Postgraduate Conference on Pharmaceutical Sciences (IPoPS) 2026, hosted by the Faculty of Pharmacy, Universiti Teknologi MARA (UiTM). This year marks our faculty's 25th Anniversary, a moment to celebrate a quarter-century of advancing pharmaceutical sciences and to look ahead to the future of global healthcare. Our theme, Digital Transformation, highlights how technologies such as artificial intelligence, digital health monitoring, data analytics, and precision medicine are reshaping research, practice, and innovation across the pharmaceutical, nutraceutical, and cosmeceutical sciences.

To foster international collaboration and scholarly exchange, IPNaCS&IPoPS2026 are proud to partner with distinguished co-organising institutions: Tokyo University of Science, IMU University, Chulalongkorn University, Universitas Indonesia, University of Santo Tomas, and Josai University. Together, our shared commitment unites renowned speakers, researchers, industry professionals, and postgraduate scholars.

On behalf of the faculty, I extend my heartfelt gratitude to all speakers, organisers, partners, sponsors, and participants. I hope IPNaCS&IPoPS2026 inspires meaningful discussions and new collaborations that will shape the future of pharmaceutical sciences.

Welcome Message

Message from the Conference Chairperson

It is my honour to welcome delegates, researchers, industry partners, and collaborators to IPNaCS&IPoPS2026. Inaugurated in 2012 through the vision of the late Professor Dr Aishah Adam, former Dean of the Faculty of Pharmacy, these conferences continue to be guided by her legacy of scientific excellence and global collaboration.

Building on past successes—including the 7th IPNaCS in Yogyakarta, hosted by Universitas Indonesia, and the 10th IPoPS in Tokyo, hosted by Tokyo University of Science, both in 2024—bringing the two events together in Malaysia marks a major milestone.



Professor Dr Mizaton
Hazizul Hasan

This joint conference unites our longstanding partner institutions: Tokyo University of Science, Universitas Indonesia, Chulalongkorn University, University of Santo Tomas, Josai University and International Medical University. This year's joint conference welcomes a rich and diverse international participation, with Indonesia and Japan contributing the largest groups of overseas presenters, alongside researchers from Iraq, the Philippines, Singapore, Russia, and the United States. This breadth of representation underscores the global relevance of our themes and the strength of our collaborative networks.

With diverse global participation, this year's compiled abstracts highlight cutting-edge advancements across pharmaceutical nanotechnology, cosmeceuticals, drug delivery, biotechnology, digital health, pharmacology, precision medicine, and regulatory science. Together, these contributions showcase the expanding frontiers of pharmaceutical innovation and the collective strength of our international research community. I extend my deepest gratitude to all contributors, partner institutions, and Dean, Prof. Dato' Dr Abu Bakar Abdul Majeed for their unwavering support.

To all our international delegates, *selamat datang*.

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Conference **Schedule**

Day 1 - Thursday, 13 August 2026

Time	Session
08:00-09:00	Registration
09:00-09:30	Opening Ceremony Doa Recital and Welcome Address by the Dean, Faculty of Pharmacy
09:30-10:00	Opening Speech Opening Speech by the Vice Chancellor of Universiti Teknologi MARA (UiTM)
10:00-11:00	Tea Break & Poster Session
11:00-12:00	Keynote Lecture Dr Azuana Ramli, Ministry of Health Malaysia <i>Stewarding Connected Innovation: Advancing Digital Transformation Across the Pharmaceutical Ecosystem</i>
12:00-12:50	Plenary Session I - Suite III Prof Dr Suzana Shahar, Universiti Kebangsaan Malaysia <i>Neuroprotective Model for Healthy Aging: The Value of Nutraceutical and Lifestyle Modification</i>
13:00-14:00	Lunch & Networking
14:00-14:50	Parallel Plenary Sessions Plenary Session II - Suite III: Prof Dr Rohana Abd Ghani, Universiti Teknologi MARA (UiTM) <i>Digital Frontier in Obesity Management: Leveraging Digital Transformation for Better Outcomes</i> Plenary Session III - Suite IV: Prof Dr Kazunori Akimoto, Tokyo University of Science (TUS) <i>Data-Driven Integrative Cancer Research: Bridging Computational Discovery, Molecular Mechanisms, and Clinical Outcomes</i>
15:00-15:30 Suite III - Track 1	Parallel Invited Talks (Tracks 1-4) Track 1: Dr Ismat binti Mohd Sulaiman, Ministry of Health Malaysia <i>Connected Data, Transforming Pharma: Lessons from MyHDW and AI Potential</i>
Suite IV - Track 2	Track 2: Associate Professor Dr Kosuke Kusamori, Tokyo University of Science (TUS) <i>Development of Cell-Based Regenerative Therapy for Lymph Node Regeneration</i>
Suite V - Track 3	Track 3: Prof Wong Tin Wui, Universiti Teknologi MARA (UiTM) <i>Natural Product Discovery and Delivery: Modulation of the Gut-Liver Axis for Hepatic Diseases</i>
Suite II - Track 4	Track 4: Associate Professor Ho Ket Li, IMU University <i>AI Revolution: Transformation of Health Education</i>
15:30-17:00	Oral Presentations Tracks 1-4 run concurrently.
17:00-18:30	Tea & Networking

Day 2 - Friday, 14 August 2026

Time	Session
09:00-09:50	Parallel Plenary Sessions Plenary Session IV – Suite III: Prof Dr Varisa Pongrakhananon, Chulalongkorn University <i>Reprogramming Cancer through Integrative Omics and Computational Pharmacology: From Biomarker Discovery to Precision Targeting of Cellular Signaling</i> Plenary Session V – Suite IV: Associate Professor Ee Pui Lai, Rachel, National University of Singapore <i>Precision Design and Programming Trap-and-Kill Peptide Nanonets for Novel Antimicrobial Therapy</i>
10:00-10:30	Tea Break & Poster Session
10:30-11:00	Parallel Invited Talks (Tracks 5-8)
Suite III - Track 5	Track 5: Associate Professor Dr Shazia Jamshed, IMU University <i>Applications of Digital Health in Pharmacy Practice and Education: Current Insights and Future Directions</i>
Suite IV - Track 6	Track 6: Prof Dr Teh Lay Kek, Universiti Teknologi MARA (UiTM) <i>From Pharmacogenomics to Pharmacophytogenomics: Diversity-Driven Pathways in Precision Health</i>
Suite V - Track 7	Track 7: Prof Dr Takehisa Hanawa, Tokyo University of Science (TUS) <i>Strategies for Improving the Solubility of Poorly Water-Soluble Drugs</i>
Suite II - Track 8	Track 8: Prof Aleth Therese Dacanay, University of Santo Tomas (UST) <i>From Classroom to Connected Care: Digital Transformation of Pharmacy Education and Practice</i>
11:00-12:15	Oral Presentations Tracks 5-8 run concurrently.
12:30-14:30	Lunch & Networking
14:30-14:45	Registration
14:45-16:30	Closing Plenary & Professorial Lecture - Suite III Prof Dr Kalavathy Ramasamy, Universiti Teknologi MARA (UiTM) <i>Listen to Your Gut</i>
16:30-17:00	Tea Break
17:00-17:30	Prize-Giving Awards for Best Oral & Poster Presentations and Collaborator Recognition
17:30-18:00	Closing Remarks Next Steps & Farewell

List of **Speakers**

Plenary Speakers



Keynote: Dr Azuana Ramli

Ministry of Health Malaysia

Dr. Azuana Ramli is Malaysia's Deputy Director-General of Health (Pharmaceutical Services) at the Ministry of Health. Appointed in April 2025, the UKM-trained pharmacist and former NPRA Director leads national drug regulation, security, and supply policies.



Prof Dr Suzana Shahar

Universiti Kebangsaan Malaysia

Prof. Dr. Suzana Shahar is the Dean of the Faculty of Health Sciences at Universiti Kebangsaan Malaysia (UKM). A Professor of Nutrition and Dietetics with a PhD from the University of Sheffield, she founded the Centre of Healthy Aging and Wellness (H-CARE) and leads key national research on geriatric health, dementia prevention, and cognitive frailty.



Prof Dr Rohana Abd Ghani

Universiti Teknologi MARA

Prof. Dr. Rohana Abdul Ghani is a Professor of Medicine and Senior Consultant Endocrinologist at Universiti Teknologi MARA (UiTM). She serves as the President of the Malaysian Society of Obesity (MYOS) and leads key clinical programs for metabolic and endocrine disorders at Hospital Al-Sultan Abdullah.



Prof Dr Kazunori Akimoto

Tokyo University of Science

Prof. Dr. Kazunori Akimoto is a Professor of Medicinal and Life Sciences at the Tokyo University of Science. A medical scientist specialising in molecular oncology and medical data science, he leads research on cellular signalling mechanisms, cancer progression, and data-driven therapeutic innovations.

Plenary Speakers



Prof Dr Varisa Pongrakhananon

Chulalongkorn University

Prof. Dr. Varisa Pongrakhananon is a Professor at the Faculty of Pharmaceutical Sciences, Chulalongkorn University. A pharmaceutical scientist specializing in cancer cell biology and pharmacology, her award-winning research focuses on microtubule dynamics, lung cancer metastasis, and novel target therapeutics.



Assoc Prof Ee Pui Lai

National University of Singapore

Assoc. Prof. Dr. Rachel Ee Pui Lai is an Associate Professor at the Department of Pharmacy and Pharmaceutical Sciences, National University of Singapore (NUS). A pharmaceutical scientist and registered pharmacist, her research focuses on antimicrobial peptides, hydrogel biomaterials, and targeted therapeutic delivery systems.



Prof Dr Kalavathy Ramasamy

Universiti Teknologi MARA

Prof. Dr. Kalavathy Ramasamy is a Professor of Microbiology at UiTM's Faculty of Pharmacy and Vice-President of the Malaysian Society of Lactic Acid Bacteria. Her research focuses on using probiotics, prebiotics, postbiotics, and bioactive metabolites to prevent age-related diseases, specifically targeting the gut-brain and gut-liver axes.

Invited Speakers



**Dr. 'Ismat binti
Mohd Sulaiman**

*Ministry of Health
Malaysia*

Dr. 'Ismat binti Mohd Sulaiman specializes in health informatics, semantic interoperability, and clinical text mining.

**Assoc Prof Dr.
Kosuke Kusamori**

*Tokyo University
of Science*

Dr. Kosuke Kusamori specializes in cellular drug discovery, biomaterial science, and advanced drug delivery systems (DDS).



Prof. Wong Tin Wui

*Universiti
Teknologi MARA*

Professor Dr. Wong Tin Wui is a highly cited Malaysian pharmaceutical scientist specializing in particle design and advanced nanodrug delivery.

**Assoc Prof Ho
Ket Li**

IMU University

Dr. Ho Ket Li specializes in cancer biology, immunology, biotherapeutics, natural products, and meta-analysis tools.



**Assoc Prof Dr
Shazia Jamshed**

IMU University

Dr. Shazia Jamshed has published over 100 research papers on healthcare systems and pharmacy behavior.

Prof. Teh Lay Kek

*Universiti
Teknologi MARA*

Professor Dr. Teh Lay Kek's research centers on understanding how genetic variations in Asian populations impact drug safety and effectiveness.



**Prof Dr. Takehisa
Hanawa**

*Tokyo University
of Science*

Professor Dr. Takehisa Hanawa is recognized for his contributions to drug formulation and advanced pharmaceutical technology.

**Prof. Aleth
Therese Dacanay**

*University of
Santo Tomas*

Professor Dr. Aleth Dacanay is a respected pharmacy educator and leader advancing professional standards, training, and national pharmacist development.



Speaker **Abstracts**

KEYNOTE LECTURE

Stewarding Connected Innovation: Advancing Digital Transformation Across the Pharmaceutical Ecosystem

Dr. Azuana Ramli

Ministry of Health Malaysia

Digital transformation is reshaping every stage of the pharmaceutical ecosystem, from scientific discovery and regulatory governance to pharmaceutical services, surveillance and patient care. Malaysia's evolving experience illustrates how this transformation can be advanced while keeping responsible stewardship at its core. Key developments include the growing role of artificial intelligence in scientific discovery and regulatory oversight, the modernisation of regulatory infrastructure, and digital platforms such as MyPhIS and MyUBAT in strengthening services and patient access. Emerging risks in the digital marketplace are also driving new approaches to surveillance, product authentication, traceability and post-market safety. Looking ahead, meaningful progress will depend on better connecting technology, data, evidence and people across the pharmaceutical ecosystem, supported by collaboration across government, academia, industry and healthcare professionals. Ultimately, connected innovation must remain anchored in patient safety, equitable access, responsible data governance and public trust.

PLENARY SESSION I

Neuroprotective Model for Healthy Aging: The Value of Nutraceutical and Lifestyle Modification

Prof. Dr. Suzana Shahar

*Centre for Healthy Aging and Wellness, Faculty of Health Sciences, Universiti
Kebangsaan Malaysia 43000 Kuala Lumpur, Malaysia*

Aging is a physiologic state in which a progressive decline of organ functions is accompanied with the development of age-related diseases including cognitive decline leading to dementia. Key factors in aging included cellular senescence, molecular damage and physiological changes. The oxidative stress theory of aging is based on the hypothesis that age-associated functional losses, such as cognitive decline are due to the accumulation of RONS-induced damages. Antioxidant therapy could positively affect the natural history of several diseases, but further investigation is needed to evaluate the real efficacy of these therapeutic interventions. The role of functional foods was considered as a potential treatment of dementia and Alzheimer's disease through inhibition of acetylcholinesterase as well as similar treatments based on herbs, spices and antioxidants rich foods. Tropical rich fruits and traditional herbs and vegetables are rich sources of antioxidants and polyphenols; have shown to be beneficial in improving oxidative stress and cognitive function among aging populations. Prudent dietary patterns with inclusion of these functional foods, together with lifestyle modifications, have the potential in advancing neuroprotective model for older adults for healthy longevity. Their effects, however, are not attributable only to antioxidant properties. They are also rich sources of dietary fibres that have a role in short chain fatty acids metabolism in the gut brain axis pathways. Larger and longer duration of clinical trials are needed to provide stronger evidence of food and lifestyle-based model for neuroprotection towards healthy longevity. Among the lifestyle-based model, multidomain intervention has been proposed for dementia risk reduction. In Malaysia, this model, ie. WE RISE, AGELESS Trial and WE CARE have been successfully conducted and showed positive findings among older adults at risk of dementia. In conclusion, although larger clinical trials are needed, the present scientific evidence could be tabulated as neuroprotective model for healthy aging.

PLENARY SESSION II

Digital Frontier in Obesity Management: Leveraging Digital Transformation for Better Outcomes

Prof. Dr. Rohana Abd Ghani

Universiti Teknologi MARA (UiTM), Malaysia

The global obesity epidemic presents an unprecedented public health and economic challenge. Traditional lifestyle interventions often suffer from poor long-term adherence and limited scalability. To address this, the digital frontier in obesity care leverages telehealth, mobile health (mHealth) applications, wearable devices, and artificial intelligence (AI) to optimize clinical outcomes and expand universal access. Personalised digital therapeutics dynamically tailor interventions to individual biological and behavioural needs. Remote monitoring, smart scales, and AI-powered conversational chatbots provide continuous, real-time behavioural coaching. Stand-alone, fully autonomous digital programs utilizing video feedback, emotional eating coping strategies, and friendly social competition can achieve significant weight losses of up to 8.4% at 12 months. Furthermore, pairing these digital behavioural strategies with emerging obesity management medications (OMMs) maximizes metabolic outcomes, supports medication adherence, and preserves muscle mass. AI has a significant potential to transform predictive analytics and diagnostics. Supervised learning models utilizing electronic health records achieve over 85% accuracy in identifying early childhood obesity risks. Additionally, computer vision and deep learning has been able to automate body composition analyses through CT, MRI, or 3D scans, enabling precise visceral fat quantifications and cardiometabolic risk stratifications beyond simple BMI metrics. Despite these advantages, several key barriers impede widespread clinical integration. Challenges include high user attrition after three to six months, data privacy and security vulnerabilities, demographic and algorithmic bias, and a lack of long-term prospective clinical validation. Robust multidisciplinary collaboration and ethical frameworks are critical to successfully integrating these digital innovations into routine care and achieving equitable health outcomes globally.

PLENARY SESSION III

Data-Driven Integrative Cancer Research: Bridging Computational Discovery, Molecular Mechanisms, and Clinical Outcomes

Prof. Dr. Kazunori Akimoto

*Department of Molecular Medical Sciences, Faculty of Pharmaceutical Sciences, and
Research Division of Medical Data Science, Research Institute for Science and
Technology, Tokyo University of Science, Japan*

akimoto@rs.tus.ac.jp

Cancer is a highly heterogeneous disease in which clinical outcomes arise from complex interactions between tumor-intrinsic programs and the tumor microenvironment. To address this complexity, we present a data-driven integrative research framework that connects computational analysis, molecular mechanism studies, and clinical interpretation, while emphasizing that these approaches can operate both independently and complementarily. In the first part, we introduce MI-POG (Mutual Information-based Prognostic Omics Gene framework), an information-theoretic approach for unbiased genome-wide discovery of prognostic biomarkers. MI-POG captures nonlinear dependencies between gene expression and clinical outcomes, enabling the identification of clinically relevant gene sets beyond conventional Cox-based or predefined gene-panel approaches. Applications to large-scale patient cohorts demonstrate its utility in extracting prognostic signals that are difficult to detect using traditional methods. In the second part, we present an independent line of investigation focusing on cell–cell junction (CCJ) biology in breast cancer. Through multi-cohort transcriptomic analyses, we show that subtype-specific CCJ gene expression signatures stratify patient prognosis, particularly in Luminal B and Basal-like tumors. These findings reveal that coordinated regulation of adhesion-related genes reflects subtype-dependent tumor behavior and microenvironmental interactions. Importantly, this work is conceptually distinct from the MI-POG framework and is based on hypothesis-driven biological investigation of junctional systems. In the third part, we describe our ongoing research on tight junction–associated mechanisms in cancer metastasis. While tight junctions have traditionally been viewed as static structural components, our findings suggest that their dynamic remodeling contributes to metastatic dissemination. We will present emerging evidence supporting a novel molecular framework linking epithelial junction architecture to invasive and metastatic potential. Together, these studies highlight two complementary but distinct research strategies: data-driven computational discovery and mechanistic molecular investigation. By integrating insights from both approaches, we aim to advance a more comprehensive understanding of cancer biology and to contribute to the development of precision oncology strategies that bridge discovery and clinical application.

INVITED TALK – TRACK 1

Connected Data, Transforming Pharma: Lessons from MyHDW and AI Potential

Dr. Ismat binti Mohd Sulaiman

Ministry of Health Malaysia

The Malaysian Health Data Warehouse (MyHDW) connects fragmented facility data into a national health repository built for big data analytics for health system planning. MyHDW treats data as a foundational asset to move forward towards predictive analytics and clinical text mining using AI. This talk examines what that connection teaches us about building an AI-ready system, and the potential using pharmacy data to take pharmaceutical care forward. MyHDW was developed in 2016 and has since accumulated years of experience from managing data from various healthcare facilities to producing insightful information for various agencies and researchers. The aim is to provide a high-level data-driven overview of the healthcare systems across the public and private sectors. That objective requires standardised data stored and shared in the same language, a governing body to create policies and ensure safety, as well as a continuous data quality assessment. MyHDW is equipped with health data standards for diagnosis and procedures using ICD, for health system planning using a national health data dictionary, and for deeper text-based analytics using SNOMED CT and LOINC. Reflecting on the pharmaceutical landscape, the same principles apply – the need for a nationally accepted drug data standard and governance. Otherwise, pharmaceutical data cannot move safely between facilities, and precision pharmacotherapy has nothing to build on. This talk will also present several pharma data-to-AI potential value chains if we do have drug data standards and governance. First, clinical decision supports systems are strengthened with medication safety through AI-supported interaction checking and individualised dosing. Second, pharmacovigilance and drug-utilisation analytics apply algorithmic methods to detect safety signals earlier than manual review allows. Lastly, predictive and personalised therapy uses pharmacogenomic frameworks to translate gene-drug data into actionable dosing guidance. Each depends on the same precondition: connected, standardised, governed data. Data governance must exist prior to AI governance, because governance ensures validation, testing, and monitoring of AI models built on top of the data. The same stewardship that makes MyHDW trustworthy must extend to how AI performs in practice. This talk ends with how the fraternity can prepare to embark on a safe and sustainable AI journey.

INVITED TALK – TRACK 2

Development of Cell-Based Regenerative Therapy for Lymph Node Regeneration

Assoc. Prof. Dr. Kosuke Kusamori

*Faculty of Pharmaceutical Sciences, Tokyo University of Science, Tokyo, Japan
Research Institute for Science and Technology, Tokyo University of Science, Chiba,
Japan*

kusamori@rs.tus.ac.jp

Secondary lymphedema is a chronic complication of lymph node dissection or lymphatic injury, and current treatments mainly provide symptomatic relief. Regeneration of functional lymph nodes could restore lymphatic drainage and immune surveillance, but reproducible construction of lymph node-like tissues suitable for transplantation remains a major challenge. This study aimed to develop a cell-based regenerative therapeutic platform for lymph node regeneration using centrifuge-based bioengineered lymphatic tissue (CeLyT) and to evaluate its structural and functional performance in vivo. CeLyT was fabricated by layering mesenchymal stromal cells (MSCs) and lymphatic endothelial cells (LECs) through controlled centrifugation. The engineered tissues were transplanted into a mouse lymph node excision model. Lymphatic drainage, tissue integration, immune cell organization, and local immune responses were assessed by lymphatic imaging, histology, immunostaining, and functional immune stimulation. Transplanted CeLyT promoted reconstruction of lymphatic pathways and improved lymphatic flow after lymph node excision. The grafts developed lymph node-like structures containing lymphatic vessels and organized immune cell compartments. Importantly, regenerated tissues supported lymph filtration and responded to immune stimulation, indicating recovery of key lymph node functions. Histological analysis further demonstrated integration of the transplanted tissues with surrounding host lymphatic structures and formation of functional lymphatic networks. These findings suggest that engineered lymphatic tissue composed of MSCs and LECs can provide both structural and functional components required for lymph node regeneration. This cell-based regenerative strategy enables reconstruction of functional lymph node-like tissue and may provide a curative therapeutic approach for secondary lymphedema. Further optimization of tissue fabrication and transplantation conditions will support future clinical translation.

INVITED TALK – TRACK 3

Natural Product Discovery and Delivery: Modulation of Gut-Liver Axis for Hepatic Diseases

Prof. Dr. Wong Tin Wui

Non-Destructive Biomedical and Pharmaceutical Research Centre, Smart Manufacturing Research Institute/Faculty of Pharmacy, Universiti Teknologi MARA Selangor, Puncak Alam, Selangor, Malaysia

UM-UiTM Excipient Development Research Unit (EXDEU), Faculty of Pharmacy, Universiti Malaya, Lembah Pantai 50603, Kuala Lumpur, Malaysia.

Non-alcoholic fatty liver disease (NAFLD) is an inflammatory hepatic disorder associated with insulin resistance, central obesity, and metabolic syndrome which raises the risk of cardiovascular diseases and may contribute to nonalcoholic fatty hepatitis, cirrhosis, and hepatocellular cancer. Its global prevalence has increased to 34 % and is expected to reach 56 % by 2030. Physical exercise and diet are the primary approaches to control NAFLD development. There is a lack of approved or effective pharmacological treatment. The occurrence and development of NAFLD are complex adopting a pronounced relationship between gut dysbiosis and the NAFLD development, known as the "gut-liver" axis. A number of herbs have been reported to contain flavonoids, polyphenols, terpenoids, alkaloids, and polysaccharides that could mitigate NAFLD development. This presentation highlights the pharmacological basis of Tibetan medicine in combating NAFLD. The formulation strategy of Tibetan medicine will be discussed from the perspectives of therapeutic value of excipients and their liver targeting capacity as well as precision attribute guided by omics analysis. Synergism of drug discovery and delivery sciences is envisaged to promote therapeutic efficacy and reduce adverse influences which are critical for clinical applications and commercialization.

INVITED TALK – TRACK 4

AI Revolution: Transformation of Health Education

Assoc. Prof. Dr. Ho Ket Li

IMU University

Artificial intelligence is rapidly transforming health professions education. Our team in IMU University uses AI to address longstanding challenges in feedback literacy and clinical skills training. Traditional approaches, such as standardized patients and scattered feedback mechanisms, are often costly, logistically complex, and lack scalability. We introduce two AI-powered educational applications developed at IMU University: Feedback IMU and TalktoPatient. These tools aim to enhance feedback literacy and provide scalable, immersive clinical simulation opportunities for healthcare students. Feedback IMU centralizes the feedback process, allowing students to request, view, and analyze feedback from tutors, peers, and supervisors. It utilizes AI to map feedback to Malaysian Qualifications Framework domains and generate personalized improvement plans. TalktoPatient offers an AI-powered virtual patient simulation designed for safe, remote practice of clinical, communication, and decision-making skills. It supports multiple languages and disciplines, including Medicine, Nursing, Pharmacy, Nutrition and Dietetics, Psychology, and Chinese Medicine. The system features adaptive difficulty levels and AI-powered rubric-based grading of student-patient conversations. Development of these platforms required navigating several technical and pedagogical challenges. These include managing token limits, preventing AI hallucinations and persona bleed, ensuring clinical accuracy and maintaining data privacy. Despite these hurdles, both applications successfully demonstrate practical solutions for automating feedback analysis and providing accessible practice environments without the logistical constraints of human actors or expensive mannequins. These tools represent a significant advancement in health education. While AI cannot replace the nuanced guidance of human teachers, these applications effectively complement traditional teaching methods. They offer scalable, cost-effective solutions for feedback management and clinical skills training. Both platforms are freely accessible upon registration, supporting the democratization of quality health education resources.

PLENARY SESSION IV

Reprogramming Cancer through Integrative Omics and Computational Pharmacology: From Biomarker Discovery to Precision Targeting of Cellular Signalling

Prof. Dr. Varisa Pongrakhananon

*Department of Pharmacology and Physiology, Faculty of Pharmaceutical Sciences,
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Cancer is a highly dynamic disease driven by complex molecular alterations that extend beyond genomic abnormalities to include changes in protein expression, post-translational modifications, and interconnected signalling networks. Understanding these molecular events requires an integrative approach capable of linking omics discoveries with functional mechanisms and therapeutic opportunities. In this lecture includes an integrated research framework that combines multi-omics technologies, functional biology, and computational pharmacology to accelerate biomarker discovery and precision therapeutic development in lung cancer. The presentation will demonstrate how transcriptomic and proteomic analyses identify clinically relevant biomarkers, using microtubule-associated proteins as examples of novel regulators of cancer progression. Functional studies further reveal how CAMSAP2 and CAMSAP3 orchestrate distinct signalling pathways controlling metastasis, autophagy, and angiogenesis, highlighting the importance of mechanistic validation beyond biomarker discovery. Building upon these findings, how quantitative acetylomics uncovers cancer-associated alterations in protein acetylation that cannot be explained by protein abundance alone will be discussed. Integrative proteomics further reveals dynamic changes in the acetylated proteome underlying lung cancer progression and identifies HDAC6 as a central regulator of protein acetylation, representing a promising therapeutic target. Finally, how computational pharmacology, including virtual screening, molecular docking, and structure-based drug discovery, accelerates the identification of novel HDAC6 inhibitors will be illustrated. Together, these studies demonstrate how integrating multi-omics, systems biology, and computational approaches enables the discovery of actionable signalling networks and advances the development of precision therapeutics for precision oncology.

PLENARY SESSION V

Precision Design and Programming Trap-and-Kill Peptide Nanonets for Novel Antimicrobial Therapy

Assoc. Prof. Dr. Ee Pui Lai, Rachel

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Antimicrobial peptides (AMPs) are traditionally developed as molecular drugs that disrupt membranes or bind intracellular targets; here we introduce a different paradigm in which peptides are engineered as environmentally activated supramolecular materials that eradicate bacteria through localized self-assembly rather than classical biochemical inhibition. We identified β -hairpin peptides that remain disordered under physiological conditions but undergo pathogen-triggered folding upon recognizing bacterial surface motifs such as lipopolysaccharide, initiating fibrillar growth directly on the microbial envelope to form extracellular “nanonets” that immobilize pathogens while exerting bactericidal activity and endotoxin sequestration. A central finding is that nanonet architecture is a programmable design parameter: residue-level modifications tune assembly to yield dense bactericidal meshes or looser trap-only structures, demonstrating that higher-order organization, not peptide potency alone, governs antimicrobial outcome, while confinement within these assemblies suppresses bacterial motility and enhances antibiotic susceptibility. The translational potential of peptide nanonets is immense ranging from their function as medical device surface coating to anti-biofilm and anti-fouling materials, demonstrating how this platform programs on-demand, bacteria-triggered nanonet formation at the material interface, spatially coupling capture, killing, and inflammatory attenuation. Overall, we aim to develop a multimodal peptide-based platform to address surface-associated microbial infections and its associated inflammation.

INVITED TALK – TRACK 5

Applications of Digital Health in Pharmacy Practice and Education: Current Insights and Future Directions

Assoc. Prof. Dr. Shazia Jamshed

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The prompt amalgamation of digital health technologies into healthcare systems is primarily reshaping both the practice of pharmacy and the training of future pharmacists. Digital health technologies covering tele-pharmacy, artificial intelligence (AI), mobile health (mHealth), and electronic health records (EHRs) have already started transforming practice and care by refining patient well-being and medication safety, concordance and adherence, clinical workflow, and shared decision-making. The current session combines existing facts and promising forthcoming trajectories in these two interrelated domains. In practice, the concept of tele-pharmacy has surfaced as one of the most credible digital health modalities. This ground-breaking model not only strengthens access to pharmaceutical care but also reinforces chronic disease management through tele-delivered medication evaluation and treatment rationalization. mHealth applications and wearable devices are facilitating communities to dynamically contribute to their individual treatment, allowing patients to be adherence-cognizant by consistently receiving reminders and monitoring health parameters, thus simplifying a hands-on, data-driven strategy to pharmaceutical care. AI and clinical decision support systems (CDSS) correspond to another frontline redesigning pharmacy practice and clinical pharmacy. Specialty pharmacy is swiftly implementing AI, telehealth, and data analytics to rationalise multifaceted workflows, expand patient contact, and improve operational productivity as real-time patient reporting becomes a valuable tool in care programs. In pharmacy education, the priority to prepare graduates for a digitally centric healthcare setting has encouraged extensive curricular transformation. Digital health innovations like eHealth, mHealth, e-learning, telehealth, AI, telemedicine, telepharmacy, virtual reality, and augmented reality are gradually amalgamated into pharmacy education to train future practitioners for a digital healthcare environment; however, indicators on their influence on learning outcomes and skill development are evolving. One cannot deny that pharmacy programs have integrated AI directions into curricula in the hope of edification of foundational proficiencies in applying digital health technologies to practice while augmenting critical reasoning in decision-making. Nonetheless, substantial gaps remain unchanged. Also, the unification of AI, real-world data, big data, and digital therapeutics confirms a conceptual modification in the context of pharmaceutical care in precision medicine. Future trajectories essentially select impartial access to digital tools, dynamic faculty preparation, and competency framework expansion, which correspond to FIP Development Goal 20 on digital health. Collective strategies not only among pharmacy schools and professional bodies but also with technology developers will be decisive to connect the extensive capacities of digital health for pharmacy practice and education.

INVITED TALK – TRACK 6

From Pharmacogenomics to Pharmacophytogenomics: Diversity-Driven Pathways in Precision Health

Prof. Dr. Teh Lay Kek

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The Human Genome Project has established the foundation for precision health by enabling the characterisation of human genetic variation and its impact on health and disease. Pharmacogenomics has emerged as a key component of precision healthcare, providing strategies to optimise medication therapy, improve treatment outcomes, and reduce adverse drug reactions. However, despite the widespread use of herbal medicines, dietary supplements, and phytochemicals, limited understanding exists regarding how genetic diversity influences responses to these bioactive compounds. Pharmacophytogenomics represents an emerging extension of pharmacogenomics that investigates interactions between phytochemicals and genetically diverse biological targets. Since 2003, our research has contributed to the development of precision health through population genomics, pharmacogenetics, and translational bioinformatics, advancing the understanding of genetic determinants of drug response in diverse Malaysian populations, particularly in therapeutic efficacy, adverse drug reactions, and precision prescribing. Beyond pharmacogenomics, whole-genome sequencing studies of the Orang Asli population have enabled the identification of population-specific genetic profiles associated with disease susceptibility, vitamin deficiencies, and nutritional risks, supporting precision health and precision nutrition strategies. Building on this foundation, our research has expanded into pharmacophytogenomics by integrating genomics, structural bioinformatics, and molecular modelling. We have investigated phytochemical interactions with HLA-B*15:01 and HLA-B*15:02 to elucidate mechanisms underlying herb-modulated carbamazepine-induced severe cutaneous adverse reactions and evaluated genotype-dependent herb–drug interactions affecting clopidogrel response through phytochemical binding analysis of wild-type and variant proteins. Furthermore, integrating whole-genome sequencing with computational modelling has enabled the identification of genetic determinants of iron homeostasis and oxidative defence, revealing how dietary phytochemicals may interact with population-specific genetic variants to influence nutritional status and blood-related health outcomes. By integrating population genomics, computational biology, and translational science, pharmacophytogenomics provides a novel framework to understand variability in phytochemical responses across genetically diverse populations. This approach supports the future development of personalised herbal medicine, precision nutrition, and safer integration of complementary therapies into precision healthcare.

INVITED TALK – TRACK 7

Strategies for Improving the Solubility of Poorly Water-Soluble Drugs

Prof. Dr. Takehisa Hanawa^{1*}, Senri Mizobuchi¹, Naoko Ishii¹, Kazuya Nomura¹, Yuko Wada², Kaoru Oda⁴, Yayoi Kawano⁴, Kaoru Hirose¹, Narumi Maida¹, Fumiyoshi Ishii²

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It is estimated that 30–40% of currently marketed pharmaceuticals are poorly water-soluble drugs classified as Class II or Class IV under the BCS classification system. Furthermore, 70–90% of new drug candidates are considered to be poorly water-soluble, making the improvement of water solubility in pharmaceuticals an urgent priority for formulation researchers. In this presentation, we will introduce two pharmaceutical engineering methods: the formation of nanoparticles via dry milling and a method utilizing porous adsorbent carriers. In this study, a ground mixture of commercially available rebamipide (RB) tablets and HPC-SSL was prepared by dry milling to obtain a dispersoid with high dispersion stability suitable for use in a mouthwash. Subsequently, the physicochemical properties of the mixture with HPC-SSL were compared with those of the powder obtained from pre-ground RB tablets. Co-grinding with HPC-SSL transformed RB into fine particles ranging from approximately 303.6 to 361.5 nm, and a tendency to prevent particle aggregation over time in solution was observed. The sample microparticles formed with HPC-SSL avoided aggregation on the mucosal surface for 24 hours and improved oral retention. The “sealed heating (SH) method” involves heating a mixture of a porous material and sublimed API in a sealed container. In this study, we focused on mesoporous silica (MPS-4R or -2R) as the porous material. The adsorption of ibuprofen (IBU) onto MPS was investigated using the SH method. PXRD analysis revealed that the IBU crystals were amorphized by the SH method. The dissolution rate of the SHM containing 10 wt.% IBU, 10 minutes after the start of the test, was approximately 2.7 times higher than that of the IBU crystals. These results indicate that the SH method is a novel technique for amorphizing poorly water-soluble drugs and improving their solubility without using organic solvents.

INVITED TALK – TRACK 8

From Classroom to Connected Care: Digital Transformation of Pharmacy Education and Practice

Prof. Dr. Aleth Therese L. Dacanay

Faculty of Pharmacy, University of Santo Tomas

The rapid advancement of digital technologies is transforming healthcare delivery, research, and professional education, creating unprecedented opportunities to enhance wellness, promote healthy longevity, and optimize therapeutic outcomes. As healthcare systems evolve toward connected, patient-centered models of care, pharmacy education must likewise undergo a fundamental transformation to prepare graduates who are digitally competent, innovative, and collaborative. This presentation explores how digital transformation in pharmacy education serves as the critical foundation for advancing connected care and shaping the future of pharmacy practice. The presentation examines the integration of emerging technologies—including artificial intelligence, health informatics, telepharmacy, virtual simulation, digital therapeutics, wearable technologies, electronic health records, and pharmacogenomics—into pharmacy curricula and professional practice. It highlights how these innovations are reshaping teaching and learning strategies, enhancing experiential education, supporting data-driven clinical decision-making, and expanding pharmacists' roles within interdisciplinary healthcare teams. Beyond technological adoption, the presentation emphasizes the importance of cultivating future-ready competencies, including digital literacy, data analytics, critical thinking, ethical leadership, adaptability, and lifelong learning. It also discusses the growing convergence among academia, research, industry, and healthcare systems to foster connected innovation that translates scientific discoveries into meaningful patient outcomes. Drawing on global trends and institutional experiences, the presentation proposes a framework for aligning pharmacy education with the evolving demands of digital health ecosystems. It underscores the need for collaborative partnerships, curriculum innovation, faculty development, and policy support to ensure that pharmacy graduates are equipped to lead digital transformation across diverse practice settings. Ultimately, the presentation explores that the journey toward connected care begins in the classroom. By embracing digital transformation in pharmacy education, institutions can empower future pharmacists to become leaders in innovation, improve access to high-quality healthcare, advance wellness and healthy longevity, and deliver personalized therapies that respond to the evolving needs of patients and society.

CLOSING PLENARY & PROFESSORIAL LECTURE

Listen to Your Gut

Prof. Dr. Kalavathy Ramasamy

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Listen to Your Gut establishes the gastrointestinal tract as the centre of human health and the cornerstone of modern preventive medicine. Backed by scientific preclinical evidence, it demonstrates how the gut microbiota actively regulates homeostasis. It reveals the ability of specific probiotics and postbiotics in restoring gut health, thereby enhancing cognitive function and cardiometabolic health. It also unveils the future of microbiome-based interventions that is expected to shift towards precision-engineered Next-Generation Probiotics (NGPs) and stable, bioactive postbiotics. Ultimately, Listen to Your Gut is a testament that a healthy gut is the foundation for lifelong resilience, longevity, and vitality.

Oral Presentation **Abstracts**

Pharmacology & Precision Medicine (PPM)

OP-001

Integrating Network Pharmacology, Virtual Screening and Molecular Dynamics Simulations to Identify Potential Dual-Targeting Medicinal Compounds for Periodontitis and Coronary Artery Disease

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Periodontitis and cardiovascular disease, particularly coronary artery disease (CAD), exhibit significant correlation, yet current pharmacological therapies for these comorbidities often lead to adverse drug events. There is a notable deficiency in research exploring the therapeutic potential of medicinal compounds for dual-targeting treatment. The aim of this study is to identify the network pharmacology and overlapping genes to establish treatment for patients with both periodontitis and CAD. Network pharmacology was carried out to identify CAD and periodontitis-specific genes using GeneCards. Network topology, KEGG, and GO enrichment analyses were employed to determine target genes. The crystallographic structures of the target proteins were obtained from Protein Databank (PDB) for the high-throughput virtual screening of medicinal compounds from the Natural Products Database for Malaysian Medicinal Plants (NADI). The compounds with the highest binding affinity were further evaluated via molecular dynamics (MD) simulations using GROMACS to assess stability and protein-ligand interactions. Network pharmacology revealed 16 hub proteins that consist of topological significance in the network through pathway enrichment analysis using KEGG and GO Analysis. TNF- α was identified as the central protein overlapping both diseases. Virtual screening of the NADI database to TNF- α identified alanditrypinone as the highest binding affinity. MD simulations demonstrated that the TNF- α -alanditrypinone complex was stably interacting during the 200 ns timescale, exhibited by hydrogen bonds and hydrophobic contacts. In summary, TNF- α is a shared target between CAD and periodontitis. Based on computational screening and MD simulations, alanditrypinone showed strong binding affinity and stability, suggesting its potential as a dual-targeting medicinal compound.

Keywords: cardiovascular disease, periodontitis, network pharmacology, molecular docking, molecular dynamic simulation

OP-006

Glycemic, Metabolic, and Safety Outcomes of Metformin Extended-Release (XR) in Type 2 Diabetes Mellitus

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Metformin is commonly the initial medication recommended for managing T2DM; however, it can cause gastrointestinal problems. Erance associated with IR formulations may limit treatment adherence and dose optimization. Extended-release metformin (metformin XR) was developed to improve tolerability and simplify dosing. However, comprehensive clinical evidence regarding its glycaemic, metabolic, and safety outcomes remains limited. This preliminary review aims to identify and summarize existing research on the application of XR in adults with type 2 diabetes mellitus, with a focus on glycaemic and metabolic outcomes and safety profiles in a hospital-based healthcare setting. A scoping review was conducted following the methodology of the Joanna Briggs Institute and documented in accordance with the PRISMA-ScR guidelines. A literature search was conducted in PubMed (MEDLINE) and Scopus, using the Population–Concept–Context (PCC) framework. Eligible studies included both experimental and observational studies focusing on the evaluation of sustained-release metformin in adults with type 2 diabetes mellitus. Data were descriptively mapped and narratively synthesized to outline outcome domains and identify evidence gaps. A total of 24 studies met the inclusion criteria. Most of these studies focused on evaluating glycaemic outcomes, with glycated haemoglobin (HbA1c) and fasting plasma glucose (FPG) being the most frequently reported. Only a few studies included data on PPG levels. Metabolic outcomes, such as lipid profiles and blood pressure, have been inconsistently reported and included in only a limited number of studies. Safety outcomes, particularly gastrointestinal adverse events and tolerability, were addressed in several studies; however, reporting standards for these outcomes varied widely. The follow-up period ranged from a few weeks to 12 months. Metformin XR is a clinically comparable alternative to insulin for glycaemic control. However, evidence regarding metabolic outcomes and safety is limited, necessitating further studies with more comprehensive safety reporting.

Keywords: type 2 diabetes mellitus, extended-release metformin, glycaemic results, metabolic results, safety, scoping review

OP-022

Nuclear PKC λ Promotes Cervical Cancer Progression Through a TCF7L2-Dependent Transcriptional Program

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Cervical cancer is the fourth most common cancer among women worldwide. Persistent infection with high-risk human papillomavirus (HPV) is a well-established etiological factor; however, HPV infection alone does not fully explain tumour progression. Nuclear localization of atypical protein kinase C lambda (PKC λ), a cell polarity-associated serine/threonine kinase, has been reported as an independent prognostic factor in cervical cancer, yet the underlying transcriptional mechanisms remain poorly understood. To verify the hypothesis; PKC λ cooperates with TCF7L2 to drive gene expression programs that promote tumorigenesis. We investigated the role of nuclear PKC λ in cervical cancer using integrated transcriptomic, bioinformatic, and functional analyses. To identify genes regulated by nuclear PKC λ , we performed microarray analysis comparing HeLa shPKC λ cells reconstituted with PKC λ A120E (nuclear-localized PKC λ mutant). we screened candidate transcription factors commonly involved in the regulation of these genes using ChIP-Atlas. To investigate the biological significance of the PKC λ -dependent TCF7L2 regulatory mechanism in cancer progression, we employed the chorioallantoic membrane (CAM) assay. Microarray analysis identified 235 upregulated and 276 downregulated genes in HeLa cells expressing PKC λ A120E. Survival analysis using the TCGA cervical cancer cohort revealed that 80 of these genes were significantly associated with poor disease-specific survival. Based on these candidates, we defined a five-gene signature that predicted unfavourable prognosis. ChIP-Atlas analysis identified TCF7L2, a transcription factor involved in Wnt/ β -catenin signalling, as a common regulator of these genes. Functional studies demonstrated that PKC λ regulates target gene expression via TCF7L2, positively controlling *UGCG* while repressing *LGR5*. Using a CAM model, we further showed that PKC λ promotes tumour growth in a TCF7L2-dependent manner, as silencing of TCF7L2 attenuated PKC λ -driven tumour growth. Our findings identify a previously unrecognized PKC λ -TCF7L2 transcriptional axis that contributes to cervical cancer progression and may represent a potential therapeutic target, with possible relevance for risk stratification beyond HPV status.

Keywords: CESC, nuclear PKC λ , TCF7L2

OP-033

Mechanical Stimuli and Mechanosensitive Ion Channels Modulate Radiation-Induced Cellular Responses

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Radiotherapy efficacy is strongly influenced by cellular stress responses, including DNA damage response (DDR) pathways. Recent studies have suggested that mechanical stimuli and mechanosensitive ion channels contribute to cancer cell survival and stress adaptation; however, their roles in radiation responses remain incompletely understood. In this study, we investigated the effects of osmotic mechanical stimulation and mechanosensitive ion channels on radiosensitivity and DDR signalling in multiple cancer cell models. In B16 melanoma cells, hypertonic stimulation and activation of the mechanosensitive ion channel PIEZO1 significantly enhanced radiosensitivity, accompanied by increased radiation-induced cellular stress responses. Furthermore, ongoing studies using A172 glioblastoma cells demonstrated that inhibition of the mechanosensitive calcium-permeable channel TRPC6 altered radiation responses and affected cell survival following irradiation. In addition, osmotic stress conditions in A549 lung carcinoma cells modulated DDR-associated signalling pathways, suggesting that both hypertonic and hypotonic environments influence radiation-induced cellular responses. Collectively, these findings indicate that mechanical stimuli and mechanosensitive ion channels play important roles in regulating cellular responses to ionizing radiation. Our work highlights mechanotransduction pathways as potential targets for improving radiotherapy efficacy and provides new insights into the relationship between biomechanical stress and cancer radiosensitivity.

Keywords: mechanotransduction, radiosensitivity, PIEZO1, TRPC6, DNA damage response

OP-045

Combined *Pkcζ*, *CTNNBIP1*, and *ALDH1A3* Expression Is Associated with Poor Outcomes in Endocrine Therapy-Treated Luminal B Breast Cancer

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Breast cancer is a heterogeneous disease comprising multiple molecular subtypes with distinct clinical behaviours and treatment responses. Although endocrine therapy is a standard treatment for hormone receptor-positive breast cancer, therapeutic resistance remains a major clinical challenge, particularly in luminal B breast cancer. Protein kinase C zeta (*PKCζ*), an atypical PKC implicated in cancer progression, has been associated with aggressive tumour behaviour; however, its relationship with endocrine therapy response remains unclear. This study aimed to investigate the clinical significance of *PKCζ* and related molecular markers in luminal B breast cancer. Publicly available breast cancer datasets containing gene expression and clinical information were analysed, including METABRIC (n = 2,509) and TCGA Pan-Cancer Atlas breast cancer cohort (n = 1,084). Kaplan–Meier survival analysis and Cox proportional hazards models were used to evaluate survival outcomes in relation to gene expression and treatment status. High *PKCζ* expression was associated with poorer prognosis in luminal B breast cancer patients receiving endocrine therapy, with a particularly pronounced association in the aromatase inhibitor treated subgroup. Differential expression analysis between *PKCζ*^{high} and *PKCζ*^{low} luminal B tumours identified *CTNNBIP1*, a negative regulator of canonical Wnt/β-catenin signalling, as a candidate associated gene. Patients with *PKCζ*^{high}/*CTNNBIP1*^{high} tumours showed poorer clinical outcomes following aromatase inhibitor treatment. Similarly, *PKCζ*^{high}/*ALDH1A3*^{high} and *CTNNBIP1*^{high}/*ALDH1A3*^{high} tumors were associated with significantly worse prognosis. Notably, patients with *PKCζ*^{high}/*CTNNBIP1*^{high}/*ALDH1A3*^{high} luminal B tumors exhibited the poorest clinical outcomes after aromatase inhibitor treatment. Combined high expression of *PKCζ*, *CTNNBIP1*, and *ALDH1A3* is associated with poor clinical outcomes in endocrine therapy-treated luminal B breast cancer, particularly in patients receiving aromatase inhibitor therapy. These findings suggest that this molecular combination may serve as a prognostic biomarker set for identifying high-risk luminal B breast cancer patients and may support treatment stratification for endocrine therapy.

Keywords: breast cancer, luminal B, *PKCζ*, *CTNNBIP1*

OP-054

Advances in Physiologically Based Biopharmaceutics Modelling: Toward Virtual Bioequivalence Simulations

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Quantitative prediction of the in vivo performance of orally administered drug products remains a major challenge in drug development. This includes predicting food effects on drug absorption, interactions with acid-reducing agents, and bioequivalence (BE) between drug products. Accurate prediction of these phenomena is essential for the rational design of formulations and the efficient development of both innovative and generic medicines. In recent years, in silico modelling and simulation approaches have attracted considerable attention due to species differences between humans and animals in gastrointestinal physiology, as well as increasing concerns regarding animal welfare. A modelling framework designed to quantitatively predict drug absorption from pharmaceutical formulations is known as a physiologically based biopharmaceutics model (PBBM). PBBMs integrate the key processes governing oral drug absorption, including drug dissolution, precipitation, gastrointestinal solubility, intestinal permeability, gastrointestinal transit, and systemic pharmacokinetics. By mechanistically linking formulation properties with human physiology, PBBMs provide a scientific framework for translating in vitro data into clinically relevant outcomes. Despite substantial progress in the field, the development of robust PBBMs remains highly challenging, and no universally applicable PBBM currently exists. Several important hurdles remain, including: (1) virtual BE assessment as an alternative to clinical BE studies, (2) quantitative prediction of food effects, and (3) the development of PBBMs that account for special populations, such as paediatric and geriatric patients, individuals with disease conditions, and different ethnic groups. In this presentation, I will introduce several examples of recent PBBM research being conducted in the Kambayashi Research Laboratory at the Faculty of Pharmaceutical Sciences, Tokyo University of Science. Particular emphasis will be placed on our latest research toward virtual BE, including advanced approaches that integrate biorelevant in vitro experiments with mechanistic absorption models. The opportunities and current limitations of virtual BE in drug development and regulatory science will also be discussed.

Keywords: physiologically based biopharmaceutics model (PBBM), virtual bioequivalence simulations

OP-103

Beyond Glutathione S-Transferase: A Novel Sulfide-Mediated Pathway for cis Urocanic Acid Conjugation

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Urocanic acid (UCA) is a chromophore found in the stratum corneum. A certain extend of UV irradiation exposure can isomerise the UCA from trans- to a cis- form. Conflicting data on the UCA has been found in multiple studies for over ten years. According to reports, UCA is physiologically useful as a photoprotective agent in the skin, but it has also been linked to a number of skin immunological and pathological disorders, including cancer. It is worthwhile to look into a crucial mechanistic gap in our knowledge of the true function of UCA in skin physiology. The objective of this study is to revisit the cis-UCA metabolic pathway via sulfide-derivative chemicals. Briefly, the study was conducted to assess the photoprotective effect of UCA-sulfide conjugate through in vitro analysis. Next, by using a cell free system, the chemical reaction between the cis-UCA and sulfide donors, including N-acetylcysteine (NAC), glutathione (GSH), L-cysteine, sodium hydrosulfide (NaHS), and sodium disulfide (Na₂S₂) were performed and analysed by the Nuclear Magnetic Resonance (NMR) metabolomic approach. From the results, the in vitro studies showed that UVB exposure, in combination with cis-UCA, significantly reduced cell viability, although the presence of supersulfides appeared to alleviate the toxicity. Cis-UCA was found to readily react with the supersulfides with no involvement of the GST enzyme. A new compound named 4-imidazoleacrylic acid-3-thiol was discovered when the cis-UCA reacted with NaHS. Furthermore, GS(CIE) and Cys (CIE) can also be formed in the absence of GST, indicating that the reaction can occur non enzymatically. This study offers a new perspective on cis-UCA metabolism and may shed light on UCA's potential role as a photo-transmitter compound. In conclusion, cis urocanic acid can directly, GST-independently conjugate with supersulfides, indicating a new UV-dependent metabolic route that could redefine the function of cis-UCA in photosignaling and skin physiology.

Keywords: Cis-UCA metabolism, UV, HaCaT, sulfide, photosignalling

OP-105

Combined *Curcuma longa* and *Centella asiatica* Granules Attenuate HFHC-Induced Metabolic and Oxidative Dysfunction: Involvement of the SOD–MDA–Nitric Oxide Axis

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High-fat high-carbohydrate (HFHC) diets promote insulin resistance and oxidative stress, impairing nitric oxide bioavailability and contributing to hepatic, renal, and vascular dysfunction. *Curcuma longa* and *Centella asiatica* have antioxidant and metabolic properties, yet their combined effect on the oxidative stress–nitric oxide axis remains unclear. This study aimed to evaluate the effects of combined *C. longa* and *C. asiatica* granules on metabolic dysfunction, oxidative stress, nitric oxide bioavailability, hepatic/renal function, and tissue histopathology in HFHC diet-induced rats. Rats were divided into six groups: normal control, HFHC control, comparator, *C. longa* (200 mg/kg BW), *C. asiatica* (100 mg/kg BW), and their combination. Glucose, insulin, and HOMA-IR were measured. SOD, MDA, and NO were assessed as oxidative/endothelial biomarkers, alongside SGOT, SGPT, and creatinine for organ function. Liver and kidney histopathology were evaluated statistically. HFHC exposure caused marked metabolic and oxidative disturbances, with altered glucose–insulin homeostasis, elevated HOMA-IR and MDA, decreased SOD, and reduced NO. Significant differences occurred in glucose, insulin, HOMA-IR, SOD, MDA, NO, SGOT, and SGPT ($p < 0.001$), and creatinine ($p = 0.017$). Combined treatment showed consistent metabolic improvement, increased SOD, reduced MDA, restored NO, and improved hepatic/renal markers versus HFHC controls, outperforming single extracts. Histopathological changes were not significant ($p > 0.05$), though descriptive trends toward reduced hepatocellular degeneration and vascular inflammation appeared in treatment groups. Combined *C. longa*–*C. asiatica* granules attenuated HFHC-induced metabolic and oxidative dysfunction via the SOD–MDA–NO axis, with hepatic/renal improvement preceding detectable structural changes, supporting their potential as a multi-target strategy against diet-induced metabolic dysfunction.

Keywords: *Centella asiatica*, *Curcuma longa*, high-fat high-carbohydrate diet, insulin resistance, oxidative stress

Pharmacy & Digital Health (PDH)

OP-009

Trustworthy Renal Decision Support in the Artificial Intelligence Era: Reducing Alert Burden Through the PANDU Pipeline

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Artificial intelligence-enabled healthcare is expanding. Yet, high-risk medication decisions still require transparent, auditable, and clinically accountable logic. In chronic kidney disease (CKD), renal decision support must generate recommendations while also managing alert burden, data missingness, and workflow feasibility. However, the upstream process of governing evidence, converting it into implementable rules, and testing electronic health record (EHR) feasibility remains underreported. This study aimed to evaluate the PANDU pipeline as a governance-to-simulation approach for trustworthy rule-based renal decision support, with alert-burden reduction assessed as an operational outcome. We developed and tested the PANDU translational informatics pipeline using the Knowledge-to-Action (KTA) framework and translational T models. The pipeline combined expert-derived readiness gates, source appraisal using the International Centre for Allied Health Evidence (iCAHE) checklist, rule implementability assessment using Guideline Implementability Appraisal (GLIA), and retrospective EHR silent simulation. Outcomes included source governance quality, rule readiness, alert evaluability, missingness patterns, laboratory timing, duplicate suppression, laboratory-linking window sensitivity, and high-severity signal yield. Analyses were descriptive, with penalized feasibility modelling for rare high-severity signals. Source appraisal showed variation in governance quality, with Kidney Disease: Improving Global Outcomes (KDIGO) documents scoring 92.9% and the Indonesian CKD national guideline scoring 71.4%. GLIA assessed 1,063 rules, including 625 dose-adjustment rules and 438 safety rules; 603 dose rules and 416 safety rules were implementation-ready. Retrospective EHR silent simulation showed that duplicate suppression reduced alert records from 1,010 to 617, a 38.9% reduction. Expanding the laboratory-linking window from ≤ 48 to ≤ 72 hours increased evaluable alerts from 55 to 62, while Red alerts remained 12. The PANDU pipeline demonstrates a replicable approach for trustworthy renal decision support in digital pharmacy. By linking source governance, rule implementability, EHR feasibility testing, and duplicate suppression, it provides an audit-ready foundation for safer, more manageable renal medication decision support.

Keywords: clinical decision support, chronic kidney disease, digital health, medication safety, pharmacy practice

OP-049

A Mobile Health RCT on Blood Pressure Control in Hypertension

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Hypertension affects over one billion adults globally and remains inadequately controlled despite available pharmacotherapy, largely due to suboptimal medication adherence. Mobile health applications represent a scalable digital strategy to bridge this adherence gap, yet rigorous randomised evidence on their impact on blood pressure outcomes remains limited. This study aimed to evaluate the effect of CareAide®, a gamified mobile health application, on medication adherence, blood pressure control, and health-related quality of life in adults with hypertension. A six-month open-label parallel-group randomised controlled trial was conducted across three Malaysian public hospitals. A total of 332 adults with hypertension and suboptimal baseline adherence were randomised 1:1 to CareAide® or standard care. Medication adherence was assessed using the Morisky Medication Adherence Scale-8. Blood pressure was measured at baseline and six months. Health-related quality of life was assessed using the Assessment of Quality of Life-6D. Between-group comparisons used Mann-Whitney U tests with Cohen d as effect size. CareAide® produced a large adherence gain at six months (intervention 6.58 versus control 5.63; Cohen d=1.00; p<0.001), with 78.4% of intervention participants achieving medium or high adherence versus 32.9% of controls. Systolic blood pressure fell 11.8 mmHg in the intervention group versus a 5.5 mmHg rise in controls (p<0.001; Cohen d=0.657). Diastolic blood pressure also reduced significantly in the intervention group (minus 5.8 versus plus 0.5 mmHg; p=0.001). Blood pressure control (<140/90 mmHg) was achieved in 60.9% of intervention participants versus 32.7% of controls. The coping domain of the Assessment of Quality of Life-6D improved significantly in the intervention group (Cohen d=0.276; p=0.004). CareAide® significantly improves medication adherence and produces clinically meaningful reductions in blood pressure in hypertensive patients. The near-doubling of blood pressure control rates supports integration of gamified mobile health tools into routine hypertension management.

Keywords: hypertension, mobile health, medication adherence, blood pressure control, randomised controlled trial

OP-059

Patient-Reported Instruments for Assessing Inhaler Adherence in Asthma and COPD in Low- and Middle-Income Countries: A Scoping Review

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Inhaled therapy is fundamental to asthma and chronic obstructive pulmonary disease (COPD) management, yet adherence assessment in low- and middle-income countries (LMICs) requires instruments that capture not only medication-taking behaviour but also inhaler-specific use and contextual barriers. This scoping review mapped patient-reported instruments used to assess inhaler adherence among adults with asthma and/or COPD in LMICs. This review followed the Joanna Briggs Institute framework and PRISMA-ScR. Searches were conducted in Web of Science, Scopus, PubMed/MEDLINE, CINAHL via EBSCOhost, and Embase via Ovid for studies published from 2015 to 2026. Records were screened in Covidence. Extracted data were synthesized into three evidence maps: instrument characteristics, alignment with inhaler-use behaviours, and readiness for LMIC implementation. The search identified 141 records; after removing 33 duplicates, 108 records were screened, 32 full texts were assessed, and 22 studies were included. The studies covered asthma (n=12), COPD (n=7), or both conditions (n=3), representing 37 study-country occurrences. Most studies were cross-sectional (n=15). Instruments were classified as general medication adherence tools (n=12), inhaler-specific instruments (n=7), or self-developed questionnaires (n=3). The Test of Adherence to Inhalers and its variants were the most prominent inhaler-specific instruments. In the behavioural alignment map, eight instruments showed high alignment, seven moderate alignment, and seven limited alignment with inhaler-use behaviours. Only seven studies classified non-adherence patterns as erratic, deliberate, and/or unwitting, while clinical outcomes were linked to adherence assessment in 12 studies. In the LMIC-readiness map, translation was reported in six studies and cultural adaptation in five. Overall LMIC readiness was rated high in five studies, moderate in 14, and limited in three. Patient-reported inhaler adherence assessment in LMIC asthma and COPD research remains heterogeneous. Future studies should strengthen local validation, cultural adaptation, and reporting of inhaler-specific behaviours, device-related barriers, and clinical outcome linkage.

Keywords: asthma, COPD, inhaler adherence, patient-reported instruments, low- and middle-income countries

OP-075

Effect of Pharmacist-Led Inhaler Education Using the Teach-to-Goal Method Combined with the Pharcare Mobile App on Pulmonary Function and Quality of Life in Patients with Chronic Obstructive Pulmonary Disease

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Chronic Obstructive Pulmonary Disease (COPD) is a progressive lung disease and one of the leading causes of death worldwide. Inhaled medications are the cornerstone of pharmacological treatment for patients with COPD. However, ensuring the correct inhaler technique remains a challenge for healthcare professionals involved in patient care, including pharmacists. This study aimed to evaluate the effect of pharmacist-led education on inhaler use delivered through a mobile application on lung function and quality of life (QoL) in patients with COPD. We conducted a quasi-experimental pre-post intervention study at Bekasi Regency General Hospital from November 2022 to April 2023. Pulmonary function and QoL were assessed using forced expiratory volume in one second (FEV₁) and the COPD Assessment Test (CAT), respectively. A total of 60 outpatients were included in the study, with 30 patients assigned to the intervention group and 30 to the control group. The control group received inhaler education using the teach-to-goal method, whereas the intervention group received the same education supplemented with the Pharcare mobile application. After 3 months of follow-up, the intervention group showed a significant improvement in CAT scores (14.73 ± 4.93 , $p < 0.001$) and were also observed in both groups compared with baseline ($p < 0.05$). Multivariate analysis showed that education delivered through the Pharcare mobile application was associated with a 4.67-point improvement in CAT score (95% CI, 1.242–44.850), but was not associated with improvement in FEV₁ ($p > 0.05$). The addition of the Pharcare mobile application to conventional inhaler education was associated with improved quality of life in patients with COPD but did not significantly improve pulmonary function. Mobile app-assisted inhaler education may be a useful strategy to enhance patient-reported outcomes and warrants further evaluation in larger, randomized studies.

Keywords: COPD, inhaler adherence, mobile app, FEV₁, QoL

OP-111

Efficacy and Durability of Pharmacist-Led Diabetes Therapy Management Interventions on Medication Adherence, Glycemic Control, and Quality of Life in Outpatients with Type 2 Diabetes Mellitus: A Systematic Review of Randomized Controlled Trials

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Type 2 Diabetes Mellitus (T2DM) represents a global health emergency requiring sustained therapeutic adherence and optimal glycaemic control to prevent microvascular and macrovascular complications. Pharmacist-led diabetes therapy management (DTM) has evolved as a primary collaborative strategy; however, comprehensive quantitative and contextual evidence synthesizing its impact on medication adherence, glycaemic biomarkers (HbA1c, Fasting Plasma Glucose [FPG], hypoglycaemia incidence), and Quality of Life (QoL) specifically within a Randomized Controlled Trial (RCT) framework is urgently needed to establish clinical standards of care. Registered in accordance with PROSPERO guidelines and reported following PRISMA 2020 standards, a systematic search was conducted across PubMed/MEDLINE, Scopus, Web of Science, Cochrane CENTRAL, and ScienceDirect for articles published up to 2026. Study screening utilized stringent PICOS criteria: Outpatients aged >18 years with T2DM (P); Pharmacist-led/managed DTM programs (I); Standard care by general practitioners/specialists (C); Medication adherence, glycaemic control (HbA1c, FPG, hypoglycaemia), and QoL outcomes (O); and RCT study design (S). Data extraction, risk of bias assessment, and structured synthesis of clinical outcomes and anomalous collocation patterns were systematically conducted. The search identified 838 records; after removing 185 duplicates, 653 records were screened, 46 full texts were assessed, and 29 studies were included. Among the 29 included candidate studies, 11 primary RCTs comprising a total of 1,143 T2DM outpatients fulfilled all PICOS criteria (with 18 non-RCT studies synthesized as secondary contextual evidence). Pharmacist interventions significantly improved medication adherence across various validated instruments (MARS, PDC >80%; $p<0.05$) and corrected covert non-adherence undetected by manual pill counts. Glycaemic control demonstrated statistically and clinically significant improvements, with HbA1c reductions ranging from -0.55% to -1.80% ($p<0.001$) alongside substantial FPG decreases. Efficacy was most pronounced in primary care facilities and resource-limited community settings compared to tertiary specialist clinics, indicating a facility-based ceiling effect. Furthermore, pharmacist care significantly suppressed hypoglycaemia incidence—particularly in geriatric populations ($p<0.01$)—and produced significant QoL improvements (DQOL, EQ-5D, SF-36, DTSQ). Contextual analysis of anomalous collocations revealed that intensive pharmacist involvement successfully reversed paradoxical QoL declines in complex insulin-treated patients and demonstrated that single-use monitoring technology (isCGM/mHealth) combined with pharmacist counselling achieved glycaemic improvements comparable to high-cost continuous monitoring. Pharmacist-led DTM interventions significantly improve glycaemic control, therapeutic adherence, and health-related quality of life in adult T2DM outpatients, offering maximal clinical benefit in primary care and underprivileged populations. Granting prescribing and dose-adjustment authority to pharmacists and integrating digital monitoring devices into collaborative care models are strongly recommended for implementation in clinical pharmacy practice and health policy.

Keywords: type 2 diabetes mellitus, pharmacist-led intervention, medication adherence, glycaemic control, quality of life, systematic literature review

Pharmaceutical Technology (PT)

OP-002

Development of Nanoemulgel Containing *Ixora coccinea* Leaves Extract for Antioxidant Property

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Nanotechnology improves the effectiveness of lipophilic drugs with poor and low bioavailability. The present study aimed to develop a nanoemulgel incorporating nanoemulsion containing the ethanol extract of *Ixora coccinea* leaves and assess its antioxidant property. Ethanol extract of *I. coccinea* were obtained using cold maceration method and assessed its antioxidant property by DPPH assay. Nanoemulsion containing extract was formulated and the particle size was determined with Zetasizer. Two nanoemulgels (NEG1 and NEG2) incorporated with nanoemulsion containing *I. coccinea* extract were formulated and evaluated including the stability study. In vitro drug release study was conducted using Franz Diffusion cell method. The antioxidant activity of the formulated nanoemulgels was also assessed. The results revealed that the ethanol extract of *I. coccinea* exhibited dose-dependent antioxidant activity, with 92.66% at 100 mcg/ml. The particle size of the most stable nanoemulsion was determined to be 152.1 ± 0.04 nm, with a zeta potential of 33.9 ± 0.07 mV. The formulated nanoemulgels (NEG 1 and NEG 2) were homogenous, spreadability (6.26 ± 0.44 gcm/s) with the viscosity of 4515.22 ± 55.81 cP. The % drug release for NEG 1 and NEG 2, were found to be 91.83 ± 0.11 and 82.74 ± 0.05 , respectively. The pH value ranged between 5.5 to 6.5. NEG1 showed higher antioxidant efficacy than NEG2 and the extract tested. The findings reveal that the formulated nanoemulgel (NEG1) incorporating the stable nanoemulsion loaded with *I. coccinea* extract has the highest antioxidant activity and is a promising formulation with satisfactory drug content, pH, viscosity, spreadability and stability.

Keywords: *Ixora coccinea* leaves, ethanol extract, nanoemulsion, antioxidant, nanoemulgel

OP-010

**Fish Oil Alginate Microspheres Produced via Wave-Based Drop-On-Demand Jetting
Electrospray: Improving Stability and Palatability**

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Despite the proven health benefits of fish oil to the human body, the unpleasant aftertaste and short shelf-life limit the public acceptability as supplementation. Microencapsulation of the fish oil into microspheres is an approach to overcome these limitations. This study developed controllable-size fish oil alginate microspheres using pulsed electrospray to enhance fish oil stability and palatability. The microspheres were optimized, characterized, and evaluated for stability and antioxidant activity over 6 months. Their anticancer activity was assessed using HCT116 and 3T3 cell lines, while a human panel test evaluated taste masking effectiveness. The optimal formulation exhibited an average particle size of $344.75 \pm 10.84 \mu\text{m}$, AR of 1.03 ± 0.02 , EE of $98.98 \pm 0.72\%$, and LC of $22.60 \pm 0.62\%$. The microencapsulation resulted in minimal fish oil loss in the stomach, leading to enhanced masking of the fishy taste. The FTIR analysis and DPPH assay revealed that the chemical properties of the fish oil and its antioxidant efficacy were preserved after microencapsulation and remained stable after 6 months of storage. The fish oil microspheres showed a superior anticancer effect compared to the free ones. In addition, the panellists of the palatability test of the fish oil microspheres showed an excellent taste masking ability. Hence, this study developed controllable-size fish oil microspheres via pulsed electrospray technique with enhanced fish oil stability, efficacy, and palatability, suggesting their potential utility as dietary supplements to fulfil the daily recommended intake of fish oil.

Keywords: fish oil, alginate, microspheres, taste masking, drop on demand, electrospray

OP-017

Development and Optimisation of a Mucoadhesive *In Situ* Nasal Gel of Budesonide for Enhanced Intranasal Delivery

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Allergic rhinitis is a prevalent inflammatory disorder, where effective intranasal drug delivery is limited by rapid mucociliary clearance and short residence time. *In situ* nasal gels offer a promising strategy to enhance drug retention and local bioavailability. To develop and optimise a mucoadhesive thermoresponsive *in situ* nasal gel of budesonide for improved nasal residence time and permeation. Budesonide *in situ* nasal gels were prepared using poloxamer 407 (P407) and iota carrageenan (ι -Cg) via the cold method. A 3^2 full factorial design was applied to optimise gelation temperature and viscosity under non-physiological (25 °C) and physiological (34 °C) conditions. The optimised formulation was evaluated for gelation time, mucoadhesive strength, physicochemical properties, compatibility (DSC, FTIR), and *ex vivo* permeation using goat nasal mucosa. The optimised formulation (F8; 0.22% w/v ι -Cg and 18% w/v P407) exhibited a gelation temperature of ~34 °C, viscosity <200 mPa·s at 25 °C and >10,000 mPa·s at 34 °C, ensuring ease of administration and effective gel formation. Rapid gelation occurred within 60 s, with mucoadhesive strength of 9492.12 dyne/cm². DSC and FTIR confirmed compatibility without chemical interaction. *Ex vivo* permeation showed enhanced drug delivery with a steady-state flux of 3.51 μ g/cm²/h and an enhancement ratio of 1.49 compared to the control. The developed mucoadhesive *in situ* nasal gel demonstrated favourable physicochemical properties, strong mucoadhesion, and improved permeation. This system represents a promising approach to enhancing the intranasal delivery of budesonide for the management of allergic rhinitis.

Keywords: budesonide, *in situ* gel, Poloxamer 407, thermoresponsive gel, nasal drug delivery

OP-023

Guiding Peptide Nanonet Formation and Antibacterial Activity through Aromatic Hydrophobicity Engineering for Sepsis Control

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Precise control over peptide nanonet architecture is critical for advancing antibacterial nanonets as a novel strategy to combat antibiotic resistance. In our previous work, we rationally designed novel β -hairpin peptide motifs in which strategic incorporation of aromatic residues (Trp or Phe) enabled the formation of a “tight nanowrap”, where nanofibers interweave directly on bacterial surfaces to form highly entangled nanonets that trap, kill, and restrict the spread of *E. coli* and *P. aeruginosa*. Building on this foundation, we sought to establish a design principle to further enhance peptide potency by fine-tuning the physicochemical properties of the key aromatic scaffold, an aspect for which no clear guiding rules currently exist. Using a panel of unnatural Trp and Phe analogs, we uncovered that antibacterial nanonet formation is governed not by absolute hydrophobicity, but by a narrow amphipilicity window that enables surface-templated assembly on bacterial membranes. Exceeding this window drove premature bulk peptide aggregation, spatially decoupling self-assembly from bacterial surfaces, while increased hydrophilicity suppressed supramolecular assembly and antibacterial activity altogether. Experimental data and molecular simulations collectively demonstrated that aromatic hydrophobicity is essential for membrane interaction and π - π stacking, stabilizing nanofiber assembly on bacterial membranes and leading to membrane damage. Notably, W3 and W4, positioned within this optimal narrow window, achieved a further two-fold enhancement in antibacterial potency while maintaining biocompatibility and robust tight nanowrap morphology against clinically relevant Gram-negative pathogens. Importantly, these peptides preserved tight nanowrap bacteria function in whole sheep blood, underscoring robustness in physiologically complex environments. Together, these findings reveal that aromatic residues function as regulators of membrane templated supramolecular assembly rather than simple membrane anchors, thereby establishing physicochemical boundaries for programming biologically localized supramolecular materials.

Keywords: aromatic β -hairpin peptides, nanonets, membrane assembly, antibacterial activity

OP-028

Development and Characterization of Prednisolone-Loaded Solid Lipid Nanoparticles for Transdermal Application

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The conventional administration of prednisolone for long-term therapy is often associated with drawbacks and poor patient compliance due to frequent dosing and high-dose regimens. A transdermal drug delivery system, capable of delivering the drug directly through the skin into the bloodstream, offers a promising alternative for targeted delivery of prednisolone. In this study, we aimed to develop prednisolone-loaded solid lipid nanoparticles (SLN) using an emulsification homogenization-ultrasonication approach, leveraging their ability to enhance drug stability and sustained drug release for transdermal application. The first series of SLN formulations was developed by varying drug-to-lipid ratios and ultrasonication time. Subsequently, a two-step process combining homogenization and ultrasonication was introduced, with a series of controlled parameters, including homogenization speed and duration, as well as ultrasonication time, temperature, and power. The results suggested that formulation H3, homogenized at 10,000 rpm for 4 minutes and ultrasonicated at 60 °C and 60 W for 4 minutes, resulted in a particle size of 319.00 ± 56.15 nm, a polydispersity index of 0.48 ± 0.022 , a zeta potential of -19.86 ± 9.56 mV, and a high entrapment efficiency of $98.25 \pm 0.49\%$. The formulation exhibited a sustained release of drug at $30.71 \pm 7.16\%$ after 24 hours and was governed by overlapping mechanisms involving diffusion and matrix-related processes. The incorporation of prednisolone into the lipid matrix was confirmed by attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR), differential scanning calorimetry (DSC), and transmission electron microscopy (TEM). Overall, this study establishes a systematic framework for optimizing prednisolone-loaded SLN and highlights the critical influence of processing conditions on nanoparticle performance for potential transdermal application.

Keywords: prednisolone, solid lipid nanoparticles, homogenization, ultrasonication, transdermal drug delivery

OP-029

Rapidly Dissolving Microneedle Patch Integrated with Benidipine-Loaded Transethosomes for Enhanced Transdermal Delivery and Bioavailability

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Benidipine (BEN) is an antihypertensive drug with low oral bioavailability (20-30%) due to poor aqueous solubility and extensive first-pass metabolism, limiting its therapeutic efficacy. This study aimed to develop and evaluate a novel transdermal delivery system comprising BEN-loaded transethosomes (BEN-TE) integrated into hyaluronic acid-based dissolving microneedles (DMNs) to enhance skin permeation and bioavailability. BEN-TEs were prepared using the thin-layer rotary evaporation method to produce unilamellar lipid vesicles and optimized using a Box-Behnken design. The optimized BEN-TE demonstrated a vesicle size of 168.3 ± 1.73 nm, entrapment efficiency of $96.42 \pm 0.36\%$, and transdermal flux of 20.21 ± 0.35 $\mu\text{g}/\text{cm}^2/\text{hr}$. Differential scanning calorimetry (DSC), attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR), and X-ray diffraction (XRD) analyses confirmed the amorphous state of BEN in the optimized formulation. Field emission scanning electron microscopy (FESEM) and high-resolution transmission electron microscopy (HRTEM) revealed smooth, spherical, nanosized vesicles with uniform morphology, while confocal laser scanning microscopy (CLSM) confirmed enhanced permeation of lipid vesicles into deeper skin layers. Dissolving microneedles containing 40% hyaluronic acid and loaded with the optimized BEN-TE formulation exhibited adequate mechanical strength, effective insertion into rat skin, and rapid dissolution. *Ex vivo* permeation studies demonstrated a significantly higher permeation coefficient for BEN-TE-DMNs compared with BEN-TE alone or BEN-DMNs alone. *In vivo* pharmacokinetic studies showed that the relative bioavailability of BEN-TE-DMNs increased by 1.86-fold compared with the marketed oral BEN tablet and 3.45-fold compared with BEN-DMNs. In conclusion, the integration of transethosomes with dissolving microneedles represents a promising strategy for enhancing the transdermal delivery and bioavailability of benidipine.

Keywords: Benidipine, dissolving microneedles, transethosomes, transdermal delivery, bioavailability

OP-037

Physicochemical and Microbiological Stability of Specialised Parenteral Nutrition Formulations for Patients at Risk of Refeeding Syndrome

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Current commercial parenteral nutrition (PN) bags often fail to meet the low-calorie, high-electrolyte requirements needed to safely manage patients at risk of refeeding syndrome (RS). This study aimed to establish the stability profiles of two novel, specialised PN formulations (Regimen E and W) designed specifically for RS risk management. Two low-volume, high-protein, and high-electrolyte formulations were compounded (n=20 per regimen) and stored at 2-8°C with light protection. Stability was evaluated at five time points (Days 1, 7, 14, 21, and 28), including assessment 7 days after adding micronutrients (vitamins and trace elements). Physical stability was assessed via visual inspection, pH, mean droplet diameter (MDD) using dynamic light scattering (DLS), and zeta potential. Chemical stability of amino acids and electrolytes was analysed using Liquid Chromatography-Mass Spectrometry (LC-MS) and Ion Chromatography. Microbiological stability was verified via membrane filtration sterility testing. Statistical significance was determined using the Mann-Whitney U test ($p < 0.05$). Both regimens remained physically stable; MDD ranged from 204.5 ± 0.7 nm to 216.2 ± 0.5 nm, zeta potential from -36.9 ± 0.1 mV to -33.4 ± 0.1 mV ($p > 0.05$ across all points) and pH remained stable between 6.17 to 6.85. Chemical analysis showed both formulations were stable for 14 days, with nutrient variations $<15\%$ from baseline. Both regimens were sterile for 28 days of storage. For administration, Regimen E and W remained sterile for 48 and 24 hours, respectively. Specialised PN formulations for RS (Regimen E and W) are physically, chemically, and microbiologically stable for refrigerated storage up to 14 days, including 7 days post-additive inclusion. These findings support the clinical implementation of stable, pre-defined PN regimens to improve safety for patients at risk of refeeding syndrome.

Keywords: refeeding syndrome, parenteral nutrition, formulation stability, physicochemical stability, microbiological stability

OP-040

Microalgae-derived Extracellular Vesicles as Sustainable Nanocarriers: Advances in Production, Isolation and Biomedical Applications

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Extracellular vesicles (EVs) are increasingly recognized as superior biogenic nanocarriers. However, mammalian-derived EVs face significant bottlenecks, including prohibitive production costs, ethical constraints, and limited scalability. Microalgae-derived extracellular vesicles (mEVs), also known as nanoalgosomes, have emerged as a high-yield, sustainable alternative, due to their renewable cultivation, high yield, and intrinsic bioactivity. Despite their potential, the technical landscape regarding their engineering and therapeutic validation remains fragmented. This scoping review systematically maps the current evidence on the production, isolation and biomedical applications of mEVs, with emphasis on pharmaceutically relevant species, specifically *Tetraselmis chuii*, *Haematococcus pluvialis*, *Acutodesmus*, and *Arthrospira platensis*. Following the Joanna Briggs Institute (JBI) methodology and PRISMA-ScR guidelines, relevant literature on focusing on mEV bioprocessing, isolation strategies, nano-preparation, and their biomedical applications was systematically synthesized. Analysis identified key species enriched with therapeutic lipidomic and antioxidant profiles. While differential ultracentrifugation remains common, Tangential Flow Filtration (TFF) demonstrates strong potential for scalable production due to improved vesicle preservation and reduced mechanical stress. mEVs exhibit exceptional biocompatibility and cellular uptake, supporting their application as dual-action nanocarriers where the innate algal cargo synergizes with loaded therapeutics. Emerging applications range from bone-targeted delivery to inhalable antioxidant therapy. Nanoalgosomes represent a disruptive green nanomedicine platform. However, further optimization of mEV biophysical properties, standardized isolation protocols, and scalable production strategies is essential to facilitate their clinical translation and future therapeutic implementation.

Keywords: extracellular vesicles, nanoalgosomes, drug delivery, green nanomedicine, scoping review

OP-096

Artificial Intelligence in Pharmaceutical Quality Assurance: Applications, Regulatory Challenges, and Future Directions

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The increasing complexity of pharmaceutical manufacturing and the rapid digital transformation of healthcare have accelerated the adoption of artificial intelligence (AI) in pharmaceutical quality assurance. This narrative review evaluates current AI applications in pharmaceutical quality assurance, examines major regulatory and implementation challenges, and discusses future directions for AI within Good Manufacturing Practice (GMP)-compliant quality systems. Recent peer-reviewed publications and regulatory documents on AI, pharmaceutical manufacturing, quality assurance, Quality by Design, and GMP were reviewed to identify common applications, implementation barriers, and emerging regulatory perspectives. The literature indicates that AI can support real-time process monitoring, predictive quality management, intelligent deviation detection, automated visual inspection, and risk-based decision-making. AI may also improve process understanding and manufacturing consistency. However, successful implementation depends on robust model validation, data integrity, explainability, cybersecurity, lifecycle management, and continuous human oversight. AI has strong potential to strengthen pharmaceutical quality assurance, but responsible adoption requires validated systems, effective governance, and alignment with evolving GMP and regulatory expectations.

Keywords: artificial intelligence, pharmaceutical quality assurance, good manufacturing practice, regulatory compliance, quality by design

Cosmetic Science & Innovation (CSI)

OP-003

Facial Pigmentation in Bangladeshi Adults: Prevalence and Associated Factors

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Facial pigmentation is a frequent dermatological condition that has an impact on both cosmetic appearances and quality of life, especially in populations that are exposed to a lot of ultraviolet light. Pigmentary disorders are prevalent in South Asia, although epidemiological data on facial pigmentation in Bangladeshi adults are scarce. Understanding its prevalence and associated variables is imperative for developing effective preventative measures and dermatological treatment. The purpose of this study was to evaluate the prevalence of facial pigmentation among Bangladeshi adults and evaluate its relationship with socioeconomic factors, sunlight exposure, protective sunscreen use, and Fitzpatrick skin type. A cross-sectional observational study was conducted among 475 individuals from Kishoregonj City, Bangladesh, including both urban and semi-urban areas. Data were gathered using standardized questionnaires and clinical examinations. Demographics, solar exposure patterns, sun protection techniques, cosmetic use, and lifestyle characteristics were all documented. The severity of facial pigmentation was determined using the Kesty Hyperpigmentation Scale (KHS), and skin phototype was graded using the Fitzpatrick Skin Type Scale. Descriptive statistics, bivariate analysis, and multivariate logistic regression were used to find relevant factors influencing pigmentation severity. 68.2% of the participants were female, with 60.0% ranging between the ages of 18 and 30. Facial pigmentation was obvious in 57.1% of patients, and most instances lasted more than a year (61.6%). However, just 4.1% received treatment. Fitzpatrick skin type III was the most prevalent, and pigmentation intensity was significantly associated with skin type ($p < 0.001$). Daily sun exposure of more than two hours was substantially linked with increasing pigmentation severity. Sunscreen and other photoprotective measures were connected to decreased severity. Facial pigmentation is very prevalent among Bangladeshi adults and is significantly linked to ultraviolet exposure and skin phototype. To lower its prevalence, more photoprotection awareness and early dermatological treatments are required.

Keywords: facial pigmentation, sun exposure, Fitzpatrick skin type, Kesty Hyperpigmentation Scale, sunscreen use

OP-035

Formulation Development and Stability Evaluation of an Astaxanthin Nanoemulsion Lotion

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Astaxanthin is a potent xanthophyll carotenoid with therapeutic benefits for skin health, but is limited by poor aqueous solubility. Therefore, a stable vehicle for topical astaxanthin preparation is essential to protect the bioactive compound from environmental degradation and ensure effective delivery. While various dosage forms exist, water-based astaxanthin nanoemulsion lotions remain underexplored. This study aimed to develop, optimize and characterize the physicochemical and microbial stability of a water-based astaxanthin nanoemulsion lotion for potential therapeutic use in wound care. Optimization was conducted in two phases: Phase 1 evaluated six formulations (F1–F6) with varying viscosity enhancer concentrations (Repoly at 1–5% w/w), Phase 2 developed three formulations (F7–F9) at an optimal 2.2% w/w Repoly with various astaxanthin concentrations (1, 3 and 5% w/w). Six-month accelerated stability testing was conducted at 5 °C, 25 °C / 65% RH and 40 °C / 75% RH, evaluating organoleptic properties, pH, globule size, PDI, zeta potential, viscosity and microbial contamination test. The target specifications for the optimized formulation were a pH below 6.3, a PDI below 0.7, a globule size below 500 nm, a zeta potential more than -30 mV, and a viscosity of 11–13 Pa·s. F9 (5% w/w astaxanthin, 2.2% w/w Repoly) was identified as the optimized formulation, with a pH of 6.2, globule size of 239.2 nm, PDI of 0.24, zeta potential of -54.7 mV and viscosity of 11.5 Pa·s. At 5°C, it demonstrated good stability over 6 months with minimal changes in pH (6.3 to 6.1), globule size (151.9 to 192.8 nm), PDI (0.19 to 0.23), zeta potential (-77.1 to -58.9 mV) and viscosity (12.5 to 12.1 Pa·s). The lotion maintained its bright orange color, remained odorless and showed no microbial contamination, warranting further *in vivo* evaluation.

Keywords: Astaxanthin, nanoemulsion, lotion, stability study, formulation development

Pharmaceutical Chemistry (PC)

OP-004

Anti-Metastatic Effects of Andrographolide and Its Derivatives in Triple-Negative Breast Cancer (MDA-MB-231)

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Breast cancer continues harming women globally with a high mortality rate every year. There were 2.26 million new cases of breast cancer reported in 2020 alone by the World Health Organization. This problem is due to a lack of effective targeted treatments for the deadliest type of breast cancer, known as triple-negative breast cancer. This research aims to ferment andrographolide with selected fungi and analyze the biotransformed metabolites using high-performance liquid chromatography profiles. Fermentation of andrographolide with selected fungi allows the biosynthesis of secondary metabolites. Biotransformed andrographolide derivatives may offer higher bioavailability and greater anti-metastatic potential against triple-negative breast cancer cells than andrographolide. The fungi used for microbial transformation fermentation were *Beauveria bassiana* and *Aspergillus niger*. Fermentation of andrographolide with selected fungi was carried out for 12 days, followed by extraction with ethyl acetate. High-performance liquid chromatography profiles of biotransformation extracts were compared with the starting material and controls, revealing *Aspergillus niger* as the most promising source of new andrographolide derivatives. MTT assay was conducted to assess andrographolide and its derivative on MDA-MB-231 cell viability and toxicity. Longer exposure and higher concentrations reduced cell viability, with IC50 values showing a similar trend. Female BALB/C mice were pretreated with immunosuppressants before being injected with the tumor to allow better tumor engraftment. Further treatment with andrographolide and its derivative will be administered for 24 days, after which blood and tissue samples will be collected for liquid chromatography-mass spectrometry metabolomic analysis to investigate metabolic changes associated with the inhibition of tumor metastasis. This study will be beneficial to the healthcare and pharmaceutical sectors, as it contributes valuable information to aid in the development of new anti-metastatic treatments for aggressive types of breast cancer.

Keywords: microbial transformation, andrographolide, *Aspergillus niger*, *Beauveria bassiana*, anti-cancer

OP-007

Drug Discovery and Natural Products: Microbes as Hidden Architects

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Natural products have long been a cornerstone of drug discovery, and microorganisms, especially bacteria and fungi—play a crucial role in this process. Many clinically important drugs originate from microbial metabolites, highlighting the strong connection between microbiology and pharmaceutical innovation. This work highlights the dual significance of microbial systems in the isolation of secondary metabolites and in the microbial transformation of structurally complex molecules such as steroids and terpenes. Microbial-derived secondary metabolites, obtained from endophyte fungal sources, represent a rich reservoir of bioactive compounds with therapeutic potential, including antimicrobial, anticancer, and anti-inflammatory activities. In addition to direct metabolite production, microorganisms offer powerful and sustainable platforms for structural modification through biotransformation. Steroids and terpenes, known for their intricate frameworks and pharmacological relevance, can be selectively modified by microbial enzymes under mild conditions, enabling regio- and stereospecific transformations that are often difficult to achieve via conventional synthetic methods. These transformations include hydroxylation, oxidation, reduction, and glycosylation, leading to the generation of novel derivatives with enhanced biological activity, improved solubility, and reduced toxicity. The integration of secondary metabolite isolation with microbial biotransformation expands the chemical space available for drug discovery while supporting green chemistry principles. This approach not only accelerates lead optimization but also provides insights into metabolic pathways and enzyme specificity. Overall, exploiting microorganisms as both producers and biocatalysts presents a promising and sustainable strategy for the development of next-generation therapeutics. In summary, microorganisms remain a powerful and largely untapped source of new drugs. Their ability to produce diverse and bioactive natural products continues to drive innovation in drug discovery, particularly in the search for new antibiotics and treatments for complex diseases.

Keywords: microbial transformation, endophytes, steroids, terpenes, antimicrobial

OP-014

**Green Synthesis of Copper Oxide Nanoparticles Using *Annona muricata* Leaf Extract:
Optimisation, Characterisation and Antimicrobial Activity**

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Antimicrobial resistance is recognized as one of the top ten global public health threats, largely driven by the misuse and overuse of antibiotics. Copper (II) oxide nanoparticles (CuO NPs) have attracted attention as broad-spectrum antimicrobial agents, however, their clinical application is limited by toxicity concerns. Conjugation with short antimicrobial peptides such as Aurein 1.2 may improve antibacterial selectivity and reduce toxicity as it is a cationic 13-amino-acid peptide with an amphipathic α -helical structure that disrupts bacterial membranes and exhibits broad-spectrum antimicrobial activity. In this study, CuO NPs were synthesized via green synthesis approach to minimize the use of toxic reagents and energy-intensive processes. Three Aurein 1.2 variants, namely the native peptide (GLFDIHKIAESF), N-terminal cysteine-modified peptide (CGLFDIHKIAESF), and C-terminal cysteine-modified peptide (GLFDIHKIAESFC), were conjugated with CuO NPs at various concentrations. The synthesized nanoparticles and conjugates were characterized using UV-visible spectroscopy, zeta potential measurement, FTIR and FESEM-EDX. The synthesized CuO NPs exhibited a surface plasmon resonance peak at 362 nm with an average particle size of approximately 96.2 nm, and a negative zeta potential of -32.60 ± 2.0 mV. FESEM analysis revealed predominantly spherical nanoparticles with minor rice grain-shaped structures. FTIR analysis confirmed surface capping by plant-derived metabolites and showed a characteristic Cu–O stretching peak at 525.25 cm^{-1} . EDX analysis verified the elemental composition of Cu and O, while XRD confirmed the monoclinic crystalline phase with an average crystallite size of 29.43 nm. Following conjugation, noticeable shifts in UV-visible absorption spectra and FTIR bands were observed, indicating successful peptide–nanoparticle interactions through electrostatic attraction and Cu–S coordination bonding. FESEM-EDX analysis further demonstrated relatively uniform morphology, increased carbon content and the presence of nitrogen signals, confirming successful peptide conjugation. Overall, the findings confirmed the successful conjugation of Aurein 1.2 variants with CuO NPs. Further antimicrobial and cytotoxicity evaluation are warranted.

Keywords: antimicrobial peptide, antimicrobial resistance, Copper (II) oxide nanoparticles, green synthesis, peptide–nanoparticle conjugates

OP-036

Computational Investigation of Kaempferol, Myricetin, Catechin, Mangiferin, and Naringin for Antimalarial Drug Development

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Malaria continues to pose a significant worldwide health threat, especially in tropical areas, and the swift development of drug-resistant *Plasmodium* strains demands the identification of novel antimalarial compounds. This work sought to assess the antimalarial efficacy of five flavonoid compounds kaempferol, myricetin, catechin, mangiferin, and naringin were extracted from *Pyrrhosia longifolia*. Molecular docking was performed against the *Plasmodium falciparum* dihydrofolate reductase–thymidylate synthase (PfDHFR-TS; PDB ID: 1J3I) to see how well it binds and interacts with other molecules. To learn more about the compounds' electronic characteristics, stability, and reactivity, we used Density Functional Theory (DFT) computations and Molecular Electrostatic Potential (MEP) investigations. Also, drug-likeness and ADMET (absorption, distribution, metabolism, excretion, and toxicity) predictions were made to see if the medicine would work well in the body. The docking data showed that all of the compounds interacted well with the active site of PfDHFR-TS, with naringin having the strongest binding affinity. DFT and MEP analyses corroborated the electronic stability and reactivity of the examined flavonoids, whereas drug-likeness and ADMET predictions suggested satisfactory pharmacokinetic profiles. In general, these results point to naringin, a flavonoid from *Pyrrhosia longifolia*, as a good lead candidate for further research as an antimalarial drug.

Keywords: *Pyrrhosia longifolia*, docking, DFT, MEP, drug-likeness

OP-038

Metabolic Fingerprinting of Medicinal Plants and Fungi: NMR-Based Strategies for Uncovering Neurotherapeutic Agents

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Natural products remain a promising source of neuroprotective agents, yet rapid chemical profiling of complex extracts remains challenging. This study employed high-resolution ¹H NMR and 2D HSQC spectroscopy combined with in vivo neurotoxicity models to identify bioactive secondary metabolites from medicinal plants (*Foeniculum vulgare*, *Curcuma longa*, *Syzygium aromaticum*, *Murraya koenigii*, *Nigella sativa*) and the entomopathogenic fungus *Beauveria bassiana* and *Aspergillus Niger*. NMR analysis revealed key chemical entities: curcuminoids and artemisinin (turmeric); flavonoids, tannins, and saponins (clove); carbazole alkaloids (*M. koenigii*); phenolic acids and coumarins (fennel). DPPH assays confirmed potent antioxidant activity across extracts. In lead and aluminum-induced neurotoxicity mouse models, these extracts significantly normalized dysregulated expression of oxidative stress markers (SOD1, Prdx6) and amyloid precursor protein isoforms (APP695, APP770), reduced serum corticosterone, improved spatial memory in Morris water maze, and restored cortical neuronal morphology. Notably, *B. bassiana* and *Aspergillus Niger* produced distinct metabolite profiles depending on growth medium PDB yielding organic acids, quinines, and 2-pyridone alkaloids, while specific media favored oosporein production demonstrating medium-driven metabolic plasticity. Our findings establish NMR-based metabolomics as a powerful, quantitative platform for guiding natural product drug discovery. This approach enables rapid identification of neuroprotective chemical entities and optimization of culture conditions for desired metabolites, accelerating the development of plant- and fungi-derived therapeutics for neurodegenerative disorders.

Keywords: NMR metabolomics, neuroprotection, natural products, secondary metabolites, oxidative stress

OP-039

Design and Synthesis of Antiviral Small Molecule Drugs Inspired by Antimalarial Drugs

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The pandemic of COVID-19 caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) remains serious concern in the world. Although some effective vaccines have been developed, only a few anti-SARS-CoV-2 drugs have been approved for the clinical use, and hence oral therapeutics against SARS-CoV-2 and other emerging and re-emerging infectious diseases are highly required. Since 2020, we have designed and synthesized ca. 500 derivatives based on the structure of amodiaquine, which had been approved as an antimalarial drug, as a seed compound. Among these synthesized compounds, some compounds were found to exhibit low toxicity and potent antiviral activity. Importantly, these compounds have almost same antiviral activity against alpha to epsilon mutants of SARS-CoV-2. Besides, antiviral activity was observed against model mice infected with SARS-CoV-2 after oral administration of our drugs. In this presentation, these results will be reported as well as the inhibitory effect of quinoline-based drugs on the production of inflammatory cytokine from macrophage activated via Toll-like receptors.

Keywords: pandemic, SARS-CoV-2, emerging and re-emerging infectious diseases, antiviral agents, small molecule drugs, prevention of cytokine storm

OP-048

Synthesis of Macrocyclic Oligosaccharides by One-pot Glycosylation of Glycosyl Donor/Acceptors and Complex with Metal Cations.

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Cyclic oligosaccharides such as cyclodextrins (CDs) are attractive and useful host molecules for the improvement of stability and solubility of organic guest molecules and drugs in aqueous phase or metal cations in organic phases. Therefore, efficient and powerful methods for supplying macrocyclic oligosaccharides are desired. In our laboratory, one-pot polyglycosylation of monosaccharides bearing both properties of glycosyl donors and acceptors have been developed. This method is expected to provide an effective approach for the simultaneous synthesis of cyclic oligosaccharides and so far, trimeric and tetrameric cycloligosaccharides have been mainly obtained. The object of this work is to synthesize large cyclic oligosaccharides using thiogalactosides bearing modified protecting groups and to evaluate the host-guest interaction with metal ions and organic guest molecules. Herein, we report on the one-pot synthesis of cyclic oligosaccharides by using galactosyl donor/acceptor substrates, in which the cyclic trimer was obtained as a main product as well as tetramer, pentamer, and hexamer. The recrystallization of some products was carried out to confirm the structure of cyclic oligosaccharides and the structure of a 1:1 complex of the deprotected cyclic trimer with sodium ion was disclosed by X-ray crystallography analysis. Currently, one-pot glycosylation in the presence of metals and additives that are expected to function as templates for the synthesis of larger cyclic oligosaccharides is conducted and these results will be reported in this paper.

Keywords: glycosylation, cyclic oligosaccharides, one-pot synthesis, glycosyl donor/acceptors, metal complexes

OP-050

Phosphate Monoester Hydrolysis by Metallosupramolecular Phosphatases Functionalized with Lewis Acidic Moieties in Two-Phase Solvent Systems

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Protein phosphorylation and dephosphorylation reactions of intracellular molecules catalyzed by enzymes such as kinases and phosphatases are essential reactions in a vast number of cellular functions such as intracellular signal transduction in living systems. The design and synthesis of artificial enzyme mimics are important research topics in bioorganic and bioinorganic chemistry. We previously reported on the hydrophobic and amphiphilic supramolecular complexes formed by the self-assembly of hydrophobic and amphiphilic bis(Zn²⁺-cyclen) complexes (cyclen = 1,4,7,10-tetraazacyclododecane), functionalized barbitol derivatives and Cu²⁺ that accelerate the hydrolysis of mono(4-nitrophenyl) phosphate (MNP) in two phase solvent systems. It was found that the MNP hydrolysis in two phase solvent systems obey Michaelis-Menten kinetics, suggesting these metallosupramolecular systems are good models of natural enzymes. Besides, amphiphilic metallosupramolecular complexes were found to hydrolyze MNP in a catalytic manner, possibly because their Cu₂(μ-OH)₂ active sites are restored by the release of inorganic phosphate, which is a product of the MNP hydrolysis, to the aqueous phase. In this work, we hypothesize that MNP, which coordinates to the Cu₂(μ-OH)₂ core in supramolecular complexes, could be activated either by Lewis acidic units such as alkali metal (Li⁺, Na⁺ and K⁺)-benzocrown ether complexes or by boronophenyl moieties around Cu₂(μ-OH)₂ active site. The findings suggest that metallosupramolecular phosphatase functionalized with a benzo-12-crown-4-Li⁺ complex shows a higher level of activity in the MNP hydrolysis of a two-phase solvent system compared with that of our previous supramolecular phosphatases in terms of hydrolysis activity and catalytic turnover. These results will be presented.

Keywords: metallosupramolecular complexes, self-assembly, artificial phosphatases, Lewis acid, hydrolysis of phosphate monoester

OP-061

Synthesis of Three 2'-Aminomethoxy Chalcone Derivatives and In Vitro Hemin Polymerization Inhibition Activity Testing

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Malaria remains a health problem, especially in tropical regions. The presence of resistance to several malaria drugs has now become a concern for the development of new drugs with better efficacy and low levels of resistance. This study conducted the synthesis of three 2'-amino methoxy chalcone derivatives (**2a-2c**) using the microwave method using 2'-amino methoxy acetophenone as the starting material and 3 variations of methoxy benzaldehyde at positions 2,4 (**2a**); 2,5 (**2b**) and 3,4 (**2c**). The results of the structural characterization of the three synthesized compounds through IR spectra, as well as ¹H-NMR and ¹³C-NMR showed that the three structures obtained were in accordance with the target structure. In vitro antimalarial activity testing was carried out through inhibition of β -hematin formation. Of the three chalcone derivative compounds, compound **2c** has a better IC₅₀ value of 52.21 μ g/mL compared to **2b** (71.39 μ g/mL) and compound **2a** (106.09 μ g/mL), while the comparator hydroxychloroquine sulfate has an IC₅₀ value of 458.52 μ g/mL. From these results, the methoxy position in the chalcone compound can affect the inhibitory activity of hemin polymerization in the formation of β -hematin.

Keywords: malaria, amino-chalcone, synthesis, hemin polymerization, inhibition

OP-100

From Ocean to Oncology: ML-Driven Discovery of Marine PIM-1 Kinase Inhibitors with Multiscale Validation

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Marine natural products represent an immense source of chemically diverse scaffolds with considerable potential for anticancer drug discovery. In this study, an integrated computational workflow combining machine learning (ML), structure-based virtual screening, molecular dynamics (MD) simulations, and pharmacokinetic prediction was established to identify novel inhibitors of PIM-1 kinase, an oncogenic serine/threonine kinase associated with tumor progression and therapeutic resistance. A benchmark dataset of experimentally characterized PIM-1 inhibitors retrieved from ChEMBL was used to construct and compare eleven ML-based QSAR classification models. The highest-performing model was subsequently employed to prioritize compounds from the Comprehensive Marine Natural Products Database (CMNPD). Fifteen compounds with the highest prediction of confidence were subjected to molecular docking, and candidates were selected based on conservation of critical binding interactions within the ATP-binding pocket rather than docking scores alone. The binding stability of the selected compounds was further examined through 100 ns MD simulations and MM/GBSA binding free energy analyses, followed by in silico ADMET evaluation. Among the screened compounds, CMNPD11687 exhibited the most favorable overall computational profile, maintaining a stable binding mode during MD simulations (average RMSD: 4.10 ± 1.67 Å), favorable binding free energy ($\Delta G = -18.45 \pm 2.70$ kcal/mol), and acceptable predicted pharmacokinetic and safety properties. Although experimental validation is still required, these findings demonstrate that integrating ML-based prioritization with multiscale molecular simulations provides an efficient strategy for exploring marine chemical space and identifying promising scaffolds for future PIM-1 inhibitor development.

Keywords: marine, PIM1, machine learning, docking, MD simulations

OP-115

Comprehensive Metabolite Profiling of Ultrasonication-Assisted Extracted Leaf, Stem, Root, and Flower Extracts of *Clidemia hirta* Via Integrated GC-MS and LC-MS Analysis and Total Phenolic Content Determination

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Clidemia hirta is a medicinal perennial shrub belonging to the Melastomataceae family, widely distributed across tropical and subtropical regions. This shrub is traditionally used as soap and for treating skin ailments, wounds, and infections. Despite its ethnomedicinal relevance, comprehensive comparative information on the chemical composition of its different tissues remains limited. The current study investigated the metabolic profiles and total phenolic content (TPC) of *C. hirta* leaves, stems, roots, and flowers using ultrasonication-assisted extraction (UAE) with solvents of varying polarities such as methanol, acetone, ethyl acetate, chloroform, and n-hexane. These extracts were analyzed using both GC-MS and LC-MS analysis techniques using NIST 2008 and METLIN database, respectively. GC-MS analysis showed a wide array of volatile and semi-volatile components, such as phenolic compounds, fatty alcohols, carotenoids, and sterols, with 2,4-bis(1,1-dimethylethyl) phenol being highly abundant in ethyl acetate fractionation. Furthermore, LC-MS analysis of methanolic fractions suggested that there were a variety of high molecular weight metabolites, such as alnusiin, delphinidin, quillaic acid, and arjunolic acid, which had shown biological significance. The compound alnusiin was reported for the first time in this plant. TPC was found to vary widely in different parts of the plant and solvent systems, with the methanolic flower extract being the most enriched (85.57 ± 5.86 mg GAE/g). In general, polar and intermediate-polarity solvents were more effective in recovering phenolics compared to non-polar solvents. Overall, the results indicate significant differences in phytochemical constituents of *C. hirta* based on the type of tissue and solvent used. This comprehensive database of metabolites is important in establishing a chemical profile of *C. hirta* that can be used in developing future work on bioassay-guided fractionation, targeted isolation of metabolites, and mechanisms for this plant material, among others.

Keywords: *Clidemia hirta*, ultrasonication extraction, phytochemicals, GC-MS analysis, LC-MS analysis, total phenolic content

Regenerative & Translational Medicine (RTM)

OP-066

Lymph Nodes Provide an Immune-Regulatory Niche for Islet Organoid Engraftment and Endocrine Function

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To determine how anatomical transplantation sites influence immune responses, graft survival, and endocrine function, and to identify an optimal niche for insulin-producing grafts using both murine and human induced pluripotent stem cell (hiPSC)-derived endocrine graft models. To determine how anatomical transplantation sites influence immune responses, graft survival, and endocrine function, and to identify an optimal niche for insulin-producing grafts. hiPSC-derived islet organoids were included to evaluate translational potential. Mesenchymal stromal cell (MSC)-based cell sheets were transplanted into subcutaneous, perivascular, adipose tissue, or lymph node sites in mice. Local immune responses, cytokine production, vascularization, and graft survival were analyzed using immunohistochemistry, flow cytometry, and bioluminescence imaging. Functional efficacy was evaluated using layered MSC sheets containing insulin-producing MIN6 cells or hiPSC-derived islet organoids in streptozotocin-induced diabetic and total pancreatectomy mouse models. MSC sheet transplantation into lymph node uniquely promoted a regulatory immune environment characterized by increased regulatory T-cell accumulation and transforming growth factor- β production while suppressing inflammatory responses observed at other transplantation sites. MSC-based grafts survived for more than 100 days within lymph nodes. In diabetic mice, lymph node transplantation of MIN6-containing sheets achieved stable long-term glycemic control. In total pancreatectomy models, lymph node grafts restored normoglycemia, whereas graft removal resulted in rapid recurrence of hyperglycemia, demonstrating graft-dependent endocrine function. Importantly, transplantation of hiPSC-derived islet organoid-containing sheets into lymph nodes maintained glucose homeostasis, induced sustained human insulin secretion, and promoted expression of endocrine maturation markers, including human C-peptide, PDX1, and NKX2.2. These outcomes were markedly superior to those observed after subcutaneous or perivascular transplantation. Lymph nodes function as immune-regulatory niches that support survival, maturation, and endocrine activity of hiPSC-derived islet organoid grafts. This strategy enables restoration of glucose homeostasis even after total pancreatectomy and may provide a clinically translatable platform for regenerative diabetes therapy.

Keywords: cell transplantation, islet organoids, immune tolerance, ectopic organogenesis, diabetes

Policy, Regulation & Ethics (PRE)

OP-024

Regulatory Governance of Over-The-Counter Medicine Distribution in Non-Pharmacy Retail Settings: A Comparative Policy Analysis of Malaysia and Indonesia

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The expansion of access to over-the-counter medicines has increasingly been adopted as a policy strategy to support self-medication and improve health system efficiency. Many countries have implemented risk-based classification systems to balance medicine accessibility with patient safety. However, evidence remains limited regarding how such risk classifications are integrated with distribution systems, professional oversight, and consumer medicine information systems, particularly in decentralized medicine distribution settings. This study aims to analyze the regulatory governance of OTC medicines in Malaysia and assess its implications for strengthening the control of OTC medicine distribution in modern retail settings in Indonesia. A qualitative comparative policy analysis was conducted using key regulatory documents from Malaysia and Indonesia. The analyzed documents comprised the Poisons Act 1952, Sale of Drugs Act 1952, and Control of Drugs and Cosmetics Regulations 1984 in Malaysia, along with Indonesia's Law No. 17 of 2023 on Health, Minister of Health Decree No. 972 of 2025, and BPOM Regulation No. 5 of 2026. This study applied a multidomain framework covering medicine classification, distribution systems, OTC restrictions, and risk control instruments. Malaysia demonstrates strong integration between risk-based classification, pharmacy-based OTC distribution, pharmacist supervision, and monitoring mechanisms. In contrast, Indonesia exhibits a more decentralized distribution model through healthcare facilities and retail outlets, with regulatory control primarily relying on product-based measures such as quantity restrictions and minimum packaging requirements in modern retail settings. This reflects the limited integration between risk-based classification and system-based distribution governance. Effective OTC medicine governance requires strengthening risk-based distribution control integrated within formal pharmaceutical distribution systems through licensed pharmaceutical wholesalers. Policy strengthening in modern retail settings should focus on restricting OTC medicine categories, standardizing dispensing procedures (SOPs), implementing small-unit packaging, and establishing referral mechanisms to pharmacies for higher-risk cases to enhance the safety of self-medication practices.

Keywords: over-the-counter drugs, regulatory governance, distribution, non-pharmacy retail

OP-055

Does Health Insurance Status Influence Patient Satisfaction with Pharmaceutical Services? A Servqual-Based Study Among Indonesian Outpatients

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Patient satisfaction is a key indicator of healthcare service quality and is frequently used to evaluate pharmaceutical services in hospitals. Differences in health insurance status may influence patients' perceptions and expectations regarding the quality of care received. This study aimed to compare the satisfaction levels of National Health Insurance (NHI) beneficiaries and self-paying outpatients toward pharmaceutical services at Bangkinang Regional General Hospital, Indonesia, using the SERVQUAL model. An analytical observational study with a cross-sectional design was conducted among outpatients who had received pharmaceutical services. A total of 110 respondents were recruited through purposive sampling, consisting of 55 NHI beneficiaries and 55 self-paying patients. Data were collected using a validated SERVQUAL questionnaire covering five dimensions: tangibles, reliability, responsiveness, assurance, and empathy. The data were analysed using the Chi-square test. The results showed that the majority of respondents were satisfied with the pharmaceutical services provided, with satisfaction rates of 80.0% among NHI beneficiaries and 74.6% among self-paying patients. Minor differences were observed in the reliability and responsiveness dimensions; however, statistical analysis revealed no significant difference in overall satisfaction levels between the two groups ($p > 0.05$). In conclusion, pharmaceutical services at Bangkinang Regional General Hospital were generally perceived positively by both insured and self-paying outpatients, and health insurance status was not significantly associated with patient satisfaction. These findings suggest that the quality of pharmaceutical services was delivered equitably regardless of patients' insurance coverage.

Keywords: patient satisfaction, pharmaceutical services, SERVQUAL, health insurance, outpatients

OP-056

Tuberculosis Knowledge, Attitude, And Practice Among Community Pharmacists in A High-Burden Tuberculosis Setting in Indonesia

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Community pharmacists' (CPs) involvement could improve tuberculosis (TB) case detection and treatment. To effectively involve CPs in TB control, they must have adequate knowledge, attitudes, and behaviors. Given the large number of pharmacies and the high burden of TB in Bandung, we aim to analyze the knowledge, attitude, and practice (KAP) of CPs in Bandung. A cross-sectional study was conducted among CPs in Bandung City using a self-administered questionnaire to assess their characteristics and KAP regarding tuberculosis (TB) and its management. Descriptive statistics were used to summarize participant characteristics and KAP levels, while multivariable linear regression analyses were performed to identify factors associated with KAP. A total of 72 participants were included. Overall, CPs had above-average knowledge and attitude (0.71 ± 0.23 and 0.77 ± 0.09 , respectively), and average practice (0.41 ± 0.07) based on the scoring system and cut-off points. Knowledge gaps were identified related to TB transmission through sharing eating utensils, the Mantoux TST test for active pulmonary TB, TB treatment, drug side effects on liver function, and drug interactions. Good attitudes were found for most aspects except for financial benefit. Below-average practices were found in TB medication swallowing and treatment monitoring. Education level and type of pharmacy were associated with knowledge, working at another place besides pharmacy was associated with attitude, while gender, type of pharmacy, and last time attending TB training were associated with practice. There was a weak but statistically significant positive correlation between knowledge and attitude scores ($r_s=0.294$, $p=0.012$) and between attitude and practice ($r_s=0.337$, $p=0.004$). Community pharmacists in Bandung demonstrated generally adequate knowledge and positive attitudes regarding TB, but important gaps in knowledge and practice remain. Targeted and continuous TB education and training may improve pharmacists' capacity and enhance their contribution to TB control efforts.

Keywords: tuberculosis, community pharmacist, knowledge, attitude, practice

OP-081

Exposure to 1,2-Dichloropropane Induces DNA Damage and Anti-Apoptotic Effect *in Vitro* and Alteration of Related Pathways *in Vivo*

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It had been reported in 2012 that several employees at an offset proof printing industry in Japan had developed bile duct cancer after chronic exposure to chlorinated organic solvents at high concentrations. The survey of the working environment indicated that 1,2-dichloropropane (DCP) was the common substance to which patients with occupational cholangiocarcinoma had been exposed. The mechanism of 1,2-DCP-induced cholangiocarcinoma is still not fully understood. The aim of this study was to determine the potential of 1,2-DCP to induce DNA damage, a hallmark of carcinogenicity, and suppress apoptosis in human cholangiocytes, and identify biological processes and target genes potentially driving its carcinogenic effects. MMNK-1 cholangiocytes, alone or co-cultured with THP-1 macrophages, were exposed to 0, 0.1, and 0.4 mM DCP for 24 hrs. Apoptosis was evaluated by TUNEL staining. A Western blot was performed to authenticate DNA damage and apoptosis. Mice were also exposed to 250 mg/kg 1,2-DCP (oral, 5 days/week for 5 weeks) for SNT analysis. The percentage of cholangiocytes exhibiting dense (≥ 55) γ -H2AX immunostaining was significantly greater in co-cultures than in monocultures. However, only 1,2-DCP decreased the number of γ -H2AX pan-nuclear-positive and TUNEL-positive cholangiocytes in co-cultures with macrophages, but not in monocultured cholangiocytes. Co-treatment with a pan-caspase inhibitor decreased the number of γ -H2AX pan-nuclear-positive cholangiocytes in each group by about 50%, suggesting that apoptotic signaling is involved, at least in part, in the induction of γ -H2AX pan-nuclear-positive cholangiocytes. Gene ontology analysis in apoptosis regulation, inflammation, xenobiotic metabolism, oxidative stress, and DNA damage response. It can be included that 1,2-DCP induced DNA damage, showed proliferative and anti-apoptotic effects, which are also recognised as key characteristics of carcinogens, and it alters pathways related to apoptosis, cell death, inflammation and DNA damage that may lead to chemical-induced liver injury and carcinogenesis.

Keywords: 1,2-dichloropropane, cholangiocarcinoma, DNA damage, carcinogenesis, anti-apoptotic effect

OP-082

The Role of the Transcription Factor Nrf2 in Lung Inflammation Induced by Exposure to Amorphous Silica Nanoparticles in Mice

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Amorphous silica nanoparticles (SiO₂NPs) are widely used in industrial and biomedical applications, raising concerns about their potential health risks. Previous studies have shown that amorphous SiO₂NPs exposure induces pulmonary inflammation in vivo and macrophage cytotoxicity in vitro. Oxidative stress is considered one of the major mechanisms underlying nanoparticle-induced toxicity, and nuclear factor erythroid 2-related factor 2 (Nrf2) is a key regulator of antioxidant defense and inflammatory responses. This study aimed to investigate the role of Nrf2 in pulmonary inflammation induced by amorphous SiO₂NPs exposure. Male C57BL/6JJcl wild-type (WT) and Nrf2-null mice were administered 30-nm amorphous SiO₂NPs (0, 3, 10, or 30 µg/mouse) by pharyngeal aspiration and evaluated at 24 h and 14 d after exposure. Inflammatory cell counts in bronchoalveolar lavage fluid (BALF), histopathological evaluation of H&E-stained lung tissue based on inflammation scores, and pulmonary mRNA expression analysis by RT-qPCR were performed. SiO₂NPs exposure dose-dependently increased the numbers of total cells and macrophages in BALF, with significant increases observed at 30 µg in Nrf2-null mice but not in WT mice at 14 d after exposure. On the other hand, there were no genotype-dependent differences in SiO₂NPs-induced changes in BALF inflammatory cell counts at 24 h after exposure. Basal levels of peribronchial and perivascular inflammation scores were higher in Nrf2-null mice than in WT mice at both 24 h and 14 d after exposure. Basal expression levels of *tnfa* were higher in Nrf2-null mice than in WT mice at 14 d after exposure. Basal expression levels of *Sod1*, *Cat* and *Nqo1* were lower in Nrf2-null mice than in WT mice at both 24 h and 14 d after exposure. These findings suggest that Nrf2 suppresses SiO₂NPs-induced increases in BALF macrophages at 14 d probably through upregulation of antioxidant gene expression and suppression of pro-inflammatory cytokine gene expression.

Keywords: nanoparticles, pulmonary toxicity, inflammation, Nrf2

OP-083

Bridging the Adherence Gap in Tuberculosis Care: The Emerging Role of Pharmacist-Led Interventions

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Tuberculosis (TB) remains a leading cause of infectious mortality worldwide, with poor treatment adherence contributing to treatment failure, relapse, drug resistance, and ongoing transmission. Although pharmacists have demonstrated benefits in improving medication adherence, their role in TB care remains underutilised and has not been comprehensively synthesised. To map and synthesize global evidence on pharmacist-led interventions that improve medication adherence and treatment outcomes among patients with TB. A scoping review was conducted according to the PRISMA-ScR guideline. Literature published between 2010 and 2025 was searched in PubMed, Scopus, Web of Science, ScienceDirect, and ProQuest using combinations of keywords related to tuberculosis, pharmacists, pharmaceutical care, medication adherence, and pharmacist-led interventions. Original studies involving pharmacist participation in TB management were included and analyzed descriptively. Of 1,247 records identified, 935 remained after duplicate removal, 76 full-text articles were assessed, and 41 studies met the inclusion criteria. The studies represented 18 countries, predominantly in Asia (43.9%), and were mainly conducted in hospitals (43.9%) and primary healthcare centers (31.7%). Common intervention components included patient education and counseling (85.4%), adherence monitoring (78.0%), adverse drug reaction management (63.4%), reminder and follow-up systems (51.2%), family engagement (43.9%), and interprofessional collaboration (58.5%). Medication adherence improved in 34 of 41 studies (82.9%), alongside improvements in patient knowledge, medication safety, and treatment completion. However, intervention heterogeneity and limited evidence from high TB-burden countries revealed important implementation gaps. Pharmacist-led interventions demonstrate considerable potential to improve medication adherence and treatment-related outcomes in TB care. However, the lack of standardized, context-specific implementation models highlights the need for stronger integration of pharmacists into multidisciplinary TB programs to support scalable, evidence-based interventions and strengthen global TB control.

Keywords: tuberculosis, pharmacist-led intervention, medication adherence, pharmaceutical care, patient-centered care

OP-097

Genetic Modification: A Legal and Ethical Perspective

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Genetic modification is a technology with immense promise. Countries are racing to acquire the latest secrets of this technology, as it has the potential to treat diseases, enhance and increase agricultural and livestock production. Nevertheless, these technologies have ignited controversy, raising legal and ethical questions centred on the sanctity of nature, the dignity of humankind, and the possibility of biological variation. One of the primary objectives of this study is to conduct a genuine and critical evaluation of the ethical and legal aspects related to genetic modification technologies. This evaluation will objectively weigh the factors driving the adoption of new technologies against the safeguards that countries and international organizations seek to impose on technology providers, ensuring their compliance with agreed-upon international standards. This study employs a normative legal research methodology. A thorough review was carried out of existing legal doctrines. This involved a comprehensive criticism of bioethical literature, international human rights treaties, and comparative national regulations in the field of biotechnology. This analysis assesses established ethical frameworks, such as utilitarianism and deontology, in light of current domestic and international legal guidelines. The results reflect a profound difference in the approach to international governance. While somatic cell therapy enjoys broad legal consensus under medical and pharmaceutical legislation, microbial genome editing and human enhancement remain highly restricted due to the risks of unintended mutations and socioeconomic inequalities. Genetic modification offers significant potential for advancing both human well-being and environmental sustainability. However, its future applications necessitate regulatory frameworks that are adaptable and globally compatible. This approach ensures that scientific progress aligns with fundamental human rights and ethical principles.

Keywords: genetic modification, bioethics, human rights, international governance, regulatory frameworks

Poster Presentation **Abstracts**

Pharmacology & Precision Medicine (PPM)

PP-011

Knowledge, Attitude, and Practice of Non-Small- Cell Lung Cancer (NSCLC): Targeted Therapies Among Pharmacists

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Targeted therapy for Non-Small Cell Lung Cancer (NSCLC) represents a significant advancement in cancer treatment. As pharmacists play a crucial role in treatment decision-making, understanding their knowledge, attitude, and practice (KAP) is vital for optimal patient care. This study aimed to assess the knowledge, attitude, and practice of Malaysian pharmacists regarding targeted therapy for NSCLC and to examine the association between participant characteristics and their KAP levels. A cross-sectional study using a self-administered questionnaire was conducted among 166 pharmacists across various healthcare settings in Malaysia. The questionnaire comprised demographic data and items assessing knowledge (factual awareness), attitude (perception and beliefs), and practice (engagement or recommendation of therapy). Descriptive statistics, Mann-Whitney U, Kruskal-Wallis tests, Chi-square tests, and logistic regression were used for analysis. As the data were not normally distributed, non-parametric tests were employed. The majority of respondents were female (53%), Malay (50.3%), and bachelor's degree holders (64.1%). The median knowledge score was 15 (IQR: 14–16), attitude 9 (IQR: 9–9), and practice 9 (IQR: 9–9). Knowledge was significantly associated with the pharmacy field ($p=0.001$). Attitude scores showed significant differences with the pharmacy field and family history of cancer. Practice scores were significantly higher among Chinese ethnicity, postgraduates, hospital pharmacists, and those with a family history of cancer. Logistic regression identified pharmacy field, education level, and race as significant predictors of high KAP scores. Correlation between KAP domains was weak and not statistically significant. While pharmacists demonstrated moderately high knowledge and positive attitudes, gaps remain in practice related to NSCLC-targeted therapies. Demographic factors such as race, field, and education significantly influenced KAP levels. Greater efforts are needed in education, training, and policy to enhance pharmacists' roles in precision oncology.

Keywords: non-small cell lung cancer (NSCLC), targeted therapy, pharmacist, knowledge attitude practice (KAP), oncology pharmacy, genetic testing

PP-013

**Antioxidant, Anti-Inflammatory, and Cytoprotective Activities of *Baeckea frutescens* Branch
Ethanollic Extract**

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Inflammation and oxidative stress contribute significantly to the progression of chronic diseases such as cancer, atherosclerosis, and rheumatoid arthritis. Natural products rich in bioactive compounds have gained attention as potential therapeutic agents with fewer adverse effects. This study investigated the phytochemical composition, antioxidant, anti-inflammatory, and cytoprotective activities of an ethanollic extract of *Baeckea frutescens* branches (BFBE) using RAW 264.7 macrophages and zebrafish embryo models. Phytochemical screening and LCMS/MS profiling identified the presence of hydrolysable tannins, flavonoids, saponins, quinic acid, castalagin, tetrahydro formononetin sulphate, and apigenin-7- β -O-glucuronide. Antioxidant activity was evaluated using DPPH and FRAP assays, where BFBE demonstrated concentration-dependent free radical scavenging activity with an IC₅₀ value of 86.16 μ g/mL and notable ferric reducing capacity. In RAW 264.7 macrophages, BFBE showed low cytotoxicity and significantly reduced hydrogen peroxide-induced oxidative damage by preserving cell viability, morphology, intracellular reactive oxygen species (ROS) levels, and superoxide dismutase (SOD) activity. Anti-inflammatory activity was assessed through nitric oxide (NO) inhibition, ELISA, and Western blot analyses. BFBE significantly suppressed NO production with an IC₅₀ value of 7.34 $\pm$ 1.03 μ g/mL and downregulated inflammatory mediators including COX-2, iNOS, IL-6, IL-1 β , and TNF- α . In zebrafish embryos, BFBE exhibited low developmental toxicity and effectively reduced ROS levels while improving embryo survival and morphology under oxidative stress conditions. Overall, BFBE demonstrated potent antioxidant, anti-inflammatory, and cytoprotective activities, supporting its potential as a natural therapeutic agent for oxidative stress- and inflammation-related diseases. Further studies are recommended to elucidate its molecular mechanisms and evaluate its therapeutic efficacy in vivo.

Keywords: *Baeckea frutescens*, antioxidant, anti-inflammatory, oxidative stress, cytoprotective activity

PP-026

TRPML1 Inhibition Reduces Viability and Migration in HCT116 Colorectal Cancer Cells

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Colorectal cancer remains one of the leading causes of cancer-related mortality, urgently necessitating the identification of novel therapeutic targets. Transient receptor potential mucolipin 1 (TRPML1) is a lysosomal Ca^{2+} channel integral to autophagy and cellular metabolism, yet its specific role in this malignancy remains poorly defined. This study aimed to evaluate the antineoplastic efficacy of TRPML1 inhibition on the viability and migration of HCT116 colorectal cancer cells using the selective inhibitor ML-SI1. HCT116 cells were treated with varying concentrations of ML-SI1 under normal and serum-starved conditions, with cell viability assessed via MTS assays at 24, 48, and 72 hours to estimate the half-maximal inhibitory concentration (IC_{50}). Additionally, cell migration was evaluated over 48 hours using a scratch assay. Results demonstrated that ML-SI1 reduced cell viability in a concentration- and time-dependent manner, yielding an estimated IC_{50} of $72.1 \mu\text{M}$ ($R^2 = 0.96$, $p = 0.0091$). While cell viability remained robust ($>75\%$) at 24 hours, a more pronounced reduction occurred at 48 hours, particularly at higher concentrations ($70 \mu\text{M}$) under normal serum conditions. Prolonged 72-hour exposure revealed a biphasic effect, where lower ML-SI1 doses ($25\text{--}45 \mu\text{M}$) under starvation induced an increase in metabolic activity ($>120\%$, $p = 0.0037$ and $p = 0.0096$, respectively), whereas high-dose treatment ($70 \mu\text{M}$) resulted in a marked loss of viability. Furthermore, ML-SI1 substantially attenuated cellular migration. Untreated cells achieved approximately 60% wound closure at 48 hours, whereas IC_{15} , IC_{25} , and IC_{50} treatments restricted closure to approximately 28%, 27%, and 12%, respectively ($p < 0.0001$). Ultimately, these findings demonstrate that TRPML1 inhibition significantly disrupts HCT116 cell survival and migration. This highlights a complex cellular response likely driven by the disruption of crucial lysosomal pathways under metabolic stress, suggesting that TRPML1 warrants further investigation as a pharmacological target for mitigating colorectal cancer progression.

Keywords: TRPML1, ML-SI1, colorectal cancer, HCT116, cell viability

PP-063

Genotoxic Effects of 4-[(6-Iodo-2-phenylquinazolin-4-yl)oxy]benzoic Acid in MCF-7 and MDA-MB-231 Breast Cancer Cells Under Normoxic and Hypoxic Conditions

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Tumour hypoxia drives breast cancer progression, metastasis, therapeutic resistance, and poor clinical outcomes, partly through the overexpression of carbonic anhydrase IX (CA IX), a key regulator of pH homeostasis. As a result, CA IX has emerged as a promising therapeutic target in hypoxia-adapted tumours. This study evaluated the genotoxic effects of 4-[(6-iodo-2-phenylquinazolin-4-yl)oxy]benzoic acid (CAA6), a newly synthesised quinazoline-based CA IX inhibitor, in MCF-7 and MDA-MB-231 breast cancer cells under normoxic and cobalt chloride-induced hypoxic conditions using the alkaline comet assay. Cells were treated with increasing concentrations of CAA6 (25–150 μ M), and DNA damage was assessed using tail length, tail moment, olive tail moment, and percentage DNA in tail. CAA6 induced a concentration-dependent increase in DNA damage in both cell lines under normoxic and hypoxic conditions. Under hypoxia, tail length increased from 9.80 ± 0.60 μ m in untreated cells to 32.85 ± 1.51 μ m at 150 μ M, while tail moment rose from 2.90 ± 0.20 to 17.14 ± 0.52 ; percentage DNA in tail also increased from $6.37 \pm 0.67\%$ to $32.69 \pm 1.64\%$. Similar trends were observed under normoxic conditions. Doxorubicin, used as a positive control, produced the greatest DNA damage, confirming assay validity. Overall, these findings show that CAA6 causes significant DNA damage in breast cancer cells and support the further development of quinazoline-based CA IX inhibitors for hypoxia-associated breast cancer.

Keywords: carbonic anhydrase IX, quinazoline, comet assay, hypoxia, breast cancer

PP-073

Asiatic Acid Suppresses CCRF-CEM Cell Proliferation Through TNF α -Associated Signalling: *In Vitro* and *In Silico* Evidence

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Asiatic acid (AA), a bioactive pentacyclic triterpene derived from *Centella asiatica*, has demonstrated anti-cancer activity in multiple malignancies. This study evaluated the anti-proliferative effects of AA against the human T-cell acute lymphoblastic leukaemia (T-ALL) cell line CCRF-CEM and investigated its potential interaction with tumour necrosis factor- α (TNF α). CCRF-CEM cells were treated with AA (0–40 μ M) for 48 and 72 hours, and cell viability was assessed using the MTT assay. TNF α protein expression following AA treatment was determined by Western blot analysis, with GAPDH used as the loading control. To further explore the molecular basis of AA activity, *in silico* molecular docking was performed to evaluate the binding affinity and interaction profile of AA with TNF α . AA significantly inhibited CCRF-CEM cell proliferation in a concentration- and time-dependent manner, with a marked reduction in viability observed between 10 and 20 μ M. The IC₅₀ value at 48 hours was 13 μ M. Western blot analysis showed a significant increase in TNF α protein expression following AA treatment ($p < 0.001$), whereas the clinical comparator, cytarabine, reduced TNF α expression. Molecular docking predicted a strong interaction between AA and TNF α , with a binding energy of -9.4 kcal/mol. The AA–TNF α complex was stabilized by a conventional hydrogen bond involving the LEU157 residue, thereby supporting a favourable binding conformation. AA exerts potent anti-proliferative effects against CCRF-CEM cells and is associated with increased TNF α expression, suggesting the involvement of TNF α -related signalling in its anti-leukaemic activity. The strong binding affinity observed in molecular docking further supports TNF α as a potential molecular target of AA. These findings highlight AA as a promising lead compound for the development of targeted therapies for T-cell acute lymphoblastic leukaemia and other haematological malignancies.

Keywords: Asiatic acid, T-cell acute lymphoblastic leukaemia, CCRF-CEM, TNF α , molecular docking, anti-proliferative activity

PP-074

E498A and R499G Variants of PCSK9 Altered PCSK9-LDLR Binding: Insight from Molecular Dynamics Simulations.

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Cardiovascular disease (CVD) remains a leading cause of death globally, with familial hypercholesterolemia (FH) being a significant genetic contributor due to elevated low-density lipoprotein cholesterol (LDL-C) levels. FH is primarily caused by mutations in genes involved in cholesterol metabolism, especially the LDLR gene, which encodes the LDL receptor (LDLR)—a crucial mediator of plasma LDL-C clearance. The protein PCSK9 regulates this pathway by binding to LDLR and promoting its lysosomal degradation, thereby reducing receptor availability. Pathogenic mutations in PCSK9 can enhance this degradation, worsening hypercholesterolemia and CVD risk. However, the molecular mechanisms underlying LDLR-PCSK9 interactions remain incompletely understood because a full-length LDLR structure is not yet available. This study aimed to investigate the structural dynamics of LDLR-PCSK9 interactions and evaluate the effects of two novel PCSK9 variants, E498A and R499G, identified in clinically diagnosed Malaysian FH patients. Molecular dynamics (MD) simulations were performed on LDLR complexes with wild-type PCSK9 and the E498A and R499G variants over a 500 ns simulation period. The simulations revealed that the R499G complex induced the most substantial structural changes in LDLR. In both the wild-type and E498A complexes, the PCSK9 prodomain formed consistent interactions with LDLR's β -propeller region, whereas both variants disrupted local residue interactions. Aligned with the observation, the MM/GBSA showed E498A had the strongest binding affinity (-63.81 kcal/mol), followed by wild-type (-33.07 kcal/mol) and R499G (-24.21 kcal/mol). These findings highlight the importance of structural flexibility in modulating LDLR-PCSK9 interactions and suggest that PCSK9 variants may differentially influence FH pathogenicity. The study provides valuable structural insights into cholesterol regulation and establishes a foundation for future investigations using complete LDLR models.

Keywords: low-density lipoprotein receptor, proprotein convertase subtilisin/kexin type-9, low-density

PP-087

Fluorescence-Based Detection of the HbE Mutation Using PAMless CRISPR/Cas12a

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β -Thalassemia is one of the most prevalent inherited blood disorders in Malaysia, with a carrier rate of approximately 4.5–5.0%. Current prenatal diagnostic methods such as chorionic villus sampling and amniocentesis are accurate but invasive and carry certain risk. Non-invasive prenatal testing (NIPT) using cell-free foetal DNA (cffDNA) is safer, but it is still expensive and requires specialized equipment. This study aimed to develop and evaluate a fluorescence-based CRISPR/Cas12a diagnostic platform for detecting the β -thalassemia HbE variant (rs33950507), serving as a proof-of-concept for future point-of care biosensor development. A synthetic positive control containing the HbE mutation was amplified using an optimized PCR protocol. Target detection was performed using the EnGen Lba Cas12a system with a PAM activated mutation-specific guide RNA. In this study, a FAM/BHQ1-labeled aptamer reporter was used to validate Cas12a-mediated collateral cleavage through fluorescence detection. Fluorescence images were captured using an inverted fluorescence microscope, and fluorescence intensity was quantified using a Tecan multimode microplate reader (excitation: 485 nm; emission: 520 nm) across different DNA concentrations. The results showed that the CRISPR/Cas12a assay successfully differentiated positive and negative samples. The positive control exhibited strong green fluorescence due to Cas12a-mediated collateral cleavage of the FAM/BHQ1 reporter, whereas the negative control showed minimal fluorescence, indicating the absence of target-induced Cas12a activation. Quantitative fluorescence analysis across serial DNA concentrations further confirmed target recognition, with the positive control consistently producing higher fluorescence intensity than the negative control. These findings demonstrate the feasibility of CRISPR/Cas12a for sensitive detection of the HbE mutation and provide proof-of-concept for developing a portable aptamer-based colorimetric biosensor. Following further optimization, this platform has the potential to enable rapid, cost-effective, and non-invasive prenatal screening for β -thalassemia.

Keywords: β -thalassemia, CRISPR/Cas12a, fluorescence detection, cell-free fetal DNA, prenatal diagnosis

PP-091

Metabolic Dysfunction-Associated Fatty Liver Disease Drives Morphological Changes in Liver Sinusoidal Endothelial Cells

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Metabolic dysfunction-associated fatty liver disease (MAFLD) is one of the leading causes of chronic liver disease worldwide, characterised by accumulation of fat in the liver accompanied by metabolic abnormalities including obesity, type 2 diabetes mellitus and other metabolic dysregulations. Although the pathogenesis of MAFLD has revealed several mechanisms that occur mostly at the hepatocytes, the contribution of liver sinusoidal endothelial cells (LSECs) as key regulators in the disease pathogenesis has not yet been fully studied. This study aimed to investigate the role of LSECs in MAFLD progression in a diet-induced MAFLD rat model. Male Sprague Dawley rats were randomly assigned to a control group fed with standard maintenance diet and a MAFLD group fed with fatty liver diet (Altromin C 1063) for 12 weeks. Anthropometric variables including body weight, body length and abdominal circumference were recorded weekly. At the end of study, all animals were sacrificed and blood was collected for lipid profile. Liver tissues were perfused-fixed for scanning electron microscopy (SEM) to evaluate ultrastructural morphology of LSECs using ImageJ software. All data were analysed using SigmaPlot v16. Result showed that the rats fed with fatty liver diet developed a metabolically dysfunctional phenotype evidenced by significant increases in body weight, abdominal circumference and BMI, serum total cholesterol ($p < 0.05$) and triglyceride ($p = 0.05$) compared to controls. SEM analysis revealed a marked LSEC defenestration, characterised by a significant reduction in fenestration frequency ($p < 0.05$) and liver porosity ($p < 0.05$). However, there are no changes in fenestration diameter for both groups. These findings confirm the successful establishment of diet-induced MAFLD. Furthermore, the observed loss of LSEC fenestrations supports the hypothesis that endothelial dysfunction is an early pathological event in MAFLD and highlights LSECs as a potential therapeutic target for preventing disease progression.

Keywords: metabolic dysfunction-associated fatty liver disease (MAFLD), liver sinusoidal endothelial cells (LSECs), fenestration; scanning electron microscopy, fatty liver diet

PP-092

Time Course Analysis of Behavioural and Molecular Changes in Intrahippocampal Streptozotocin-Induced Female Sporadic Alzheimer's Disease Rat Model

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Sporadic Alzheimer's disease (sAD) accounts for approximately 90% of Alzheimer's disease (AD) cases. Females represent nearly two-thirds of these AD cases, highlighting sex-related vulnerability; however, preclinical AD models, especially for sAD, remain predominantly developed in male animals, limiting sex-specific translational relevance. The commonly used intracerebroventricular streptozotocin (STZ) model exhibits variable reproducibility in females, resulting in insufficient characterization of AD-like pathology. The intrahippocampal STZ (IH-STZ) model provides improved specificity by directly targeting the hippocampus, a key region involved in AD-related cognitive dysfunction, but its applicability in female models remains unclear. Therefore, this study aimed to establish a female sAD model using bilateral IH-STZ injection by characterizing the behavioural and molecular changes at early latency periods (7 days (L7) and 14 days (L14)). This will determine the optimal latency period or induction time for AD-like pathological development. Behavioural assessments evaluating anxiety-like behaviour, locomotion, recognition memory, and spatial learning and memory were paired with immunoblot analysis of amyloid precursor protein (APP) and tau oligomers in the hippocampus and prefrontal cortex. Both STZ L7 and STZ L14 showed increased anxiety-like behaviour ($p < 0.05$), impaired recognition memory ($p < 0.05$), and delayed spatial learning with reduced memory retention ($p < 0.05$), followed by reduced cortical APP expression ($p < 0.05$) compared to control. However, only STZ L14 exhibited increased anxiety-like behaviour with hyperlocomotion ($p < 0.05$), which corresponded to elevated hippocampal APP expression and tau oligomerization ($p < 0.05$). In conclusion, bilateral IH-STZ administration induces sAD-like behavioural impairments and region-specific APP and tau pathology in female rats, indicating the 14-day post-injection interval is the optimal latency period for establishing a female sAD model, thus providing a relevant preclinical platform for investigating sex-specific mechanisms and evaluating potential therapeutic interventions for AD.

Keywords: sporadic Alzheimer's disease (AD), intrahippocampal, streptozotocin (STZ), female rat model, cognitive impairment.

PP-099

Acute Oral Toxicity, Biochemical, and Histopathological Evaluation of *Pterygoplichthys pardalis* Flesh as a Sustainable Pet-Diet Ingredient

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The invasive suckermouth catfish (*Pterygoplichthys pardalis*) poses a significant ecological threat to Malaysian freshwater ecosystems due to its rapid reproduction and lack of natural predators. Utilizing its nutrient-rich biomass as an alternative protein source for pet food offers a sustainable solution, but comprehensive safety data are required. This study evaluated the acute oral toxicity profile of *P. pardalis* flesh homogenate in female Sprague-Dawley rats, focusing on behavioral, biochemical, hematological, and histopathological outcomes. Following OECD Guideline 423, rats (n=3 per group) were administered a single oral dose of 2000 mg/kg of the homogenate, while control groups received normal saline. Animals were systematically monitored over a dual-phase observation window of 24 hours and 14 days. No mortality, abnormal clinical signs, neurological symptoms, or significant behavioral changes were observed in any treatment group. While mild, non-toxic weight changes (<10%) occurred, there were no statistically significant differences in body weight or relative liver and kidney weights between the treated and control groups ($p>0.05$). Blood biochemistry assessments (ALT, AST, BUN, creatinine) and full hematological profiling showed all parameters remained within normal physiological ranges with no significant treatment related variations. Furthermore, histopathological examination verified by an expert pathologist revealed preserved hepatic and renal architectures with normal hepatocytes, intact glomeruli, and renal tubules, showing no signs of necrosis, degeneration, or inflammation across both observation phases. Consequently, the approximate lethal dose (ALD) and no-observed-adverse-effect level (NOAEL) were established to be greater than 2000 mg/kg. These findings demonstrate that *P. pardalis* flesh homogenate is non-toxic under acute exposure, confirming its safety and high potential as a sustainable, high-quality protein replacement for novel pet-diet applications.

Keywords: *Pterygoplichthys pardalis*, acute oral toxicity, histopathology, blood biochemistry, pet food ingredient

PP-101

Potentially Inappropriate Medications Across Heart Failure Phenotypes in a Real-World Outpatient Setting in Indonesia

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Heart failure (HF) encompasses distinct phenotypes, HF with reduced ejection fraction (HFrEF) and non-HFrEF each warranting phenotype-tailored pharmacotherapy. Yet potentially inappropriate medications (PIMs), a critical safety concern in chronic HF, remain poorly characterized across these phenotypes in real-world settings, limiting the precision of medication safety strategies for this heterogeneous population. To evaluate the prevalence, types, and determinants of PIMs across CHF phenotypes, informing phenotype-specific approaches to prescribing safety. A prospective cross-sectional study was conducted at a hospital-based outpatient service in Indonesia. Adult CHF patients receiving pharmacotherapy for ≥ 1 month were included. PIMs were assessed using the American Heart Association (AHA) drug list (all patients), Beers Criteria and Medications Associated with Geriatric Syndromes (MAGS) (aged ≥ 65 years), and Potentially Inappropriate Prescribing in HFrEF (PIP-HFrEF) criteria (HFrEF only). Medications identified by multiple instruments were counted once. Associations were examined using Chi-Square and Mann-Whitney U tests. Among 104 patients enrolled (non-HFrEF: $n=54$; HFrEF: $n=50$), mean age was 61 ± 11 and 60 ± 9 years, respectively. The non-HFrEF group had significantly higher proportions of older age and obesity than HFrEF ($p=0.019$ for both). PIMs were identified in 48 patients (46.2%), predominantly among those with older age (>65 years; $p=0.001$) and higher comorbidity burden (Charlson Comorbidity Index ≥ 5 ; $p<0.001$). PIMs were detected in 25, 26, 28, and 13 patients by the AHA list, Beers Criteria, MAGS, and PIP-HFrEF criteria, respectively. Metformin, warfarin, and isosorbide dinitrate were the most frequently implicated agents. PIMs affected nearly half of CHF outpatients regardless of phenotype, indicating that PIM risk is largely phenotype-independent and underscoring the need for standardized, phenotype-informed medication review across the HF spectrum. Further studies evaluating the clinical impact of PIMs on adverse outcomes are warranted.

Keywords: cross-sectional study, heart failure, Indonesia, polypharmacy, potentially inappropriate medications

PP-102

Fruit Waste Bioconversion for Bioactive Fermentation: Linking Phytochemical Composition with Biological Activities

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Globally, approximately 1.3 billion tonnes of food waste are generated annually, with fruit and vegetable waste accounting for nearly 45% of this total. In Malaysia, more than 39,000 tonnes of municipal solid waste are produced daily, of which over 30% comprises food waste. Fruit waste, particularly discarded fruit peels, represents an abundant source of bioactive phytochemicals that remain largely underutilised despite their potential pharmaceutical and nutraceutical value. This study investigates the phytochemical composition, antioxidant activity, and cytotoxicity of five fruit waste fermentation formulations produced from banana (*Musa acuminata* × *balbisiana*), orange (*Citrus sinensis*), rockmelon (*Cucumis melo*), and their combinations (rockmelon + orange; rockmelon + orange + banana) using fruit peel waste as the fermentation substrate. The formulations were prepared through a three-month molasses-based fermentation process. High-Performance Liquid Chromatography (HPLC) analysis identified major phytochemicals, including ascorbic acid, gallic acid, and tannic acid, which may contribute to the observed biological activities. Antioxidant activity was evaluated using the DPPH radical scavenging assay at concentrations ranging from 0.625 to 10 mg/mL. All formulations exhibited concentration-dependent antioxidant activity, achieving radical scavenging activities of approximately 86–91% at 10 mg/mL, indicating strong free radical neutralisation capacity. Cytotoxicity was assessed using the MTT assay in HaCaT human keratinocyte cells at concentrations ranging from 0.625 to 10 mg/mL. The banana fruit waste fermentation demonstrated moderate cytotoxicity ($IC_{50} = 7.85 \pm 0.18$ mg/mL), whereas no IC_{50} values were observed for the individual orange and rockmelon fruit waste fermentations. The combined formulation of rockmelon, orange, and banana fruit waste exhibited enhanced cytotoxic effects ($IC_{50} = 8.28 \pm 0.07$ mg/mL), suggesting possible synergistic interactions among the fermented bioactive constituents. Overall, these findings suggest that fruit waste fermentation may be explored as a source of bioactive compounds for future biomedical applications, including topical formulations.

Keywords: fruit waste bioconversion, bioactive fermentation, phytochemical profiling, antioxidant activity, cytotoxicity

PP-107

An Integrated Transcriptomic and Machine Learning Approach for Identifying Doxorubicin-Responsive Exosomal miRNAs

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Doxorubicin is an effective chemotherapeutic for lung cancer, but it often leads to cardiotoxicity and skeletal muscle wasting. Tumour-derived exosomes play a key role in systemic communication by transferring microRNAs (miRNAs) to distant organs. However, the changes in exosomal miRNAs induced by doxorubicin and their roles in cardiac and skeletal muscle injury remain poorly understood. To identify doxorubicin-responsive exosomal miRNAs from A549 lung cancer cells using small RNA sequencing, bioinformatics, and machine learning, and to prioritize candidate miRNAs and pathways linked to cardiac and skeletal muscle injury. Exosomes were isolated from untreated and doxorubicin-treated A549 cells and characterized by scanning electron microscope and Western blotting. Exosomal miRNAs were extracted and sequenced using the Illumina small RNA platform. Differential expression was analysed using the OASIS-DESeq2 pipeline. A reverse-genomics approach integrated target prediction databases (TargetScan, miRDB, miRTarBase) with functional enrichment analyses (KEGG, Gene Ontology, and Reactome) to identify cardiac and skeletal muscle targets. Machine learning feature prioritization was performed using LASSO and Random Forest. Selected miRNAs were validated by qPCR. All samples yielded high-quality sequencing libraries. Significant differential expression was observed following 10 μ M doxorubicin treatment. miR-133 was significantly upregulated and emerged as the primary muscle-associated candidate. Reverse-genomics identified validated targets in PI3K-Akt, FoxO, TGF- β , cardiac muscle contraction, cardiomyopathy, apoptosis, mitochondrial function, extracellular matrix remodelling, and skeletal muscle differentiation. Machine learning consistently ranked miR-133 as the top features, with miR-21-5p, miR-22-3p, and miR-486-5p as abundant secondary candidates. An integrated approach utilizing small RNA sequencing, bioinformatics, reverse genomics, and machine learning identified p-hsa-miR-133 as a key exosomal candidate associated with cardiac and skeletal muscle injury. These findings provide a foundation for further functional validation and support the development of exosomal miRNAs as biomarkers and therapeutic targets for chemotherapy-induced toxicity.

Keywords: exosomes, microRNA, doxorubicin, lung cancer, cardioncology

PP-109

Extraction-Dependent Activity of *Caesalpinia bonducella* Seed Extracts: Correlation Between Phytochemical Composition with Antioxidant and Anti-Inflammatory Activities

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Caesalpinia bonducella is a medicinal plant traditionally used for inflammatory conditions, with several studies reporting promising antioxidant and anti-inflammatory properties. However, variations in extraction methods and phytochemical composition may affect its biological activity, warranting further investigation. This study investigated the phytochemical composition, antioxidant (DPPH and FRAP assays) and anti-inflammatory (MTT and Griess assays) activities of aqueous and hexane seed extracts of *C. bonducella*. Qualitative screening showed the presence of terpenoids, alkaloids, phenols, and phytosterols, whereas quantitative screening revealed that A5E had the highest phytochemical content. However, all extracts exhibited weak antioxidant activity as indicated by their high EC₅₀ values with low bioactivity. In contrast, notable anti-inflammatory activity was observed in LPS-stimulated RAW 264.7 macrophages, with HE showing the strongest inhibition of nitric oxide production while maintaining high cell viability. Correlation analysis revealed negative correlations among phytochemical content, antioxidant EC₅₀ values, and nitric oxide production, suggesting an interconnectedness between antioxidant and anti-inflammatory effects. The presented results suggest that lipophilic compounds in *C. bonducella* contribute to limited antioxidant but significant anti-inflammatory capacities. Further studies should focus on identifying active compounds and validating their mechanisms in vivo.

Keywords: *Caesalpinia bonducella*, phytochemical composition, antioxidant, anti-inflammatory mechanism, lipophilic compounds

PP-112

Water Kefir, Rice Water, and Rice Water Kefir as Potential Anti-Aging Agents: *In Vitro* Superoxide Dismutase (SOD-Like) Activity

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Skin aging, a biological process accelerated by oxidative stress from increased reactive oxygen species (ROS), may be prevented by natural antioxidant defenses like superoxide dismutase (SOD). However, the potential anti-aging properties and SOD-like activity of water kefir, rice water, and rice water kefir remain underexplored, prompting this study to assess the *in vitro* SOD-like activity of these substance as possible anti-aging agents. The method of autooxidation of pyrogallol was experimentally studied in the laboratory. We evaluated the following samples: Water Kefir (WK), Hot Rice Water (HRW), Cold Rice Water (CRW), Hot Rice Water Kefir (HRWK), Cold Rice Water Kefir (CRWK). The activity of SOD-like was measured using a microplate reader at 420 nm, and the results were analysed using the ANOVA and Games-Howell as a follow-up test at 95% confidence level. Results demonstrated that SOD-like activity was exhibited by all samples. The highest activity was obtained in vitamin C as a positive control (2.355 ± 0.100 U/mL) followed by HRWK (2.005 ± 0.065 U/mL), CRWK (1.920 ± 0.033 U/mL), HRW (1.611 ± 0.009 U/mL), CRW (1.162 ± 0.200 U/mL), and WK (0.595 ± 0.151 U/mL). The SOD-like activity of rice water fermented with kefir starter was higher than that of the rice water without fermentation. The hot rice water kefir is the most potential as a natural ingredient in the anti-aging product.

Keywords: anti-aging, rice water, rice water kefir, SOD-like, water kefir

PP-116

Adaptation and Validation of Diabetic Foot Ulcer Scale – Short Form (DFS-SF) in Indonesian

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Diabetic foot ulcers (DFUs) severely impair patients' quality of life (QoL), highlighting the urgent need for a culturally adapted version of the internationally validated Diabetic Foot Ulcer Scale–Short Form (DFS-SF) in Indonesia. This study aimed to translate and culturally adapt the DFS-SF into Indonesian and to evaluate its psychometric properties. Conducted from January 2023 to February 2024, the study followed a prospective two-phase design. In the first phase, the original DFS-SF was translated into Indonesian (DFS-SF-Ina) following the standardized Mapi Research Trust linguistic validation protocol, comprising forward translation, back translation, expert panel review, cognitive debriefing, and finalization. In the second phase, psychometric evaluations were performed at three Rumat Clinic branch locations (Pengumben, Buaran, and Cipondoh). Construct validity was evaluated using Pearson's corrected item–total correlation (threshold $r > 0.278$), while internal consistency reliability was assessed using Cronbach's alpha (Cronbach's $\alpha > 0.70$). DFS-SF-Ina was administered to 50 patients with DFUs, with the EQ-5D serving as the comparator instrument. Participants had a mean age of 57 years, were predominantly male (52%), and most presented with stage 1 ulcers (46%). DFS-SF-Ina demonstrated moderate-to-strong correlations with EQ-5D scores, satisfactory item–total correlations (0.502–0.885), good convergent and discriminant validity (corrected item–scale correlation $r > 0.4$ in most domains), and high internal consistency (Cronbach's $\alpha > 0.7$). Higher ulcer severity was significantly associated with lower QoL scores. In conclusion, DFS-SF-Ina is the first linguistically and culturally validated DFU-specific QoL instrument for the Indonesian population. Its strong psychometric performance supports its use in clinical practice and research, enabling standardized QoL assessment and facilitating cross-cultural comparisons in DFU outcomes.

Keywords: diabetic foot ulcers, quality of life, Indonesian version, validity, reliability

PP-120

Abdominal Obesity and Oxidative Stress-Induced DNA Damage

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Abdominal obesity in young adults is a growing global public health concern due to lifestyle changes and shifting dietary habits during this transitional life phase. It causes physiological imbalance in the regulation and normal functioning of adipose tissue, leading to other medical issues such as hyperglycaemia and dyslipidaemia. These conditions trigger the production of oxidative stress, which is worsened by a decrease in antioxidant defense systems in obese patients. DNA damage is any modification to the structure of DNA and mainly caused by oxidative stress. This study was conducted to evaluate abdominal obesity and the relationship with oxidative DNA damage in the students of Faculty of Pharmacy (n=152) in Universiti Teknologi MARA (UiTM), Malaysia. Abdominal obesity was assessed by measuring waist-to-hip circumference ratio (WHR). The subjects were then divided into two groups based on their WHR value (normal and high WHR value). DNA damage was assessed using Comet Assay. Urinary 8-hydroxy-2 deoxyguanosine (8-OHdG) as a biomarker for DNA oxidative damage was measured using ELISA method. The mean WHR of subjects ranged from 0.68 to 0.98. Fourteen of the subjects (9.21%) had WHR values above the recommended level. No significant difference ($p>0.05$) in the tail length, tail moment, olive moment and the percentage of DNA in tail were observed between groups, although greater value of DNA damage parameters was seen in the group with high level of WHR. There was also no correlation between WHR and DNA damage ($p>0.05$). The mean of 8-OHdG levels was 4.485 ± 1.649 ng/mL and no correlation with WHR ($p>0.05$). In summary, there was no significant difference in the levels of DNA damage between groups with no correlation between DNA damage, 8-OHdG and WHR ($p>0.05$). Further study should be conducted to understand the mechanism/s that contributed to this condition.

Keywords: abdominal obesity, oxidative stress, DNA damage

PP-123

**Identification of FDA-Approved Drug Candidates Targeting the Dengue Virus NS2B-NS3
Protease Through *In Silico* Screening**

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Dengue virus infection remains a major global health burden, with no approved antiviral therapy currently available. The highly conserved NS2B-NS3 protease, which is essential for viral polyprotein processing, represents an attractive target for antiviral drug development. In this study, 2,470 FDA-approved compounds were screened against the DENV-3 NS2B-NS3 protease using molecular docking to identify candidates with drug repurposing potential. The ten highest-ranked compounds were subsequently subjected to independent docking validation, and their binding interactions were analysed. The results show that binding affinities clustered within a narrow range (−7.60 to −8.90 kcal/mol) across both docking approaches, such that binding score alone did not clearly distinguish the top candidates. Interaction analysis, however, revealed clear differences: argatroban (−8.18 kcal/mol) established multiple conventional hydrogen bonds with both His51 and Ser135, while oxatomide (−7.87 kcal/mol) was the only compound predicted to form a catalytically relevant interaction with Asp75, through a conventional hydrogen bond supported by a π -anion interaction. Nilotinib, despite the strongest predicted affinity (−8.90 kcal/mol), formed only a weak halogen interaction with Asp75, which does not constitute a catalytically relevant interaction. These findings demonstrate that binding affinity alone does not necessarily reflect the formation of catalytically relevant interactions at the active site and highlight argatroban and oxatomide as promising repurposing candidates for dengue NS2B-NS3 protease inhibition.

Keywords: molecular docking, dengue antiviral, NS2B-NS3 protease, FDA-approved compounds, drug repurposing

PP-126

Molecular Docking and ADME-Based Evaluation of Kratom Alkaloids Targeting the 5-HT_{2A} Receptor

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The psychoactive plant *Mitragyna speciosa* (kratom) contains a structurally diverse group of alkaloids that are traditionally associated with opioid-mediated effects. However, increasing evidence suggests that kratom exhibits complex polypharmacological activity involving multiple central nervous system (CNS) receptors beyond the classical opioid system. This study investigated the molecular interactions of 33 kratom alkaloids with the 5-hydroxytryptamine 2A serotonin receptor (5-HT_{2A}R), a receptor implicated in mood regulation, cognition, perception, and behavioural modulation. *In silico* pharmacokinetic profiling and drug-likeness evaluations were performed using SwissADME, followed by molecular docking simulations using MolModa to assess binding affinities and characterize ligand interactions within the orthosteric binding pocket of the receptor. ADME analysis demonstrated that 97% of the evaluated alkaloids (32 out of 33) complied with Lipinski's Rule of Five with high gastrointestinal absorption, and 63.6% (21 out of 33) possessed favourable blood-brain barrier permeability, with strictosidine identified as the only compound with limited permeability and multiple violations. Docking results present successful binding of all 33 alkaloids to the orthosteric binding pocket of the 5-HT_{2A}R. Among the evaluated compounds, mitralactonine exhibited the strongest binding affinity (-11.1 kcal/mol), followed by mitragynine (-10.4 kcal/mol) and tetrahydroalstonine (-10.0 kcal/mol). Binding stabilization was primarily mediated through interactions with key residues including Asp155, Val156, Leu228, Leu229, Phe339, and Phe340 involving extensive hydrophobic contacts and hydrogen-bonding networks. Collectively, these findings suggest that kratom alkaloids possess significant structural compatibility with serotonergic receptor systems, supporting the hypothesis that kratom pharmacology extends beyond opioid receptor activity. This study provides additional molecular insight into the potential contribution of serotonergic pathways to the complex CNS effects of kratom.

Keywords: *Mitragyna speciosa*, kratom alkaloids, 5-HT_{2A} receptor, molecular docking, ADME prediction

Pharmacy & Digital Health (PDH)

PP-030

Multi-Horizon Forecasting of Cardiovascular Medication Utilisation During Flood Disasters: A Comparison of XGBoost, LSTM and Temporal Fusion Transformer

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Flood disasters disrupt healthcare delivery and medication supply, threatening continuity of care for patients requiring long-term cardiovascular treatment. Accurate forecasting of medication utilisation can therefore support disaster preparedness and resource planning in public healthcare facilities. This study compared the forecasting performance of local, subgroup, and global machine learning models across four horizons (T= 7, 14, 21, and 28 days) for cardiovascular medication utilisation during flood disasters. Daily dispensing data for amlodipine, perindopril, and simvastatin from 10 public hospitals and 14 primary care clinics in flood-prone areas of Malaysia (2017–2022) were used as the training set. XGBoost was employed as the machine learning algorithm where three pooling strategies were used: local models trained on a single facility and medication pair, subgroup models pooling all medications across all hospitals, and global models pooling all medications across all facilities. The models also incorporated flood-related and environmental predictors from flood reports, hydrological indicators, and climate data. Hyperparameters were optimised using Optuna, with robust scaling applied to manage utilisation fluctuations. Performance was evaluated using sMAPE and MASE across overall, flood, and non-flood scenarios. Global models consistently outperformed subgroup and local models. The Global T28 configuration achieved the lowest overall error (sMAPE 12.07%). Additionally, accuracy improved at longer forecast horizons across all configurations, with local model sMAPE decreasing from 25.60% at T7 to 18.88% at T28, while global model sMAPE decreased from 21.00% to 12.07%. Although most configurations did not outperform the naïve benchmark overall (MASE >1.000), global models achieved superior performance during flood periods (flood MASE 0.7683–0.9586). The results showed that multi-facility pooling and longer forecast horizons improved forecasting during flood disasters. Pooling across facilities may mitigate local data sparsity and improve signal detection during crises, while longer forecast horizons may reduce the impact of short-term dispensing volatility.

Keywords: XGBoost, medication forecasting, flood disaster, cardiovascular medication utilisation, disaster preparedness

PP-106

Adverse Effects of SSRIs: Prevalence and Associated Factors in Patients with Mental Disorders at a Specialized Regional Hospital Indonesia

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Mental disorders such as depression and anxiety show emerging prevalence in Indonesia. However, local data on the adverse effects of Selective Serotonin Reuptake Inhibitors (SSRI) as one of antidepressant commonly used remain limited. This study aimed to evaluate the adverse effect profiles of SSRIs in patients with mental disorders. A cross-sectional study was conducted with secondary data source from 128 medical records of patients diagnosed with mental disorders who initiated SSRI therapy in 2024 at a mental specialised regional hospital in Jakarta, Indonesia. Adverse effects were assessed using three standardized tools: the Naranjo Algorithm (causality), Hartwig-Siegel Scale (severity), and Schumock-Thornton method (preventability). The data were analysed using SPSS. Majority of the patients were female (71.9%), young adults from 18 to 29 years old (53.1%), and diagnosed with depression (71%). Most of the SSRI used were sertraline (57%) with trihexyphenidyl (42.2%) and aripiprazole (35.9%) were shown as the highest frequent of therapy combined. The study revealed 166 cases of adverse effects experienced by 59% patients with the most common manifestations were gastrointestinal and neurological symptoms. The adverse effects were mild in 85% of cases, and 64% were deemed preventable. Female gender ($p=0.015$; $OR=2.6$) and potential drug interactions ($p=0.033$; $OR=3.8$) significantly correlated with adverse effect occurrence (95% CI). The high frequency of adverse effect found underscores the need of pharmacist intervention to educate and monitor patients using SSRI.

Keywords: antidepressant, adverse effect, SSRI, mental disorder

PP-110

Development of a Mobile Application for A Self Care in Atrial Fibrillation Patients Undergoing Follow-up at A Malaysian Teaching Hospital

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Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia worldwide and is associated with increased risk of stroke, heart failure, and reduced quality of life. Despite effective treatments, many patients have limited knowledge of AF, its medications, symptom monitoring, and self-care practices. This study aimed to develop a user-friendly, web-based mobile health application providing AF education, symptom monitoring, and lifestyle guidance, and to perform face and content validation of the application. A non-experimental development study was conducted in two phases. First, a web-based application was developed using Wix Studio, guided by evidence-based clinical guidelines and literature, in both Bahasa Malaysia and English using patient-friendly language. Second, content validation was performed by five healthcare experts with at least five years of cardiology-related experience, while face validation was conducted among 20 AF patients. Content validation assessed accuracy, relevance, clarity, and completeness using the Content Validity Index (CVI), while face validation assessed language clarity, design, ease of use, and usefulness using a four-point Likert scale. Data were analysed descriptively using Microsoft Excel. Content validation showed overall mean scores ranging from 2.83 to 3.07, with Item-level CVI values ranging from 0.60 to 1.00 and a Scale-level CVI (S-CVI/Ave) of 0.75. Face validation demonstrated favourable acceptance, with mean scores ranging from 3.45 to 3.70 for individual items and 3.47 to 3.67 across the four domains. Feedback supported further refinement, particularly in visual design, language consistency, and interactive educational features. The developed application demonstrated good acceptance among patients and satisfactory content validity, supporting its potential as a supportive educational tool to enhance self-care among AF patients. Further studies are recommended to evaluate its usability and effectiveness in improving patient knowledge and self-care behaviours in larger populations.

Keywords: atrial fibrillation, self-care, patient education, mHealth, validation, validation.

PP-114

**Development of an Evidence-Based ABCDE Heart Failure Management Protocol for
Healthcare Professionals in Malaysia: A Modified Delphi Study**

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Heart failure (HF) remains a major cause of morbidity and mortality worldwide. Despite evidence-based clinical guidelines are available, variation in clinical practice and the absence of a simplified competency-based protocol may hinder optimal HF management of clinical settings in Malaysia. An evidence-based workflow using the ABCDE framework was developed to provide a practical memory aid for healthcare professionals. This study applied a modified Delphi approach to develop expert consensus of five domains HF evidence-based care. To develop and achieve expert consensus on an evidence-based ABCDE heart failure management protocol for healthcare professionals in Malaysia using a modified Delphi approach. A three-round modified Delphi study was conducted between January 2025 and July 2026. An expert panel comprising seven senior clinical pharmacists and five cardiologists across different practice locations and all geographic regions in Malaysia evaluated competency statements developed from international and local HF guidelines and published literature. Participants rated each statement using a five-point Likert scale. Consensus was predefined as $\geq 80\%$ agreement, a median score ≥ 4 , and an interquartile range (IQR) ≤ 1 . Items not meeting consensus were revised based on qualitative feedback and reassessed in subsequent rounds. 12 out of 32 invited experts completed all three Delphi rounds (100% retention). In Round 1, 15 of 20 protocol components (75.0%) achieved consensus. Following revision, three additional components reached (100.0%) consensus in Round 2, while the remaining two components were further refined. In Round 3, all remaining components achieved (100.0%) consensus, resulting in full agreement on the final ABCDE heart failure management protocol. Expert feedback enhanced the protocol's clinical relevance, clarity, applicability, and alignment with current evidence-based HF management. Expert consensus was achieved on all five domains competencies an evidence-based ABCDE heart failure management protocol tailored to Malaysian clinical practice. ABCDE framework provides a practical, evidence-based protocol to guide standardized, safe, and holistic heart failure care delivery by healthcare professionals in Malaysia.

Keywords: heart failure, Delphi study, ABCDE framework, evidence-based practice, Malaysia

PP-117

The Knowledge, Attitude and Practice Towards Artificial Intelligence (AI) Among Undergraduate Pharmacy Students in UiTM

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Artificial Intelligence (AI) has rapidly transformed higher education and training of health professionals. However, there is limited empirical evidence regarding the AI literacy among pharmacy students in Southeast Asia. This study aimed to evaluate the level of knowledge, attitude and practice (KAP) regarding AI, determine the associations with sociodemographic characteristics, and assess correlations among KAP domains among undergraduate pharmacy students in Malaysia. A cross-sectional survey was conducted among 289 undergraduate pharmacy students at Universiti Teknologi MARA (UiTM) using a validated, self-administered online questionnaire. Data were analysed using Chi-Square tests and Spearman's rank correlation. Participants demonstrated good knowledge (78.6%, n = 227), a neutral attitude (76.5%, n = 221), and good practice (54.8%, n = 158) towards AI. Academic year was significantly associated with both knowledge ($p = 0.041$) and attitude ($p = 0.004$). There are significant positive correlations between knowledge and attitude ($r = 0.143$, $p = 0.015$) as well as between attitude and practice ($r = 0.473$, $p < 0.001$). While undergraduate pharmacy students possess adequate knowledge and actively utilize AI tools, their overall attitude remains neutral. Curricular efforts towards integrating AI into pharmacy education are crucial in cultivating more positive, ethical and responsible AI usage among the students and future healthcare providers.

Keywords: artificial intelligence, pharmacy students, higher education, curriculum development, Malaysia

PP-119

Treatment Patterns and Clinical Outcomes in Women with Gestational Diabetes Mellitus: A Retrospective Study at a Malaysian Tertiary Hospital

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Gestational Diabetes Mellitus (GDM) is a frequent complication during pregnancy, marked by impaired glucose tolerance, leading to significant maternal and neonatal risks. When lifestyle modifications are insufficient, pharmacological interventions, particularly insulin and metformin, are often required. This study aimed to characterize GDM management strategies and evaluate their associations with maternal and neonatal outcomes at Hospital Al-Sultan Abdullah (HASA). This retrospective observational study involved patients diagnosed with GDM who received antenatal care and delivered at HASA between January 2023 and December 2024. Data were collected from electronic health records, focusing on patient demographics, type of management, glycaemic control, and clinical outcomes. Descriptive statistics and Fisher-Exact tests were used to assess associations between treatment types and glycaemic control or outcomes. Among 196 patients, diet control was the most common management strategy (58.3%), followed by metformin (25.2%) and combination therapy (16.5%). Initial treatment of GDM favoured diet control (60.7%). Significant associations were found between GDM management and glycaemic control for FBG ($p = 0.001$), 2HPP ($p = 0.019$), and HbA1c ($p = 0.006$), with diet control showing the best outcomes. Insulin use was significantly associated with an increased risk of preeclampsia ($p = 0.002$). No significant associations were found between treatment type and other maternal or neonatal outcomes, including mode of delivery, preterm birth, macrosomia, or NICU admission. Diet control was the most prescribed strategy for GDM in this cohort, associated with better glycaemic outcomes. Insulin use was significantly linked to an increased risk of preeclampsia, highlighting the need for careful monitoring in patients requiring pharmacological therapy. No significant differences were observed in other maternal and neonatal outcomes across the various treatment groups, which may be attributed to the limited sample sizes within certain subgroups.

Keywords: gestational diabetes mellitus, treatment patterns, pregnancy, maternal outcomes, neonatal outcomes, glycaemic control

PP-122

Intention To Adopt and Use Telepharmacy Services Among Patients with Asthma in Puncak Alam, Malaysia

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Telepharmacy, a digital health innovation, extends pharmaceutical care to remote populations and has demonstrated promise in managing chronic conditions such as asthma. Despite its potential, adoption rates in Malaysia remain suboptimal. Understanding patients' intentions is critical for developing patient-centred telepharmacy models that address specific needs and barriers. This study aimed to explore the factors influencing the intention to adopt and use telepharmacy services, and to identify the challenges and barriers faced by patients with asthma in Puncak Alam, Selangor. A qualitative descriptive study was conducted using semi-structured face-to-face interviews among 11 adult patients with asthma recruited via purposive sampling. The interview guide was developed based on the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), and UTAUT2 (Price Value construct). Interviews were audio-recorded, transcribed verbatim, translated into English by a bilingual researcher, and thematically analysed using Braun and Clarke's six-phase approach with NVivo 14 software. Data saturation was achieved by the 10th participant. Eight participants (72.7%) were unaware of telepharmacy prior to the study. Key facilitating factors included perceived benefits and usefulness, particularly improved inhaler technique guidance, time efficiency, and convenient access to pharmacists—as well as technology readiness, perceived simplicity of use, and willingness to pay for services. Major barriers were lack of awareness and exposure, need for pharmacist guidance, and technological challenges including poor internet connectivity. Participants proposed targeted strategies, including verbal briefings by healthcare providers, development of a user-friendly mobile application with live chat and strategic partnerships with health clinics. Patients with asthma perceive telepharmacy as beneficial for improving asthma management, yet significant awareness and support gaps persist. Addressing these barriers is essential for ensuring effective and equitable telepharmacy adoption. Future research should explore integration strategies within Malaysia's healthcare system across diverse chronic conditions and demographic groups.

Keywords: telepharmacy, intention to adopt, asthma, UTAUT, TAM

PP-125

Development and Expert Review of Cases on Vitamins, Mineral and Natural Health Product (VMNHP)-Related Problems

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The global use of vitamins, minerals, and natural health products (VMNHPs) is increasing, but it is not without risk. VMNHP-related problems have been identified across various healthcare settings. Despite these risks, knowledge regarding VMNHPs remains low among healthcare professionals, which is further compounded by a lack of mock cases for clinical training. The objective of this study was to develop and review evidence-based cases on VMNHP-related problems. Prior to this study, a scoping review and focus group discussion were conducted to outline key VMNHP-related problems. Ten clinical cases were developed, each incorporating a maximum of three different VMNHP-related problems. A panel of ten experts reviewed the cases using a 4-point Likert scale and evaluated agreement on the pre-identified VMNHP-related problems. In total, there are ten experts consisting of physicians (n = 3), pharmacists (n = 5), and academic pharmacists (n = 2). Case review mean scores ranged between 2.90 and 3.80 on a 4-point Likert scale. Appropriately embedded VMNHP-related problems in Case 4 scored highest (mean 3.80), while information sufficiency in Case 1 scored lowest (mean 2.90). Case 4 also scored highest for clarity (mean 3.70). Consensus on pre-identified VMNHP-related problems ranged from 80.0% to 100.0% across all ten cases. Four cases (Cases 2, 4, 6, and 7) achieved complete consensus (100.0% agreement), while Case 9 recorded the lowest agreement rate (80.0%). Ten evidence-based clinical cases on VMNHP-related problems were successfully developed and reviewed, demonstrating strong expert consensus. These reviewed cases serve as an effective educational tool to bridge existing knowledge gaps and enhance training on VMNHP-related problems. Future research should pilot-test these reviewed cases within medical and pharmacy curricula to evaluate their practical impact on improving learners' clinical decision-making regarding VMNHP-related problems.

Keywords: case-based learning, natural health product, health professional education

Pharmaceutical Technology (PT)

PP-012

Phenolic-Mediated Redox Crosstalk and Antioxidant Mechanisms of Green-Synthesized Gold Nanoparticles from Purple Carrot Root Extract
“Eco-Friendly Nanotechnology Meets Redox Biology for Antioxidant Synergy”

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Oxidative stress is increasingly recognized as a central factor in the initiation and progression of chronic diseases, including cardiovascular, neurodegenerative, and metabolic disorders. This has led to growing scientific interest in developing therapeutic approaches that are not only effective but also safe, biocompatible, and environmentally sustainable. Green nanotechnology represents a promising alternative to conventional nanoparticle synthesis by avoiding toxic reagents and employing plant-based extracts as natural reducing and stabilizing agents. Such an approach offers a dual benefit of eco-friendly synthesis while retaining the therapeutic potential of the bioactive compounds. In this study, we explored the green synthesis of gold nanoparticles (AuNPs) using the root extract of *Daucus carota* ssp. *sativus* var. *atrorubens* Alef. (Commonly known as purple carrot), a plant rich in phenolic compounds and anthocyanins known for their antioxidant activity. Two formulations, PC-GN 1:2 and PC-GN 1:4, were prepared by varying the extract-to-gold precursor ratios to evaluate how extract concentration influences nanoparticle properties. The purple carrot extracts successfully acted as both a natural reducing and capping agent, eliminating the need for hazardous synthetic chemicals. Characterization was carried out using Fourier-transform infrared spectroscopy (FTIR), transmission electron microscopy (TEM), X-ray diffraction (XRD), and zeta potential analysis. FTIR spectra confirmed the involvement of hydroxyl, carbonyl, and amine groups in the reduction and stabilization process. TEM images revealed spherical to polyhedral nanoparticles ranging from 15 to 60 nm in diameter, with PC-GN 1:4 demonstrating improved morphological uniformity and reduced aggregation compared to PC-GN 1:2. XRD analysis confirmed a crystalline, face-centred cubic structure typical of elemental gold, while zeta potential measurements highlighted greater colloidal stability in PCGN 1:4 (+25.5 to +29.7 mV) compared with PC-GN 1:2 (−11.4 mV). The antioxidant activity of both formulations was evaluated at a fixed concentration of 100 µg/mL using DPPH radical scavenging, ferric reducing antioxidant power (FRAP), and total phenolic content (TPC) assays. PC-GN 1:4 exhibited superior performance, achieving 92.4% DPPH inhibition, a FRAP value of 3.26 mM AAE, and 147.8 mg GAE/mL total phenolic content. These values were higher than those observed for PC-GN 1:2 and the crude purple carrot extract, supporting the inference that increasing extract concentration enhances nanoparticle stability and antioxidant capacity. Taken together, these results highlight the potential of purple carrot-mediated AuNPs, particularly PC-GN 1:4, as eco-friendly nanocarriers with strong in vitro antioxidant properties. Although this study is limited to chemical assays, it establishes a foundation for further exploration. Future research should focus on cytotoxicity testing, in vivo models, and functionalization for targeted drug delivery to validate the biomedical and pharmaceutical applications of these green-synthesized nanoparticles.

Keywords: green nanotechnology, gold nanoparticles, purple carrot extract (*Daucus carota*), redox crosstalk, phenolic compounds, antioxidant mechanisms

PP-018

Lipid Vesicles as Promising Carriers for Improved Transdermal Delivery of Nifedipine

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Nifedipine is an antihypertensive drug with low oral bioavailability due to extensive first-pass metabolism. One potential solution is transdermal drug delivery, which offers advantages such as improved bioavailability and enhanced patient compliance. This study aimed to formulate lipid vesicles, including transfersomes, ethosomes, transethosomes, and invasomes, as carriers to enhance the transdermal delivery of nifedipine. Nifedipine-loaded lipid vesicles, including transfersomes, ethosomes, transethosomes, and invasomes, were prepared using the thin-film hydration technique and characterised in terms of vesicle size (VS), polydispersity index (PDI), zeta potential, entrapment efficiency (EE%), morphology, Fourier-transform infrared spectroscopy (FTIR), and differential scanning calorimetry (DSC). The nifedipine-loaded lipid vesicles were further incorporated into gel formulations and evaluated for rheological properties, drug content, *in vitro* drug release, and *ex vivo* permeation. The different types of nifedipine-loaded lipid vesicles exhibited particle sizes of less than 200 nm, a PDI of less than 0.7, a zeta potential greater than ± 30 mV, and an entrapment efficiency of more than 90%. Morphological studies confirmed the formation of spherical vesicles without aggregation. FTIR and DSC studies confirmed the successful encapsulation of nifedipine within the vesicles. Rheological studies suggested that the prepared gels exhibited pseudoplastic behaviour. The *in vitro* drug release study demonstrated that the lipid vesicle-incorporated gels provided controlled drug release, while the *ex vivo* permeation study showed enhanced permeation of nifedipine from the lipid vesicle formulations compared with the plain nifedipine gel. The developed nifedipine-loaded lipid vesicles demonstrated promising potential as a transdermal delivery system for enhancing the permeation and delivery of nifedipine.

Keywords: transfersomes, ethosomes, transethosomes, invasomes, nifedipine

PP-020

Metal-Sequestering Self-Assembling Peptide Nanonets for Immobilizing and Inhibiting the Growth of *Escherichia coli*

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Transition metals serve as essential biological cofactors for ongoing bacterial viability and replication. During infection, the host innate defense system deploys nutritional immunity to systematically sequester trace transition metals, thereby inhibiting bacterial metal uptake and restricting the spread of the infection. Translating this mechanism into a biomimetic approach offers an evolution resistant paradigm. Designing multifunctional peptides capable of metal chelation and self-assembly simultaneously deprives pathogenic bacteria of critical cofactors and physically traps them, minimizing selective pressure for mutations. To address this challenge, our study characterizes the metal-binding capacity, secondary structure, fibrillization behaviour, and copper-enhanced antimicrobial potency of a series of novel engineered peptides against *Escherichia coli*. Mass spectrometry (MS) confirmed the lead Peptide J1's metal-binding function, revealing distinct mass-to-charge shifts and robust adduct peaks upon copper addition. Circular dichroism (CD) spectroscopy demonstrated that J1 adopts a beta-sheet conformation with or without copper ions (Cu^{2+}), providing the structural basis for self-assembly. Scanning electron microscopy (SEM) visualized this process, showing that upon interaction with lipopolysaccharide or *E. coli* cells, J1 rapidly self-assembles into dense antimicrobial nanonets that physically entangle the bacteria. Finally, broth microdilution assays assessed biological activity. The minimum inhibitory concentration (MIC_{90}) heatmap revealed a 2-fold improvement in J1's antimicrobial efficacy against *E. coli* in the presence of Cu^{2+} , translating its metal-sequestering and self-assembling features into improved overall antibacterial performance. In conclusion, J1 acts as an innovative multifunctional scaffold that chelates copper and self-assembles into dense nanonets upon bacterial contact. This dual-action mechanism effectively inhibits and immobilizes the pathogen, presenting a highly viable, evolution-resistant framework for combating resilient bacterial infections.

Keywords: self-assembling peptide, metal-chelating peptide, antimicrobial activity, nutritional immunity, nanonets

PP-052

**Systematic Evaluation of the Predictive Performance of Biorelevant *In Vitro* Systems
Integrated with PBBM for Human Pharmacokinetics**

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Drugs classified as BCS class IIa/IIb often exhibit pH-dependent dissolution behaviour and substantial inter- and intra-individual variability in gastrointestinal conditions, making quantitative prediction of human pharmacokinetics (PK) and rational formulation design challenging. Although several biorelevant *in vitro* systems have been developed, including the BE Checker for dissolution assessment in a single vessel and the Gastrointestinal Simulator-a (GIS-a) for simulating gastrointestinal transit across multiple compartments, a direct comparison of their predictive performance for human PK has not yet been reported. This study aimed to evaluate and compare the predictive capabilities of three *in vitro* platforms—USP apparatus II, BE Checker, and GIS-a—using model drugs representing BCS class IIa and IIb. Drug dissolution profiles, pH-shift behaviour, and medium composition were evaluated using the different *in vitro* systems and subsequently integrated into physiologically based biopharmaceutics modelling (PBBM). We systematically investigated how system-specific characteristics influence the accuracy of simulated human PK profiles and assessed predictive performance across drugs with different physicochemical properties to identify compounds with limited predictability. Integration of the *in vitro* data into PBBM enabled successful prediction of human PK profiles while revealing differences in predictive performance among the tested systems. Variations in pH-shift kinetics and medium composition were found to influence prediction accuracy. Furthermore, the study clarified the advantages of individual systems, such as the ability to capture dynamic gastrointestinal transit, and identified drug properties associated with reduced predictive performance. This study provides an objective comparison of these biorelevant *in vitro* systems and clarifies their respective strengths and limitations. The findings offer practical guidance for selecting appropriate *in vitro* systems according to drug physicochemical properties and support more rational formulation development.

Keywords: PBBM, biorelevant *in vitro* dissolution system, USP apparatus II, BE Checker, Gastrointestinal Simulator-a

PP-053

Development of a Mechanistic Model for Predicting the Bioavailability of Subcutaneously Administered Therapeutic Antibodies

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The bioavailability (BA) of subcutaneously administered therapeutic antibodies is known to vary substantially among molecules. Accurate prediction of BA is therefore critical for antibody molecular design and formulation development. Although various *in vitro* and *in silico* approaches have been reported, methodologies capable of quantitatively predicting BA across a wide range of therapeutic antibodies remain limited. This study aimed to develop a predictive method for antibody BA by constructing a mechanistic model that describes the absorption process following subcutaneous administration. A physiological model was developed to describe antibody behaviour within the interstitial tissue at the subcutaneous injection site. Focusing on transport from the injection site to the peripheral lymphatic system, this model is based on differential equations describing diffusion processes in the local tissue environment. The model incorporates the relationship between antibody concentration-dependent viscosity and diffusion coefficients, the effects of antibody charge state on diffusion behaviour, precipitation and macrophage-mediated phagocytosis at the injection site, and neonatal Fc receptor (FcRn)-mediated recycling. The developed model was applied to more than ten therapeutic antibodies to predict BA following subcutaneous administration, using intravenous pharmacokinetic data as a reference. The model predicted the BA of most therapeutic antibodies within 30% of the observed values, demonstrating robust predictive performance across diverse therapeutic antibodies. However, prediction accuracy decreased for antibodies exhibiting low observed BA. These findings suggest that additional physiological factors at the subcutaneous administration site may contribute to antibody BA. While the developed mechanistic model successfully predicts the BA of many therapeutic antibodies, further refinement is underway to investigate additional physiological parameters and identify common molecular characteristics among antibodies with low prediction accuracy.

Keywords: therapeutic antibodies, bioavailability, subcutaneous administration, mechanistic model, pharmacokinetics

PP-121

A Modified Hanging Trap for Sampling Blowflies (Diptera: Calliphoridae) In Tropical Rainforest Environments

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We describe a modification to a commercially available, low-cost hanging fly trap that was tested across various tropical habitats over two years. The original trap, similar in design to the Van Someren-Rydon model but smaller and lightweight, was improved for enhanced practicality and field performance. The modified version is cost-effective, weather-resistant, easy to deploy, ant-proof, and capable of collecting live and intact blowfly specimens. Field testing confirmed improved capture rates for medically important flies from the families Calliphoridae, Sarcophagidae, Muscidae, and Anthomyiidae. This design offers a standardized, reliable, and field-practical solution for blowfly sampling in tropical environments.

Keywords: blowfly, Calliphoridae, insect trap, tropical rainforest, sampling method

Nutraceutical Science (NS)

PP-015

Alpha-Tocotrienol Preserves Hormonal Balance and Follicular Ultrastructure in Nicotine-Treated Mice

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Nicotine exposure is known to induce oxidative stress and impair female reproductive function by disrupting hormonal balance and ovarian architecture. Alpha-tocotrienol (α-TCT), a potent member of the vitamin E family, possesses strong antioxidant and cytoprotective properties that may counteract reproductive toxicity. This study investigated the protective effects of α-TCT against nicotine-induced ovarian damage. Six- to eight-week-old female Balb/c mice were randomly assigned into five groups (n= 6 per group). Group A served as the untreated control, while Group B received of tocopherol-stripped corn oil (0.1 mL). Group C was administered nicotine alone (3 mg/kg body weight/day). Group D received α-TCT alone (30 mg/kg body weight/day), whereas Group E received concurrent treatment with nicotine (3 mg/kg body weight/day) and α-TCT (30 mg/kg body weight/day) for seven consecutive days. At the end of the treatment period, serum samples were collected for the analysis of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) levels using enzyme-linked immunosorbent assay (ELISA). Ovarian tissues were harvested for ultrastructural assessment using transmission electron microscopy (TEM). Hormonal analyses demonstrated that nicotine exposure reduced serum FSH and LH levels, indicating disruption of reproductive endocrine function. In contrast, concurrent α-TCT supplementation increased both FSH and LH levels compared to the nicotine-treated group, suggesting partial restoration of hormonal balance. Ultrastructural examination further revealed that nicotine exposure (Group C) caused marked degenerative changes in ovarian follicles, including granulosa cells shrinkage and reduced follicular distribution. However, co-treatment with α-TCT (Group E) ameliorated these adverse alterations, as evidenced by preservation of granulosa cell morphology and improved follicular integrity. These findings suggest that α-TCT exerts protective effects against nicotine-induced reproductive toxicity by improving hormonal regulation and preserving follicular ultrastructure. The study highlights the therapeutic potential of α-TCT as a protective agent against oxidative-stress mediated ovarian damage. Further studies are warranted to elucidate the underlying molecular mechanisms involved.

Keywords: alpha-tocotrienol, nicotine, ovary, granulosa cells, transmission electron microscopy (TEM)

PP-041

Postbiotics Derived from *Lactiplantibacillus plantarum* LAB12-Fermented Soymilk Attenuated Lipopolysaccharide (Lps)-Induced Neuroinflammation *In Vitro*

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Alzheimer's disease (AD) is the leading cause of dementia worldwide, with neuroinflammation as the key driver of its pathogenesis. It is increasingly recognised that persistent microglial activation contributes to oxidative damage and amyloid- β accumulation, further exacerbating neuroinflammation in AD. Unfortunately, current anti-AD therapies alleviate symptoms but do not cure the disease. This raises the need for alternative approaches. Interestingly, emerging evidence linking gut dysbiosis to neuroinflammation through the gut-brain axis supports the possibility of probiotic-derived bioactive metabolites (i.e., postbiotics) as potential neuroprotective agents. This study aimed at evaluating the protective effects of postbiotics derived from soymilk fermented by either *Lactiplantibacillus plantarum* LAB12 (probiotic lactic acid bacteria (LAB) isolated from *tempeh*) and *Lacticaseibacillus casei* strain Shirota (designated as LAB13) against LPS-induced neuroinflammation *in vitro*. LAB-derived postbiotics were profiled for total phenolic content (TPC) using the TPC assay and short chain fatty acids (SCFA) using GC/MS. Their antioxidative properties were examined using the 2,2-diphenyl-1-picrylhydrazyl (DPPH) assay. The highest subtoxic concentration of the postbiotics against HMC3 (human microglia) and SK-N-SH (human neuroblastoma) cell lines were determined using the Sulforhodamine B (SRB) assay. The protective effects of the LAB-derived postbiotics against lipopolysaccharides (LPS)-induced HMC3 and SK-N-SH that mimic neuroinflammation were assessed using the Reactive Oxygen Species (ROS) and Griess assays. LAB12-derived postbiotics yielded TPC and exhibited antioxidant activities comparable to those of LAB13-derived postbiotics. SCFA profiling revealed significantly ($p < 0.05$) increased levels of acetate, butyrate, propionate and valerate in soymilk fermented especially by LAB13. Both LAB-derived postbiotics conferred protection against LPS-induced ROS and NO in HMC3 but not SK-N-SH. LAB12-derived postbiotics exhibited comparable protective effects against ROS, but significantly ($p < 0.05$) reduced NO when compared to LAB13-derived postbiotics. The present findings on the potential protective effects of LAB12-derived postbiotics against neuroinflammation *in vitro* warrants further validation using *in vivo* models.

Keywords: Alzheimer's disease, neuroinflammation, *Lactiplantibacillus plantarum* LAB12, antioxidants, short-chain fatty acids

PP-042

Potential Role of the Gut-Brain-Muscle Axis in Exercise-Ameliorated Cognitive Frailty in D-Galactose-Induced and Naturally Aged Sprague Dawley Rats

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Cognitive frailty (CF), the co-presence of physical frailty and mild cognitive impairment (MCI) but without dementia, has emerged as a growing concern amongst the aged population given its association with increased vulnerability to adverse health outcomes. Our previous findings demonstrated that exercise improved physical and cognitive parameters in D-galactose-induced ageing (animal model 1) and naturally aged (animal model 2) rat models. This study aimed at elucidating potential mechanisms underlying these improvements in relation to alterations in serum (oxidative stress and bone metabolism), hippocampal markers (neurotrophic factor, cholinergic enzyme, inflammatory and antioxidant), caecal short-chain fatty acids (SCFA) and gut microbiota in animal models 1 and 2. Animal model 1 (male, 26-week old; n=7/group) was administered with D-galactose over 9 weeks to accelerate ageing, whereas animal model 2 (male; n=4/group) was subjected to exercise starting from 20-month old (midlife) until 23-month old (aged). Both rat models were subjected to the motorised wheel running exercise regimen (≥ 9 weeks). Caecal, blood, and hippocampal samples were collected for 16S rRNA sequencing, short chain fatty acid (SCFA) analysis, and enzyme-linked immunosorbent assay, respectively. In animal model 1, exercise significantly ($p < 0.05$) lowered serum CTX (bone breakdown marker) and MDA (oxidative stress marker) while increasing hippocampal BDNF (neurotrophic marker) but reducing IL-6 (pro-inflammatory marker). Exercise also altered the gut microbiota through pathways related to carbohydrate metabolism, nutrient transport, and microbial communication. In animal model 2, exercise significantly ($p < 0.05$) decreased serum OC (bone formation marker) and hippocampal IL-6. Exercise also enriched *Bacillota* (phylum correlated with increased butyric acid production) but reduced *Coriobacteriaceae* (genus). Functional pathway analysis further revealed enrichment of microbial metabolic pathways and SCFA metabolism. The complementary findings from both animal models supported the potential role for exercise in protecting against CF that could be mediated through the gut-brain-muscle axes.

Keywords: cognitive frailty, exercise, ageing, gut, Sprague Dawley rats

PP-043

Clinical Evidence of Probiotics, Prebiotics and Synbiotics on Gut Microbiota and Lipid Profiles of Individuals with Dyslipidaemia and Cardiovascular Diseases: A Systematic Review

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Cardiovascular diseases (CVD) are the leading cause of death globally, with dyslipidaemia as the key risk factor. Although statins remain the mainstay therapy, poor adherence limits their effectiveness. This raises the need for microbiota-targeted therapies, namely probiotics, prebiotics and synbiotics, which are increasingly recognised for their beneficial effects on lipid metabolism through modulation of gut microbiota. This systematic review appraised the clinical evidence of these interventions on gut microbiota and lipid profiles in individuals with dyslipidaemia and CVD. Literature search was performed through PubMed, Scopus, ScienceDirect and the Cochrane Library. Eligible studies were shortlisted based on inclusion and exclusion criteria. The risk of bias was assessed using the "RoB 2" tool and the NIH quality assessment tool for before-after (Pre-Post) studies with no control group. The primary outcome is gut microbiota while secondary outcomes included lipid profile, glycaemic parameters and anthropometric parameters. A total of 21 publications were shortlisted from 17,636 articles. Seventeen publications were, however, presented with high risk of bias. Synbiotic interventions from two publications with low risk of bias under a single trial reported significantly increased *Bacteroidetes* and *Bifidobacterium* spp., alongside an increase in high-density lipoprotein cholesterol. Probiotic and prebiotic interventions, on the other hand, were associated with increases in beneficial *Bifidobacterium* spp., *Faecalibacterium* spp., *Lactobacillus* spp., as well as reductions in total cholesterol and low-density lipoprotein cholesterol. Nevertheless, the overall clinical evidence of these interventions remained inconclusive due to the predominance of shortlisted studies with high risk of bias and considerable heterogeneity in study designs, interventions (e.g., probiotic strains and prebiotic types), and microbiota assessment methods. Current clinical evidence remains insufficient to establish the effectiveness of probiotics, prebiotics and synbiotics in dyslipidaemia or CVD management. This calls for well-designed and standardised clinical trials to validate their beneficial effects in this regard.

Keywords: probiotics, prebiotics, synbiotics, gut microbiota, dyslipidaemia.

PP-124

10-HDA Ameliorates Cancer-Related Fatigue Through Modulation of Neuroinflammation and Metabolic Regulation

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Cancer-related fatigue (CRF) is a debilitating syndrome that severely impairs the quality of life in cancer patients and survivors. Despite its high prevalence, effective pharmacological interventions remain limited due to its complex and multifactorial etiology. Although royal jelly is recognized as a functional dietary supplement with anti-fatigue and anti-inflammatory properties, the therapeutic effects of its bioactive compound, 10-hydroxydecanoic acid (10-HDA), has not been evaluated in preclinical CRF models. This study aimed to investigate the therapeutic potential of 10-HDA against CRF and to elucidate its underlying metabolic and neuroinflammatory mechanisms. Male BALB/c mice were intravenously inoculated with murine colon carcinoma (C26) cells to induce CRF, and they were fed either a normal diet (control) or a 1% 10-HDA supplemented diet. Fatigue-like behaviour was evaluated using treadmill endurance tests at baseline (Day -1) and post-treatment (Day 13). Blood, liver, muscle, spleen and brain specimens were collected at day 13 post-tumour inoculation and further examined. Metabolic parameters (glucose, liver glycogen, ketone levels) were assessed by biochemical analyses, and neuroinflammation was assessed by expression of inflammatory genes (*IL-6*, *IL-1 β* , *TNF- α*) in the prefrontal cortex via qRT-PCR. In addition, anti-tumour effect of 10-HDA was evaluated *in vitro* on C26 cells using CCK-8 assays. We found that daily dietary supplementation with 10-HDA significantly restored treadmill running performance in C26-bearing mice compared to controls, without altering overall body weight or food intake. 10-HDA also normalized blood glucose and ketone levels, although liver glycogen depletion remained uncorrected. Furthermore, 10-HDA significantly attenuated the C26-induced upregulation of *IL-6*, *IL-1 β* and *TNF- α* mRNA expression in the prefrontal cortex, effectively dampening neuroinflammation. These findings suggest that 10-HDA alleviates CRF through a dual mechanism involving the stabilization of energy homeostasis and the suppression of neuroinflammatory pathways, highlighting 10-HDA as a promising nutraceutical candidate for CRF management.

Keywords: cancer-related fatigue, royal jelly, neuroinflammation, tumour-bearing mice, 10-hydroxydecanoic acid

Pharmaceutical Chemistry (PC)

PP-027

Literature Review on *Solanum* Species (The Tomatoes)

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This mini project aims to review the tomato (*Solanum* species, Solanaceae family), which is used in the culinary, food, pharmaceutical, nutraceutical, cosmetic industries and traditional medicine. From the reports, their biological properties include anti-inflammatory and antioxidant effects. The objective of this investigation is to search for articles about tomatoes, including cherry tomatoes (*S. lycopersicum* var. *cerasiforme*), and perform a methodological review on the extraction of the biomolecules. From the literature, the phytochemical composition and the extraction of compounds of these tomatoes were discussed. Lycopene (C₄₀H₅₆) is recorded as the dominant carotenoid, which is 90 % of the total carotenoids in tomatoes. The community pharmacies promote some products containing these *Solanum* species by introducing the lycopene-containing soft gels. Healthy snacks are also highlighted, which consist of dried *Solanum* fruits. Meanwhile, the fruits extracts are produced as botanical drinks that include lycopene, as a natural protection for the skin. Furthermore, the pulps are marketed as the ingredients for spaghetti, available in convenience stores and supermarkets. In terms of methodology, the extraction of the carotenoids and polyphenolics (rutin, naringenin) from the fruits was established. The scientific papers published on the comparison among the conventional solvent extraction (hot & cold solvent), microwave assisted extraction, ultrasound-assisted extraction, deep eutectic solvent extraction, and supercritical carbon dioxide extraction. Gas chromatography–mass spectrometry profiling could reveal the retention of key aldehydes and esters associated with the characteristic tomato aroma. Furthermore, UV/Vis/NIR spectroscopy and machine vision could also be utilized for lycopene detection. Biotechnological research was conducted to propose that red tomatoes could serve as a natural, alternative dye to replace synthetic colorants. Finally, it is anticipated that by enriching *Solanum* fruit intake, the consumption of the beneficial bioactive compounds of this species would be greatly appreciated.

Keywords: lycopene, phenolics, review, *Solanum*, tomato

PP-031

Recent Studies on the Beetroot and the Betalains

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This presentation discusses recent studies on the red beetroot (*Beta vulgaris* L.) and its bioactive compounds, notably the nitrates and betalains (or betacyanins). These are water-soluble, nitrogen-containing plant pigments, responsible for the vibrant red and purple colours of beetroot. The most common betacyanin is betanin, a glucoside of betanidin. Comprehensive reviews have examined the occurrence of betalains, their molecular structures, the spectroscopic and fluorescent properties, antioxidant activity, and bioavailability. In addition, the applications of betalains as safe food colorants have been widely discussed, including their fate and environmental role. Publications have also presented various data demonstrating the nutraceutical potential of the beetroot by-products. A narrative review of peer-reviewed literature has also been conducted, focusing on human, animal, and *in vitro* studies evaluating beetroot's nutrient composition, functional food applications, and associated health outcomes. Furthermore, extensive reviews of the betalains provide current knowledge on both conventional and novel extraction methods, biological characteristics, purification procedures and the physicochemical properties of the red beetroot extracts. In terms of methodology, the maceration and extraction of *B. vulgaris* pigments have been carried out using Soxhlet, cold press, and ultrasonic methods. In addition, the extract can be analysed using FTIR and UV-visible spectrophotometry. Meanwhile, a betalain-based analytical device using Whatman paper, has been proposed as a viable alternative for the on-site detection of cupric and ferric cations in water. It is noteworthy that betalains are less stable than nitrates when exposed to heat, such as during cooking and food processing. This instability occurs because betalains readily degrade, whereas the nitrate structures remain relatively stable. Therefore, improving the chemical stability of betalains is crucial for applications in the food industry. In summary, increasing attention to beetroot research could pave the way for the use of this plant's natural root pigments as substitutes for synthetic acid-base indicators in titrimetric analysis.

Keywords: beetroot, betalain, review

PP-034

Antioxidant Activity of Basil (*Ocimum Basilicum*): A Systematic Review and Meta-Analysis

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Sweet basil (*Ocimum basilicum*) is a promising source of bioactive compounds with potent antioxidant properties for pharmaceutical applications. While numerous studies have evaluated its antioxidant potential, a comprehensive synthesis of this data remains limited. This study conducted a systematic review and meta-analysis to evaluate the antioxidant activity of *O. basilicum* essential oils (EOs) and extracts, comparing their efficacy against ascorbic acid, a standard antioxidant agent. A systematic literature search was conducted across PubMed, Google Scholar and ProQuest for articles published between 2000 and 2023. Studies utilizing the DPPH radical scavenging assay and reported via percent inhibition and half-maximal inhibitory concentration (IC₅₀) were included. Following PRISMA guidelines, twenty-four articles met the inclusion criteria (7 EO studies and 17 extract studies), and study quality was evaluated using the OHAT Risk of Bias tool. The systematic review confirmed significant antioxidant activity in both EOs and extracts. However, meta-analysis revealed that ascorbic acid exhibited superior potency. For percent inhibition, the overall standardized mean difference (SMD, d-value) in the forest plot was -1.27 (95% CI, -2.49; -0.05) for EOs and -12.58 (95% CI, -18.53; -6.62) for extracts. For IC₅₀, the d-values were -0.34 (95% CI, -1.47; 0.78) and 3.73 (95% CI, 1.87; 5.59) for extracts. Notably, the limited number of EO experimental trials reduced meta-analytic consistency, and funnel plot asymmetry indicated presence of publication bias across both categories. Furthermore, high statistical heterogeneity was observed due to variations among the included studies. Ultimately, while *O. basilicum* shows clear antioxidant merit, ascorbic acid remains a significantly more potent agent.

Keywords: sweet basil, *Ocimum basilicum*, 2,2-Diphenyl-1-picrylhydrazyl, ascorbic acid

PP-051

From Gastric Fluids to Mucociliary Clearance: Reviewing the Physicochemical Bottlenecks of Oral and Intranasal Antiviral Absorption

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The clinical management of influenza largely depends on oral antiviral medications. However, orally administered drugs must face several physiological barriers before reaching the respiratory tissues. In contrast, intranasal delivery offers a direct and localized approach at the viral infection site, although it also faces distinct physical limitations. This systematic review compares the physicochemical and physiological barriers influencing drug absorption through both administration routes. A comprehensive literature search was conducted using PubMed, ScienceDirect, and Google Scholar for peer-reviewed studies published between 2016 and 2026. Data extraction focused on biological degradation, residence time, and tissue-specific bioavailability parameters (C_{max} and T_{max}) of influenza therapeutics. The findings indicate that oral delivery is primarily limited by chemical and metabolic barriers. Antiviral drugs must withstand acidic gastric conditions (pH 1.5–3.5), intestinal enzymatic degradation, and extensive hepatic first-pass metabolism, which reduce respiratory tissue distribution and often require higher systemic doses. In comparison, intranasal delivery bypasses hepatic clearance but is constrained by mechanical barriers, including rapid mucociliary clearance within 15–20 minutes and the dense mucin glycoprotein network with pore sizes ranging from 20–200 nm. Structural analysis demonstrates that microparticles (1–10 µm) are trapped within mucus and rapidly cleared by ciliary action, whereas nanoparticles (<200 nm) better match mucosal pore dimensions. This nanoscale size enables improved mucus diffusion, enhanced epithelial transport, and direct cellular uptake through endocytosis. Current literature highlights increasing interest in polymeric nanocarriers, particularly chitosan, thiolated chitosan, PLGA, and sodium alginate, to improve localized drug absorption through mucoadhesion and sustained drug release. Overall, findings suggest that oral delivery is predominantly affected by chemical and metabolic degradation, whereas intranasal delivery is mainly constrained by mechanical clearance and drug carrier size. Future strategies should focus on nanoengineered biomaterials capable of penetrating or adhering to nasal mucus to prolong residence time and enhance therapeutic efficacy.

Keywords: influenza antivirals, intranasal drug delivery, oral bioavailability, mucosal permeation, polymeric nanoparticles

PP-058

Novel PARP Inhibitor-NOTA Conjugate: Synthesis, Optimisation, and Gallium-68 Radiolabelling for Cancer Theranostics

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Poly ADP-ribose polymerase-1 (PARP-1) is overexpressed in multiple carcinomas and represents a validated therapeutic and imaging target. Existing radioiodinated and radioastatinated PARP inhibitor (PARPi) conjugates suffer from in vivo serum instability and rapid deiodination or deastatination. Gallium-68 (⁶⁸Ga), a generator-produced positron emitter with a 68-min half-life compatible with small-molecule pharmacokinetics, and NOTA, a macrocyclic chelator enabling efficient room-temperature ⁶⁸Ga complexation, were selected to overcome these limitations. To synthesise a novel PARPi-NOTA precursor and optimise its radiolabelling with ⁶⁸Ga through systematic evaluation of conjugation chemistry, chelator selection, solvent systems, and coupling reagents. An Olaparib derivative was conjugated with NOTA-O^tBu (protected) or NOTA-NHS ester (pre-activated) under multiple reaction conditions. Solvents tested were DCM, toluene, and DMF; coupling reagents were DCC/DMAP and TBTU/TEA. Conjugation yields were quantified by LC-MS at dual wavelengths (254 nm and 305 nm). The protected conjugate was deprotected using trifluoroacetic acid (TFA) or 85% phosphoric acid in DCM or DMF. The deprotected precursor was radiolabelled with ⁶⁸GaCl₃ at pH 4.0–4.5 for 10 min at room temperature or 40–45°C, and characterised by radio-HPLC and radio-TLC. TBTU/TEA in DMF afforded the highest conjugation yield (>69%), outperforming DCC-based couplings (14–23%) and NOTA-NHS reactions (<32%). TFA in DCM (1:3 v/v) delivered >97% deprotection yield, while TFA in DMF was ineffective and phosphoric acid in DMF gave concentration-dependent yields. Radiolabelling gave radiochemical yields (decay-corrected) of 75±11% and 70±10%, and radiochemical purity of 81±10% and 73±11% at room temperature and 40–45°C (both n=3; p>0.05), corroborated by radio-TLC (76.1% and 73.6%). The radiolabelled product (clogP: -7.57) was more hydrophilic than its NOTA-Olaparib precursor (clogP: -3.16), consistent with net positive charge from incomplete hexadentate Ga³⁺ coordination. TBTU-mediated amide coupling in DMF followed by TFA/DCM deprotection provides a reliable synthetic route to PARPi-NOTA precursors for ⁶⁸Ga-theranostic applications in cancer.

Keywords: PARP inhibitor, Olaparib, NOTA, gallium-68, bioconjugation

PP-071

Microgreens as Emerging Functional Foods: Antioxidant Property and Phytochemical Composition of Malaysian Leafy Vegetables

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Microgreens are young seedlings of vegetables and herbs harvested at an early stage of development, typically within 7 to 21 days after germination. They are increasingly recognized for their dense nutritional profile, which often surpasses that of mature plants. This study aimed to comparatively evaluate the total phenolic content (TPC), total flavonoid content (TFC), and antioxidant activity of microgreens and their mature counterparts across eight commonly consumed Malaysian leafy vegetables: *Brassica rapa* subsp. *chinensis* (pak choi), *Ipomoea aquatica* (water spinach), *Amaranthus tricolor* (Chinese spinach), *Cosmos caudatus* (ulam raja), *Brassica juncea* (mustard green), *Coriandrum sativum* (coriander), *Lactuca sativa* (lettuce), and *Brassica oleracea* (kale). Microgreens were cultivated under controlled laboratory conditions, while mature plant samples were commercially sourced. Lyophilized plant materials were extracted using methanol, and the phytochemical and antioxidant properties were evaluated using standard assays. Results showed that microgreens exhibited significantly higher levels of TPC and TFC as well as enhanced antioxidant activity (DPPH assay) compared to their mature counterparts. These findings underscore the potential of microgreens as nutrient-dense, antioxidant-rich functional foods and support their integration into sustainable urban agriculture systems.

Keywords: microgreens, functional foods, antioxidant activity, phenolic components, nutritional profiling

PP-072

Radiolabeling Conditions and *In Vitro* Stability of Bismuth and Lead Complexes with Novel 18- and 21-Membered Benzodiazacrown Macrocycles for Targeted Alpha Therapy

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Targeted alpha therapy (TAT) is an emerging radiopharmaceutical approach that use cytotoxic and short tissue range of alpha particles for selective tumor cell destruction. Bismuth-212/213 and lead-212 are among the most clinically promising TAT radionuclides. A critical challenge in TAT radiopharmaceutical development is the availability of chelators capable of rapid, efficient radiolabelling under mild conditions while retaining the radiometal securely under physiological conditions. This study investigates the radiolabelling efficiency and in vitro stability of two novel benzodiazacrown macrocycles with amide-pendant arm named BADAM-18 and BADAM-21 with Bismuth and Lead cation. Radiolabelling was performed in 0.1 M HEPES buffer (pH 7) across a ligand concentration range of 10^{-2} to 10^{-6} M, under both room temperature and 90°C conditions. Radiochemical yields (RCY) were determined by thin-layer chromatography using systems developed specifically for these chelators. In vitro stability was assessed in fetal bovine serum at 37°C over seven days, and kinetic inertness was evaluated by metal cation challenge against physiologically relevant competitors (Cu^{2+} , Fe^{3+} , Zn^{2+} , Mg^{2+} , Ca^{2+} , Na^{+}). [^{207}Bi]Bi-BADAM-18 achieved $87 \pm 11\%$ RCY at room temperature, with no improvement upon heating, indicating rapid, temperature-independent complexation kinetics highly suited to heat-sensitive antibody conjugates. [^{207}Bi]Bi-BADAM-21 reached $>99 \pm 2\%$ RCY upon heating to 90°C, reflecting a thermally activated binding mechanism driven by the larger, more flexible cavity. Lead complexes achieved moderate yields (50–73%) at 10^{-3} M, with efficiency declining sharply at lower concentrations. In vitro, both bismuth complexes demonstrated strong serum stability: Bi-BADAM-18(RT) retained $>90\%$ integrity for two hours while Bi-BADAM-21 (heated) maintained $>90\%$ stability for four days. Both complexes showed robust resistance to endogenous metal transchelation. Lead complexes exhibited poor serum stability, attributed to the hemidirected coordination geometry of Pb^{2+} and insufficient donor group count in the current design. These results establish BADAM-18 and BADAM-21 as promising chelator for bismuth-based TAT radiopharmaceuticals. Further structural optimisation is needed to improve lead complex stability toward ^{212}Pb applications.

Keywords: targeted alpha therapy, radiopharmaceuticals, macrocyclic chelators, bismuth-212, lead-212, radiolabelling, in vitro stability, benzodiazacrown

PP-085

**Identification of Flavonoid Based Inhibitors Targeting the Dengue Virus Envelope Protein
Through Structure Based Molecular Docking & ADME Analysis**

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Dengue virus (DENV) infection remains a major global health challenge, with no approved antiviral therapy currently available. The viral envelope (E) protein, which mediates host cell recognition and membrane fusion, is a promising target for antiviral drug development. The study aims to identify potential small molecule inhibitors of the DENV E-protein through structure-based molecular docking and in silico pharmacokinetic evaluation. A library of 20 compounds was screened against the DENV E-protein crystal structure (PDB ID: 1OKE) using Auto Dock Vina. The docking protocol was validated by root-mean-square deviation (RMSD) analysis. Binding affinities, protein-ligand interactions and absorption, distribution, metabolism, and excretion (ADME) properties were subsequently assessed. Among the screened compounds, Apigenin and Naringenin exhibited the strongest binding affinities with docking scores of -9.5 and -9.2 kcal/mol respectively. Both compounds formed stable hydrogen bonds and hydrophobic interactions with key residues in the E-protein binding site, indicating favorable complex stability. ADME analysis predicted good drug-like characteristics including compliance with Lipinski's Rule of Five, high gastrointestinal absorption and acceptable aqueous solubility. Overall, Apigenin and Naringenin demonstrated promising binding affinity, favorable molecular interactions and suitable pharmacokinetic profiles supporting their potential as lead compounds for development of DENV E-protein inhibitors. These findings provide a basis for further experimental validation and optimization of flavonoid based antiviral agents against dengue viruses.

Keywords: dengue virus, envelope protein, molecular docking, flavonoids, ADME predictions

Regenerative & Translational Medicine (RTM)

PP-044

Inhibition Of Cxcr4/Cxcl12 Signalling Pathway Disrupted Migration of Proliferative Cells from Frontal to Midbrain and Further Impaired Locomotor Recovery in the 6-OHda-Lesioned Adult Zebrafish Parkinson's Disease Model

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Parkinson's disease (PD) is a progressive neurodegenerative disorder characterised by the degeneration of dopaminergic neurons (DpN) in the midbrain. Given the lack of curative treatment, current research efforts are directed towards neuroregeneration as potential PD therapy. To understand the cellular events of neuroregeneration, we established a 6-OHDA-lesioned adult zebrafish PD model characterised by DpN loss (approximately 85%) in the ventral diencephalon (vDn) (analogous to the human midbrain). Moreover, the established model presented significantly reduced locomotor activity, reflecting functional impairment following DpN loss. It appears that newly generated cells may have migrated from the olfactory bulb and telencephalon (frontal brain) to the vDn via the preoptic area (PoA) at 14 days post-lesion (dpl). The CXCR4/CXCL12 signalling pathway has been implicated in regulating this process. This study aimed at validating the PoA as the potential migratory route through which proliferative cells would migrate towards the vDn by inhibiting the CXCR4/CXCL12 signalling pathway using AMD3100 (a selective antagonist of CXCR4 chemokine receptor). Firstly, the zebrafish (male, 4 months old, n=6/group) were injected with neurotoxin 6-OHDA at the vDn. Next, 0.7-12mM AMD3100 were administered to the 6-OHDA-lesioned zebrafish via intracerebroventricular injection at the PoA on 13 dpl. Proliferative cells were labelled using 5-ethynyl-2'-deoxyuridine (EdU) (*i.p.*) on 14 dpl. Locomotor assessment was performed to evaluate the effects of AMD3100 treatment in 6-OHDA-lesioned zebrafish on 14 dpl (migration phase), 22 dpl (peak differentiation), and 30 dpl (full recovery), respectively. The highest concentration of AMD3100 (12mM) resulted in a significant accumulation of EdU-stained proliferative cells at the PoA, suggesting disrupted migration of proliferative cells towards the vDn. The locomotor recovery of AMD3100-treated 6-OHDA-lesioned zebrafish was also further compromised when compared to the 6-OHDA-lesioned zebrafish group. This framework warrants further investigation into DpN restoration and glial activation during adult zebrafish neuroregeneration.

Keywords: migration, neuroregeneration, adult zebrafish, CXCR4/CXCL12, 6-OHDA

Policy, Regulation & Ethics (PRE)

PP-016

**Anti-Money Laundering Framework and Pharmaceutical Crime in Malaysia:
A Statutory Gap Analysis**

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Pharmaceutical crimes, including the trafficking of falsified medicines, controlled substances, and unregistered products, generate substantial illicit proceeds that may constitute predicate offences for money laundering. In Malaysia, the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA) serves as the primary legislative instrument against financial crime; however, only the Dangerous Drugs Act 1952 is listed as a serious offence under its Second Schedule. Pharmaceutical statutes enforced by the Pharmacy Enforcement Division (PED) of the Ministry of Health Malaysia remain outside this scheme, leaving a critical statutory gap in addressing the financial dimensions of pharmaceutical crime. This study examines whether the existing legal framework adequately empowers the PED to investigate money laundering linked to pharmaceutical offences and identifies key statutory gaps constraining such investigations. A doctrinal methodology was adopted, supported by library-based research and semi-structured interviews with two senior PED officers. Primary sources analysed include AMLA, the Poisons Act 1952, the Sale of Drugs Act 1952, and the Dangerous Drugs Act 1952. A comparative analysis was conducted involving the United Kingdom, India, and Singapore, each of which has scheduled drug-related offences within its principal anti-money laundering legislation. Findings reveal that pharmaceutical statutes administered by the PED contain no provisions enabling money laundering investigation, and AMLA lists no offence under these statutes as a serious offence. The United Kingdom and Singapore address drug-related money laundering through general proceeds-of-crime legislation, while India directly schedules relevant offences within its Prevention of Money Laundering Act 2002, a model most compatible with Malaysia's legislative architecture. Malaysian law currently provides only a partial legal basis for addressing money laundering linked to pharmaceutical crimes. It is recommended that selected provisions enforced by the PED, particularly those concerning controlled substances, precursors, and falsified or substandard medicines, be inserted into the Second Schedule of AMLA, supported by enhanced inter-agency coordination.

Keywords: pharmaceutical crime, anti-money laundering, comparative law, legal adequacy, regulatory enforcement

PP-032

Pharmacists' Insights on Their Internship Training Experiences in Malaysian Community Pharmacies

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Community pharmacy has become an increasingly important sector for pharmacist workforce development in Malaysia, especially following the 2012 liberalisation of training pathways. However, little is known about the experiences of Provisionally Registered Pharmacists (PRPs) training in this setting. This study explored internship experiences in community pharmacies and examined their influence on professional and personal development. A qualitative study using semi-structured interviews was conducted with 17 pharmacists who had completed their PRP training in community pharmacies across Malaysia. Data was analysed thematically to capture lived experiences and identify factors influencing training outcomes and career trajectories. Placement choices were driven by personal interest, prior exposure, and perceived constraints in other sectors. Training support was more structured in chain pharmacies than in independent settings. Across sites, participants reported increased confidence, communication skills, clinical knowledge, and early leadership responsibilities. Work-life balance consistently emerged as a decisive factor in long-term career planning. Community pharmacy PRP training serves as a critical bridge between academic learning and practice, yet the quality of training remains inconsistent. These findings provide timely evidence to guide policy reforms to ensure consistent, high-quality internship nationwide implementation and enhance the expansion of community pharmacy services in Malaysia.

Keywords: community pharmacy, provisionally registered pharmacist, pharmacy internship, pharmacist training, professional development

PP-104

Content Validation of Assessment Instrument for Complex Generic Medicines Development and Registration in Malaysia

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Complex generic medicines are generic products with complex active ingredients, formulations, dosage forms, routes of administration, or drug-device combinations. Their development and registration present greater scientific, manufacturing, and regulatory challenges than conventional generics. Despite their increasing importance in improving access to affordable medicines, limited evidence is available on the challenges faced by pharmaceutical manufacturers and regulatory affairs personnel in Malaysia. A valid questionnaire is therefore needed to systematically assess these challenges. To evaluate the content validity of an instrument designed to assess challenges in the development and registration of complex generic medicines in Malaysia. A questionnaire was developed through literature review and adaptation of published survey instruments, with modifications to reflect the Malaysian regulatory context. The questionnaire comprised company demographics, technical and manufacturing challenges, regulatory and compliance issues, supply chain, quality control, human resources, financial considerations, and collaboration ecosystem. Five experts from regulatory authority, academia and pharmaceutical industry independently evaluated each item for relevance, clarity, simplicity, and comprehensiveness using a four-point scale. Item-level Content Validity Index (I-CVI), Scale-level Content Validity Index based on universal agreement (S-CVI/UA), and average (S-CVI/Ave) were calculated. Items with I-CVI below the acceptable threshold were revised based on expert feedback. The questionnaire consisted of 57 items across 5 domains. The questionnaire demonstrated excellent content validity, with an S-CVI/Ave of 0.936 and S-CVI/UA of 0.682. I-CVI values ranged from 0.667 to 1.000, with most items achieving an I-CVI ≥ 0.80 . Expert feedback mainly improved wording, terminology, and clarity while retaining the questionnaire's conceptual domains. All relevant recommendations were incorporated before pilot testing. The questionnaire demonstrated satisfactory content validity and is suitable for use in assessing the challenges in complex generic medicine development and registration in Malaysia. It provides a validated instrument for nationwide surveys and may inform future regulatory and policy improvements.

Keywords: complex generic medicines, content validity, questionnaire development, regulatory science, Malaysia

PP-113

Public Perceptions and Future Acceptance of Community Pharmacist-Led Vaccination Services in Malaysia

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Vaccination is one of the most effective public health interventions for preventing infectious diseases. Several countries have expanded vaccination services to include community pharmacists to improve accessibility and reduce the burden on healthcare facilities. However, community pharmacists are currently not authorized to administer vaccines in Malaysia. This study aims to assess public perception and future acceptance of community pharmacist-led vaccination services in Malaysia. A cross-sectional study was conducted among Malaysian age 18 years and above using a self-administered questionnaire. Data were collected between March and June 2026 using a snowball sampling method. The questionnaire assessed demographic characteristics, public perceptions of vaccination, perceptions of community pharmacists, future acceptance of community pharmacist-led vaccination services, and perceived advantages and barriers. Data were analysed using descriptive and inferential statistics in SPSS version 29. A total of 419 participants completed the survey. The majority agreed that community pharmacists should be trained and certified to provide vaccination services (92.9%) and were willing to receive vaccination from certified community pharmacists (88.8%). A positive correlation was observed between perceptions of community pharmacists and future acceptance of community pharmacist-led vaccination services ($\rho = 0.693$, $p < 0.001$). The main perceived advantages were convenient location and shorter waiting times, while the main possible barriers were concerns about pharmacists' injection skills and vaccines side effects. The Malaysian public generally demonstrated positive perceptions towards community pharmacist-led vaccination service in the future. The findings support the potential implementation of community pharmacist-led vaccination services in Malaysia while highlighting the need to address public concerns through appropriate training and education.

Keywords: community pharmacist, pharmacist, vaccination, public perception

PP-118

Knowledge and Attitudes of Community Pharmacists in Managing Dermatological Conditions: A Malaysian Cross-Sectional Study

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Dermatological conditions such as acne, psoriasis, and atopic dermatitis are common in Malaysia and significantly impact patients' quality of life. Community pharmacists (CPs) play an important role in managing dermatological conditions through counselling and promoting adherence to topical medications. However, limited evidence exists on their knowledge and attitudes in this area. The study aimed to assess the knowledge of CPs regarding dermatological conditions and topical treatments and to evaluate their attitudes toward their responsibilities in patient education, application guidance, and promotion of treatment adherence. A cross-sectional study was conducted among 356 CPs in Malaysia using a self-administered online questionnaire. Descriptive statistics were used, and associations were examined using Mann-Whitney, Kruskal-Wallis, and Spearman's correlation tests. Most CPs demonstrated good knowledge (mean = 26.26 ± 2.75) and positive attitudes (mean = 38.17 ± 4.76). Common conditions such as atopic dermatitis, psoriasis, and acne were well recognized, while knowledge gaps were observed in less common dermatological conditions. Knowledge was significantly associated with ethnicity, location, and education level, while attitude varied by gender, age, and pharmacy type (independent vs. chain), work status, and education level. A weak but significant positive correlation was found between knowledge and attitude ($p = 0.203$, $p < 0.001$). Respondents demonstrated high questionnaire-based knowledge scores and reported frequent counselling practices; however, these findings may be affected by self-selection and social desirability bias. Nevertheless, targeted training and workplace support are needed to address specific gaps and enhance CPs' role in dermatological condition management.

Keywords: community pharmacists, knowledge, attitude, dermatology, topical medication

PP-127

Disposal Trends and Cost Analysis of Blood and Blood-Forming Organ Drugs Collected from Alpro Pharmacy

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Medicines acting on the blood and blood-forming organs (Anatomical Therapeutic Chemical [ATC] Group B) are essential for managing cardiovascular and haematological conditions; however, their high costs and long-term use may contribute to medication wastage, financial loss and environmental harm. This study aimed to determine the frequency distribution and estimated financial loss associated with disposed ATC Group B medicines and to evaluate trends in their disposal volume and cost from 2023 to 2026. Unused and expired medicines collected through the Alpro Pharmacy medication disposal programme were classified according to the ATC system. Medicine details were recorded in Microsoft Excel and analysed descriptively. Unit prices were obtained from Alpro Pharmacy, DoctorOnCall and the Malaysian Pharmaceutical Information Centre (MyPriMe), and estimated costs were calculated using the average of the available prices. Of 32,584 medication units disposed of in 2026, 2,453 (7.53%) belonged to ATC Group B. Despite their relatively low disposal volume, these medicines accounted for the highest estimated financial loss among all ATC main groups, amounting to RM10,626.68 (32.41% of the total estimated loss). Within Group B, antithrombotic agents were the predominant subgroup, comprising 1,925 units (78.5%) and contributing an estimated financial loss of RM10,364.16. Trend analysis demonstrated persistent disposal of Group B medicines between 2023 and 2026, with a marked increase in estimated financial loss in 2026 despite a comparatively low disposal volume. In conclusion, the financial burden of pharmaceutical waste is influenced not only by the quantity discarded but also by the cost of individual medicines. Cost-weighted disposal monitoring and targeted strategies, particularly for antithrombotic agents, are therefore needed to promote appropriate medicine use and reduce preventable wastage.

Keywords: medication disposal, pharmaceutical waste, financial loss, blood and blood-forming organs, antithrombotic agents

PP-128

Medicine Price Transparency in Malaysia: A Cross-Sectional Analysis of Public Knowledge, Practices, and Sociodemographic Predictors

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In Malaysia's private healthcare sector, varying medicine prices lead to transparency and affordability concerns. The Price Control and Anti-Profitteering Order requires private facilities and community pharmacies to display medicine prices, but limited public awareness may affect policy effectiveness. This study assessed public knowledge and practices regarding the Medicine Price Display Policy and identified key sociodemographic predictors of these levels. A cross-sectional survey of 386 Malaysian adults was conducted using convenience sampling. Data were collected via a newly developed, content-validated questionnaire administered through Google Forms and face-to-face methods, and analyzed using descriptive and binary logistic regression. Most respondents (93.8%) demonstrated moderate-to-poor knowledge of the policy, with only 6.2% showing good knowledge. Conversely, 48.4% demonstrated good practice, while 51.6% showed moderate-to-poor practice. Regression analysis identified no significant predictors for knowledge. However, good practice was significantly predicted by gender (females: COR = 1.65, $p = 0.023$), age (45–54 years: COR = 0.45, $p = 0.014$), rural residence (COR = 0.55, $p = 0.035$), purchasing medicines five or more times (COR = 0.18, $p = 0.001$), and residing in Kedah (COR = 3.97, $p = 0.032$), Perak (COR = 4.03, $p = 0.014$), Selangor (COR = 3.16, $p = 0.038$), or Putrajaya (COR = 5.60, $p = 0.050$). Public knowledge of the Medicine Price Display Policy is critically low, yet consumer practices are moderately positive, likely driven by healthcare cost management. Strengthening public education and awareness is essential to promote informed purchasing decisions.

Keywords: medicine price display, price transparency, public knowledge, consumer practice, Malaysia

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